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Allianz Trade

How AI is rewiring global trade

Concentrating Power, Dependencies, and Supply Chains

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Content

Page 3-4
Executive Summary

Page 5-11
An AI boom built in Asia

Page 12-20
The growth in trade in services and data flows is accelerating with the diffusion of AI

Page 21-24
The world is flat: AI reconfigures the geography of trade and dependencies

Page 25-29
The race for AI dominance is increasingly also being fought through industrial policy

Executive Summary

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AI and trade are no longer separate policy domains. AI growth depends on globalized supply chains for semiconductors, computing infrastructure and digital services while trade is increasingly shaped by who controls AI infrastructure, data flows and cloud capacity.

- **Trade openness is a structural precondition for AI-driven productivity gains.** Open economies benefit disproportionately from cheaper inputs, faster innovation diffusion and AI adoption spillovers. Trade openness accounts for 23% of the variation in AI adoption across countries with highly open economies, such as Singapore, UAE and Ireland, leading in diffusion. However, while AI can significantly boost growth, its benefits are unlikely to be evenly distributed.

- **Export volumes of AI-enabling goods have surged from USD1trn in 2014 to USD3.8trn in 2025 (+280%), accounting for 15% of global trade and far outpacing the 40% growth in goods trade overall.** Asia dominates the supply side, accounting for 65% of global AI-related exports and seven of the top ten exporters, led by China (18% of AI-related exports), Taiwan (12%) and Hong Kong (11%). The composition remains concentrated in intermediate inputs (76%) and equipment (23%), reflecting deep dependence on semiconductors and data-center infrastructure. On the demand side, the US has tripled its AI-related imports since 2023, underpinned by 5,427 operational data centers, 45% of the global total. Europe's import growth of just +40% underscores a widening infrastructure gap.

- **Services imbalances could scale with AI.** ICT services trade reached USD900bn in 2024 (11% of global services trade), with Ireland alone exporting USD173bn, exceeding both the rest of the EU (USD142bn) and the US (USD108bn), and reflecting its role as a billing hub for US multinationals. However, it masks structural dependence, with the EU excluding Ireland running a USD45bn ICT services deficit (mainly with Ireland), highlighting its reliance on US digital ecosystems. And things could get worse. Hypothetically, if the AI services subscription penetration rate of US providers such as ChatGTP Plus and Claude Pro increases from 3% to 50% in a high adoption scenario in the Eurozone, annual payments to US providers could reach EUR34bn (currently EUR2.7bn), equivalent to 20% of the current EU–US goods services deficit. AI thus acts as a scalable, recurring channel that exacerbates structural EU–US digital imbalances and has the potential to lower the US overall trade deficit.

Beyond the headline data, three structural dynamics are reshaping the global AI trade order:

1. Supply chain concentration as single points of failure

The AI supply chain is concentrating rapidly at the technological frontier. Taiwan, South Korea, the US and the Netherlands dominate the production of advanced chips, high-bandwidth memory and semiconductor equipment, creating critical single points of failure with no near-term redundancy. Unlike broader goods trade, which has partially reoriented along geopolitical lines, AI-related goods remain largely unfragmented across geopolitical blocs, with semiconductors and key components still flowing globally except at the technological frontier (notably advanced US export controls) due to deep interdependencies, challenging the narrative of technological decoupling.

2. Infrastructure as geopolitical power

Control over data centers, cloud infrastructure and computing capacity is emerging as a primary source of geopolitical leverage, distinct from traditional goods trade dynamics. Europe's strategic exposure is acute. With less than 10GW of operational data-center capacity, four times below the US (60GW), and a development pipeline six times smaller, the gap is not expected to narrow. US hyperscalers already control 35% of European computing capacity and account for nearly half of the upcoming pipeline, consolidating a 70% cloud market share. Structural constraints, fragmented regulation, complex permitting processes, grid connection delays, no domestic hyperscaler and limited VC or state-backed funding reinforce this dependency. Finally, reliance on Asian hardware inputs complements the US dependence that will increase without sufficient EU investment in the area. Under that backdrop, Europe is permanently under the threat of a US "kill switch" on cloud data. In the meantime, this effective expansion of this key infrastructure is dependent on inputs sourced in Asia, notably in China (14% import ratio for AI datacenter components) and Taiwan for semiconductor, and as a result exposed to current supply-chain disruption risks along new geopolitical tensions in Middle East. Regaining control and strengthening sovereignty on AI good production but also service delivery is critical for the region to avoid suffering a twin external dependence.

3. Industrial policy: from subsidies to protectionism

Global tariffs on AI-enabling goods have fallen from 5.6% in 2015 to 2.8% in 2025, well below the 7.8% average for manufacturing overall, but non-tariff measures have surged, driven primarily by the US and China, reflecting a strategic shift from financial support toward technology protectionism. Over 3,600 AI subsidy measures targeting critical materials, semiconductors, GPUs, computing equipment and optical fibers are now in force globally, with China having nearly doubled its trade coverage over five years, followed by South Korea, Malaysia, the US and Japan. The overlap of national and multilateral governance regimes is concentrating production, infrastructure and modelling capabilities in the hands of a small number of economies. Regulatory fragmentation is becoming a binding constraint on AI services trade, with ICT trade restrictiveness already explaining 5% of the cross-country variation in AI services flows. The EU and US are advancing incompatible regulatory frameworks, creating compliance friction that reinforces existing structural imbalances.

An AI boom built in Asia

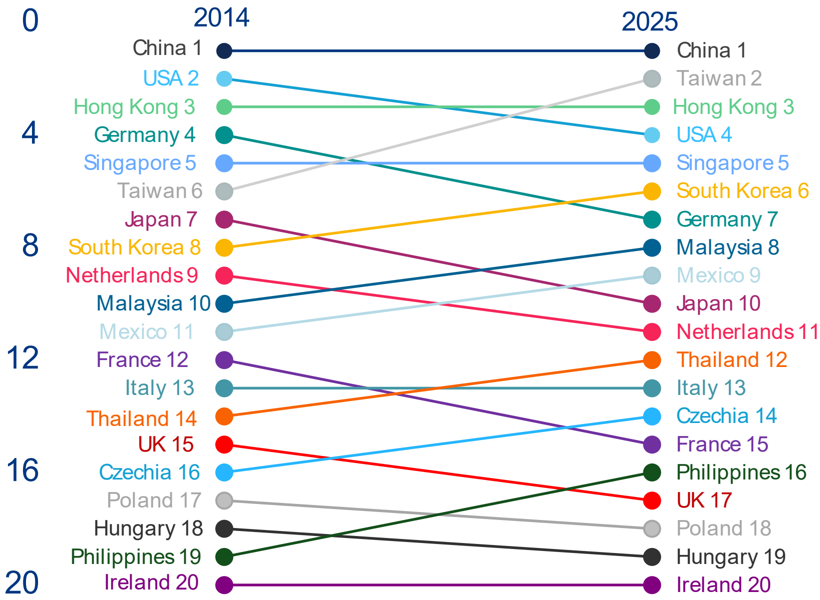
AI fundamentally depends on global trade in goods and services, with a complex, internationally fragmented value chain spanning raw materials, semiconductors, computing hardware, cloud infrastructure, data and final applications (AI-enabling goods). Trade enables access to critical inputs, such as advanced chips, specialized equipment and energy resources, while facilitating the cross-border exchange of intermediate goods, services and knowledge needed to build and scale AI systems. However, the strong geographic concentration of key stages of the AI supply chain coupled with heightening frictions tightening the exchange channel (tariffs, shutdown of the Strait of Hormuz) raises concerns about unequal access and the risk of a widening technological divide.

Over the past decade, global trade in AI-related goods has doubled, far outpacing the growth of overall goods trade and of non-AI-related goods in particular. Global AI-related goods trade has expanded rapidly over the past decade, doubling from USD1.9trn in 2014 to USD3.8trn in 2025. Meanwhile, overall global trade in goods increased more moderately, rising by +40% to USD26trn in 2025. This growth was increasingly supported by AI-related sectors, with trade in non-AI goods rising by only a third. This expansion has seen three distinct acceleration phases. The first acceleration occurred in 2017–2018 (+15% y/y and +11% y/y respectively), driven by fundamental developments in need for AI-related goods and the introduction of the transformative architecture, which underpins most large language models. A second surge followed in 2021 (+30% y/y), driven by the release of early large-scale models, including GPT-3 (2020) and DALL·E (2021) and paving the way for the emergence of generative AI in 2022, which needed more confounding technical infrastructure for model innovation and computation. However, momentum weakened in 2022 amid a challenging macroeconomic environment. Higher interest rates and elevated inflation combined with the introduction of US export controls on advanced AI chips and semiconductors, led to a marked deceleration in AI-related goods trade growth (+4% y/y in 2022), followed by a contraction in 2023 (-7% y/y), with export values falling below their 2021 peak. More recently, advances in multimodal and agentic AI systems have reignited investment momentum. As AI became increasingly available and embedded in the economic system, AI-related goods exports reached new heights in 2025

(+22% y/y). Early 2026 data, covering January and February, show that the boom continues, especially in countries such as Taiwan (+62% y/y) and South Korea (+105% y/y).

Asia has firmly established itself as the center of global trade in AI-enabling goods. The region accounts for 65% of global exports and dominates the entire value chain. The region is at the heart of an increasingly concentrated landscape, with seven of the top ten exporters representing 80% of global trade. China leads with a 18% share, followed by Taiwan (12%) and Hong Kong (11%). Singapore (7%), South Korea (6%), Malaysia (4%) and Japan (3%) also reinforce the region's strength. The US (8%), Germany (5%) and Mexico (4%) are the only non-Asian economies in this group, highlighting the strong presence of AI trade in Asia. This dominance has deepened over time as competitive dynamics have shifted in the region's favor, causing advanced economies to lose ground. Despite a slight decline in its share (-3pps), China remains the leading exporter, while Taiwan's rise from 6th place in 2014 to 2nd place in 2025 (+7pps) marks the most significant shift. This was supported by a +53% year-on-year export surge in 2025 (Figure 1). Hong Kong has also gained prominence, further reinforcing Asia's lead. At the same time, emerging players are gaining traction: Mexico recorded the fastest growth in 2025 (+62% year on year), while Malaysia, Thailand (+49% year on year) and the Philippines are expanding as alternative hubs. Together, they accounted for 11% of global trade (+3pps) and 18% of additional trade in 2025. By contrast, most advanced economies, including those in Europe, have lost ground. The US, for example, slipped from second to fourth place, reflecting a relative decline in manufacturing-based AI trade despite continued strength in high-value innovation.

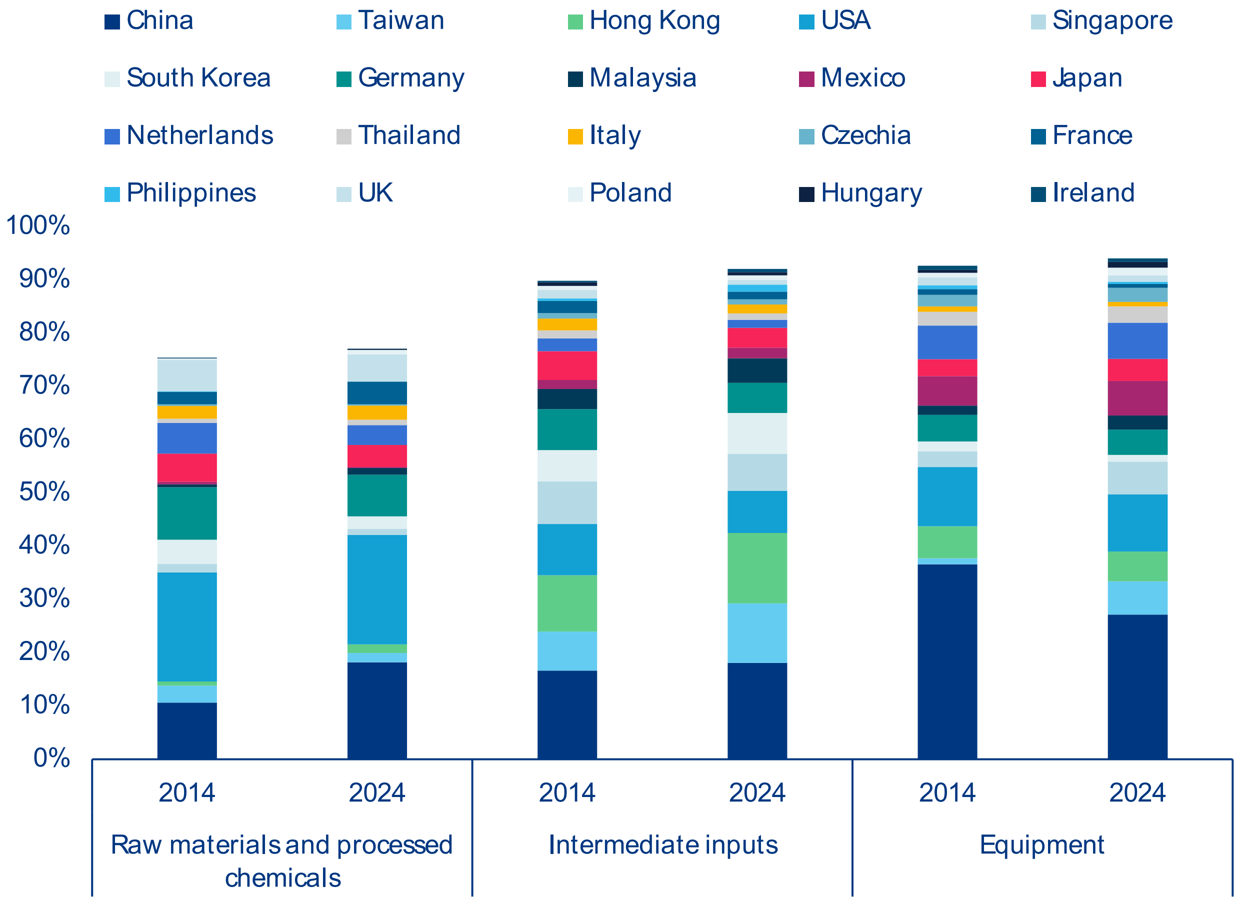
Figure 1: Top 20 AI-enabling goods exporters ranking (2014 vs 2025)



Sources: UN Comtrade, National Customs, Allianz Research

Although AI-enabling goods trade is already highly concentrated globally, the value chain reveals even greater specialization. Leading exporters cluster at distinct stages, particularly in raw materials, processed chemicals and equipment. AI-enabling goods can be categorized into three groups: raw materials and processed chemicals, intermediate inputs and equipment. Raw materials and processed chemicals constitute critical substances that are fundamental to the fabrication of AI equipment. At the aggregate level, this category represents the most diversified segment of the AI value chain. The top 20 exporters in this category account for three-quarters of global exports. This indicates significant concentration of the primary inputs for AI development, and this concentration only strengthens further upstream in the value chain, with the top 20 exporters representing over 90% of exports in the intermediate inputs (processed or partially manufactured goods) and equipment categories (specialized tools and machine components essential for AI development). China and the US have a particularly pronounced footprint at the extremes of the value chain. Together, they account for 39% of global exports of both raw materials and processed chemicals, as well as equipment (Figure 2). For intermediate inputs, the level of concentration remains high in absolute terms, but it is comparatively more evenly distributed, with China, Hong Kong and Taiwan collectively representing 43% of global exports, and each country holding a similar share. Notably, shifts in countries' global export shares of intermediate inputs have been linked to changes in the overall ranking. Taiwan and South Korea have gained +4pps and +2pps, respectively, reaching 11% and 8% of the global export share. Similarly, Hong Kong's share has increased by +3pps to 13% over the past decade. However, this has not allowed it to become the second-largest exporter, as Taiwan has made a notable entry at the high end of the value chain. It has gained +5pps to reach a 6% export share in equipment; it is the only country, alongside Singapore, to have increased its export share in this category. This has come at the expense of China, which has lost -9pps and stands at a 27% export share.

Figure 2: Decomposition of the AI value chain (% share of global trade for each category 2014 vs 2024)



Sources: UN Comtrade, National Customs, Allianz Research.

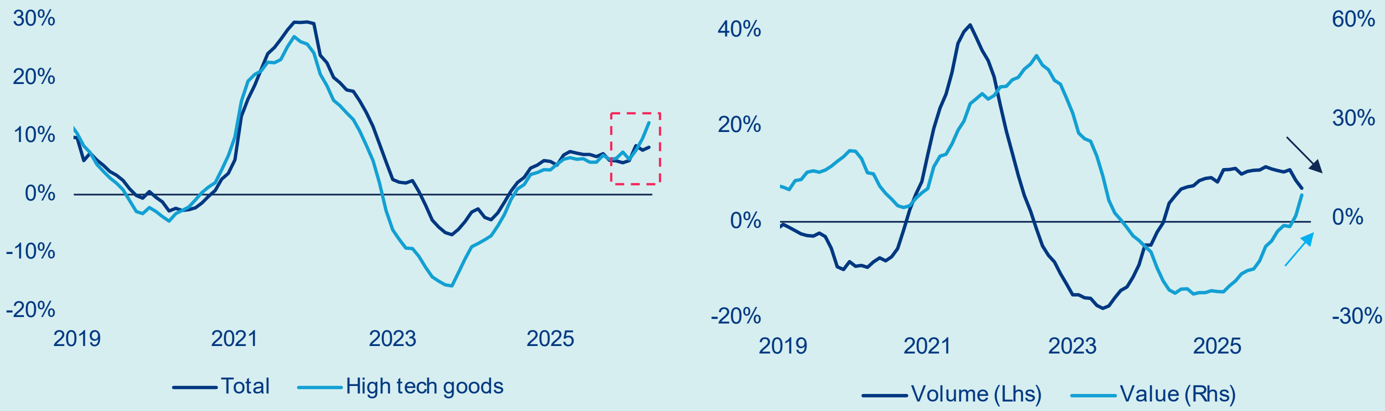
An ultra-segmented AI supply chain, with Europe holding a modest role. The global AI ecosystem is sharply segmented geographically. Upstream activities – chip design, semiconductor equipment, data-center infrastructure and telecoms – are concentrated in Western economies, with the US dominating GPU/CPU architecture and Europe and Japan leading in lithography and wafer-production machinery. Downstream, Asia takes over almost entirely: China holds a critical position in rare earth processing, battery manufacturing and electronic components; South Korea and Taiwan anchor the foundry landscape, the former specializing in memory chips and the latter as the undisputed leader in advanced logic manufacturing. Malaysia and Thailand have developed strong packaging and assembly capabilities, with Thailand increasingly focused on data-center hardware. Critical raw materials are similarly dispersed – nickel from Indonesia and Australia, cobalt from the DRC, lithium from Chile, helium from Qatar, silicon from China. Europe's position across this architecture is, overall, low to moderate, leaving the region with limited strategic leverage and meaningful exposure to supply-chain disruptions at any of its key upstream or downstream nodes.

The trade of goods related to AI is dominated by intermediate inputs and equipment, reflecting a surging demand for high-performance AI infrastructure. Although raw materials and processed chemicals account for only 2% of the trade in AI-enabling goods, exports are dominated by intermediate inputs (76% of total AI-related exports) and equipment (23%). This masks significant country-level specialization and exposure to supply risks. The UK stands out in the upstream sector, with a 7% share of raw materials (+5pps above the global average). At the intermediate stage, the strongest concentration is in Asia: South Korea and the Philippines rely heavily on this segment (95% and 91% respectively), as do Hong Kong and Taiwan, all of which are well above the global average of 75%. European exporters such as Italy (84%) and France (83%) exhibit comparable, albeit less pronounced, patterns. At the downstream stage, equipment exports are more prevalent in the Netherlands (57%), Mexico (49%), the Czech Republic (45%) and Thailand (43%). Meanwhile, China and the US have a more balanced approach, with each country having a 68% share of intermediate inputs and equipment shares of 31% and 28%, respectively.

Box: The Middle East crisis could send semiconductor prices surging

Over the past two years, two-thirds of the expansion in Asian semiconductor exports were due to price increases and one-third due to higher volumes. Risks of energy shortages from the Middle East crisis could send prices even higher, given the already tight and extremely concentrated supply. Looking at the recent results of the biggest foundry worldwide located in Taiwan (70% market share in 2025) as a proxy for the advanced-node segment of the supply chain, wafer shipments grew +28% year-on-year in Q1 2026, a robust rate by historical standards. However, revenue in USD terms expanded by +40.6% over the same period, leaving a gap of approximately 12pps that cannot be attributed to volume alone. The explanation lies primarily in a structural shift in product mix rather than in nominal price increases on comparable goods. Taiwan's production capacity base has migrated progressively toward 3nm and 5nm process technologies, which carry substantially higher per-wafer prices than the legacy nodes they are displacing in the revenue composition; advanced technologies now account for 74% of total wafer revenue despite representing a modest share of physical units shipped. However, a look at China's trade statistics suggest that inflationary effects are rather broad-based as even for the less advanced chips used for industrial purpose – above 28nm – we observe a diverging trend between export volumes (downward) and value (upward) in Q1 (Figure 3, right). Chinese data show that strong volume on AI products is masking rising disruption risks in manufacturing production (over -10% contraction of global shipments of smartphone and computer estimated this year, Figure 3, left), as well as for some electric and mechanical devices (i.e., transistors, printed circuits, HVAC, batteries) that are critical for building up the data-center fleet.

Figure 3: China exports value, total & high-tech goods (left) and China exports value & volume, semiconductor in USD, 12-month trailing yoy% (right)



Sources: China National Customs, Allianz Research

The sustained increase in semiconductor pricing is best understood as a demand-led phenomenon, with supply-side constraints serving as an amplifying mechanism. The commercial deployment of large language models from late 2022 onwards ramped up the compute requirements of hyperscale cloud operators. Since then, the transition from AI model training to large-scale inference has extended and broadened the requirement for advanced silicon across a wider range of end-users. Naturally, this is squeezing supply: Leading-edge foundry capacity requires three to five years and capital commitments in the tens of billions of dollars to come online. The imbalance between demand urgency and supply inelasticity has given foundries meaningful pricing power, with Taiwanese semiconductor producers' gross margins expanding from 58.8% to 66.2% within a single year. Supply-side cost pressures related to critical materials, notably gallium, germanium, tungsten and antimony, all subject to successive rounds of Chinese export controls since 2023, are another increasingly consequential factor, with affected material costs rising in the range of 30-50%, though they are more of a medium-term geopolitical risk for now.

The fact that the majority of AI-enabling goods exports are driven by a small group of countries also makes their trade model highly vulnerable to potential shocks. Taiwan and Hong Kong are the most exposed to an AI bubble burst, with 74% and 59% of their exports being related to AI goods respectively (Figure 4). They are followed by Singapore and the Philippines (both 47%), Malaysia (43%) and South Korea (32%). In contrast, trade in AI-enabling goods accounts for 15% of US exports, indicating that American exports are relatively well diversified. Similarly, Mexico and Thailand (both 23%), China (19%), and Japan (18%) have a moderate level of export concentration in AI-enabling goods.

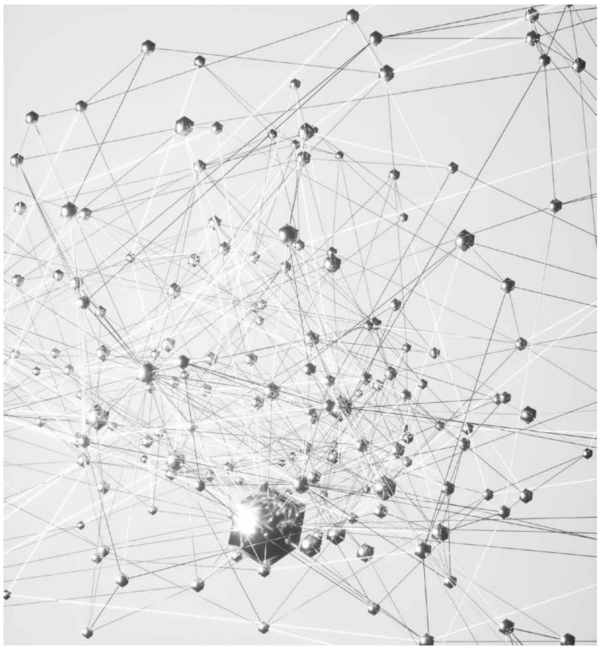
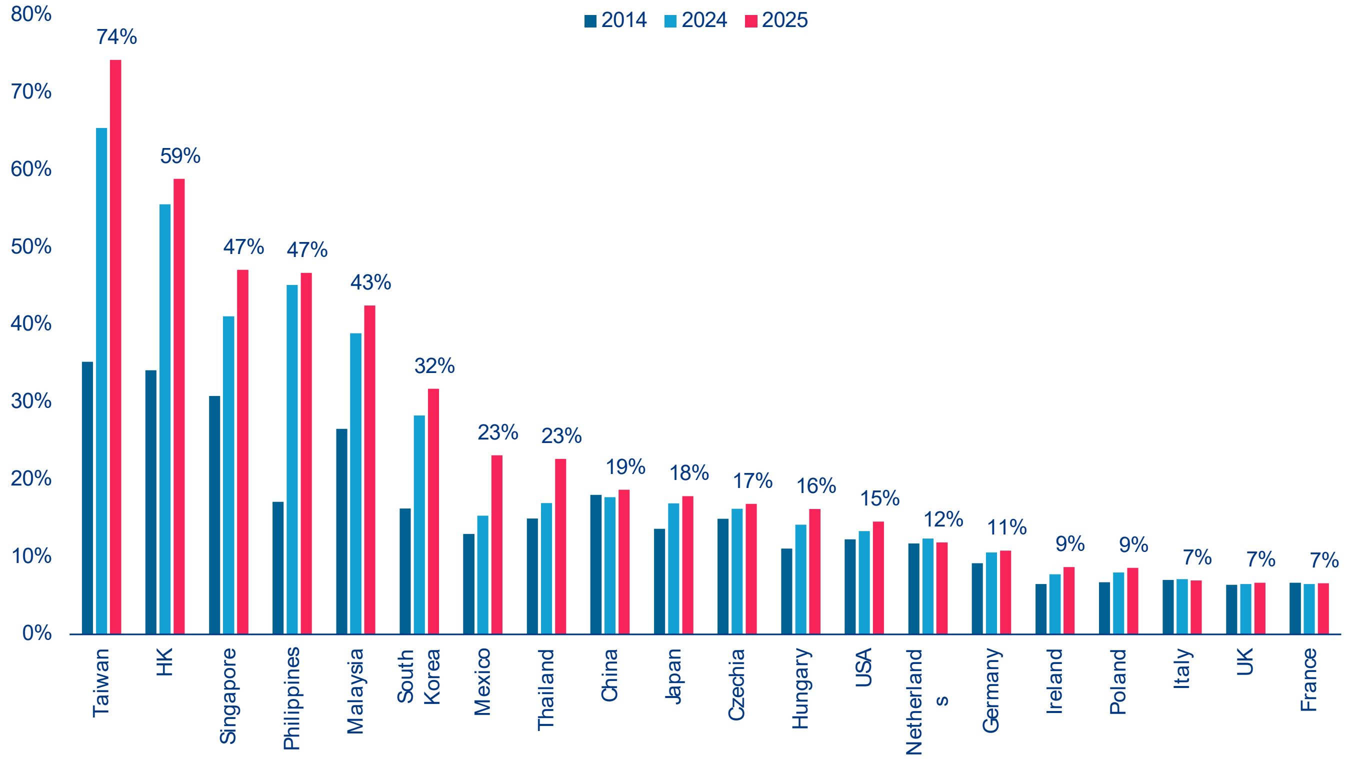


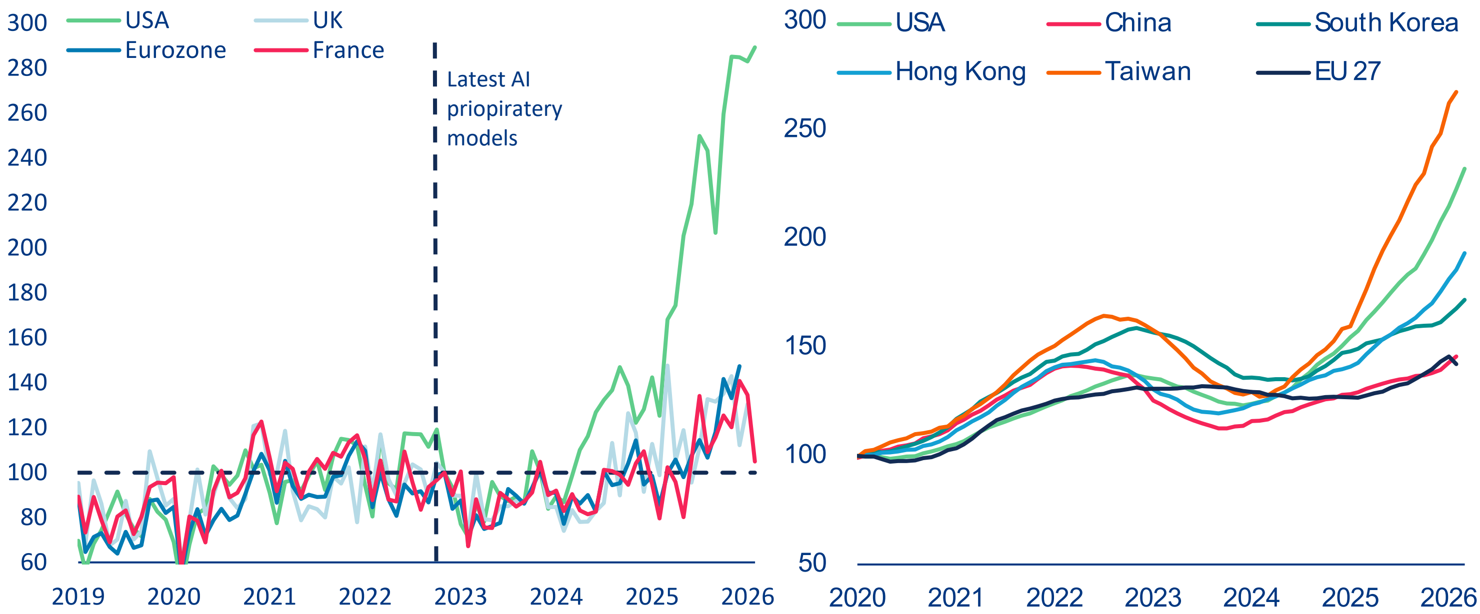
Figure 4: Share of AI-related goods (% of countries' total exports)



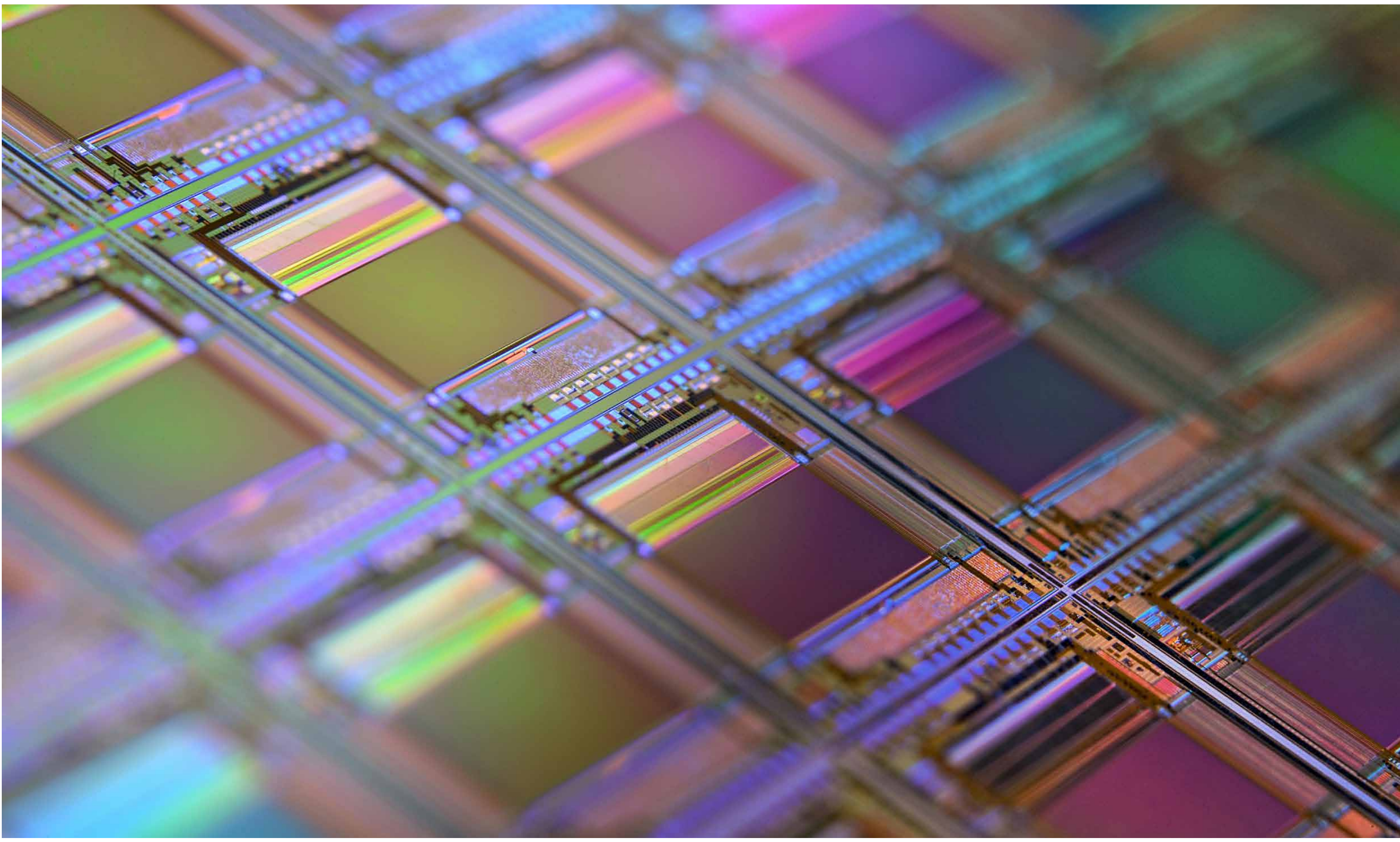
Sources: UN Comtrade, National Customs, Allianz Research

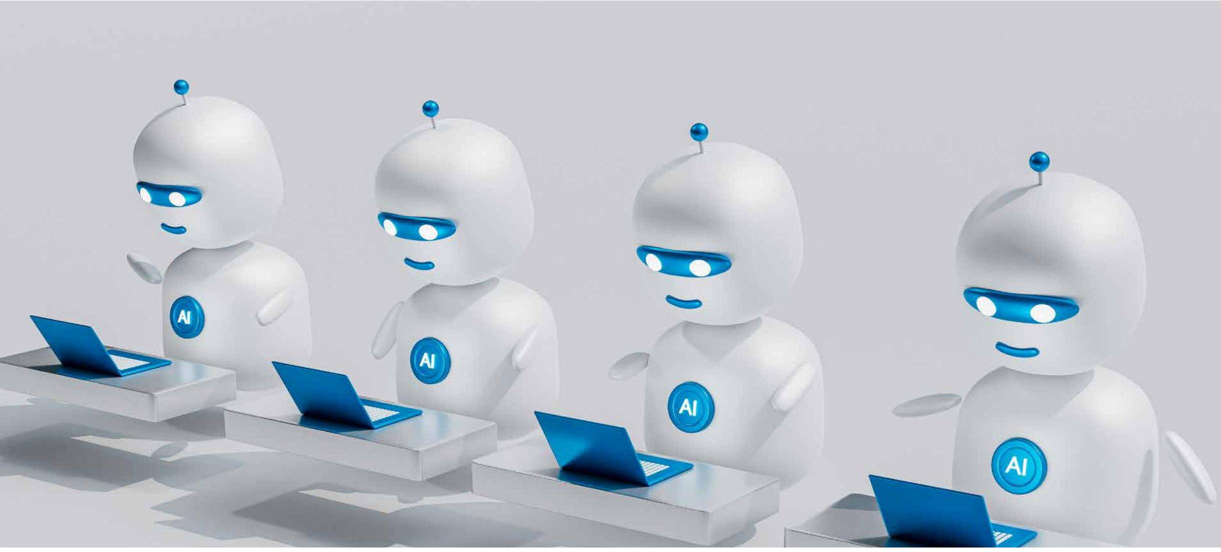
However, AI-related import demand is highly concentrated in US and partially Asia. Since 2023, Taiwan and the US have recorded the largest increases in AI-related imports, reflecting their pivotal positions in the AI supply chain: Taiwan through the import of capital equipment for semiconductor manufacturing and the US through demand linked to its dominance in AI services and data-center infrastructure. Since the mass-market deployment of large language models since 2023, the US has tripled its imports of advanced AI-related products (Figure 5, left), reflecting massive domestic investment in AI and continued reliance on foreign semiconductor supply. This surge has been reinforced by the rapid expansion of US data-center infrastructure, with 5,427 facilities accounting for 45% of global capacity in 2025 and 10 times the count of any other country. By comparison, Europe's AI-related imports have grown by only around 40% since 2023 (Figure 5, right), reflecting weaker investment in AI infrastructure, limited data-center capacity (Germany 2nd with 529, UK 3rd with 523, France 6th with 322 data-centers) and restricted access to advanced chips compared to US hyperscalers. Growth slowed further in early 2026. In Asia, beyond Taiwan, the strongest import growth has been seen in South Korea, reflecting its dual role in semiconductor production and consumers of advanced AI components, and Hong Kong, highlighting its important roles in semiconductor production and technology flows into China due to US export controls. Overall, the AI semiconductor market remains highly concentrated geographically: the US being the dominant end market for the most advanced generation of AI semiconductors, the EU and China as the other two large but structurally distinct demand pools. This creates significant exposure to shifts in demand, trade policy and regulation across a small number of economies: for AI hardware manufacturers any deterioration in US demand, tightening of trade and tariff policy, or regulatory disruption in Europe or China would have outsized consequences for the suppliers currently running at capacity to serve them.

Figure 5: US imports growth from selected AI products have skyrocketed, index (left) and Imports of overall AI-related products, in USDbn (right)



Sources: Trade Map, Allianz Research



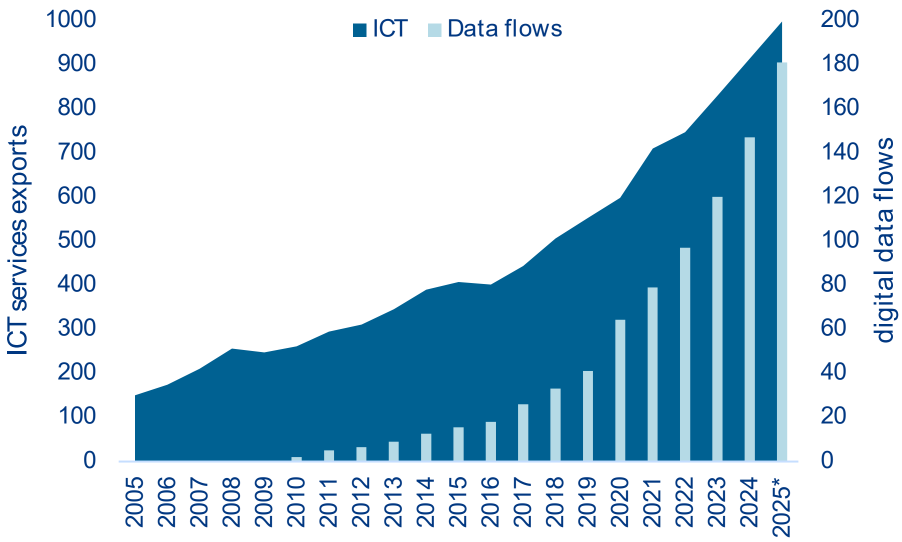


The growth in trade in services and data flows is accelerating with the diffusion of AI

Data flows and AI-related services trade are the backbone of transforming economic models into the future. It differs structurally from the trade of goods in several key respects. Firstly, it is deeply integrated into broader digital and intra-firm flows, with much of its value being captured through cloud services, software licensing, data infrastructure and model access, which are frequently bundled together rather than being traded as individual services. This helps to explain the outsized role of hubs such as Ireland, which function less as traditional exporters and more as invoicing and coordination centers for global technology firms, particularly US multinationals. Consequently, bilateral imbalances, such as those between the EU excluding Ireland (''EU26'') and Ireland, understate the true extent of dependence on US-based AI ecosystems. Indeed, the EU26 entity runs a large ICT services trade deficit of USD45bn, mostly explained by the deficit with Ireland, at close to USD55bn. India and the UK are far behind in terms of ICT exports to the EU and the US comes fourth, exporting less USD5bn directly to the EU26. Secondly, AI services act as the operating system for the trade of AI goods: advanced chips, servers and hardware only generate value when combined with cross-border flows of computing power, software and continuous model updates. This makes services a prerequisite rather than an adjunct to the trade of goods. Thirdly, the high concentration of trade reflects platform-based business models rather than pure comparative advantage, with a small number of economies capturing disproportionate value through intellectual property, data and digital infrastructure. Finally, the trade of AI services is increasingly shaped by regulatory and data governance regimes, which influence where services are hosted, billed and exported from. This reinforces fragmentation across jurisdictions. Together, these features imply that not only is AI-related services trade concentrated, it is also foundational, determining how value is distributed across the global AI economy.

An essential for the development and distribution of AI-systems are cross-border data flows. As AI systems rely on large, diverse and continuously updated datasets that only generate value when combined with software, hardware and expertise, the flow of these data internationally is essential for AI development and adoption. As data is non-rivalrous, it can be reused by multiple users and applications, creating strong economies of scale and scope. Free data flows enable firms to combine information across markets, build more accurate and generalizable models and accelerate innovation. By contrast, restrictions limit scale, raise costs, reduce interoperability and slow diffusion. The growing importance of this issue is reflected in broader trends: data-driven services such as computing, telecommunications and finance now account for almost half of the global trade in services. ICT services have evolved rapidly over time making up 0.53% of nominal global GDP in 2015 and have increased by 56% to 0.85% in 2025, while cross-border data volumes have grown more than tenfold from 15.5 zettabytes in 2015 to 181 zettabytes in 2025 (Figure 6). Together, these dynamics demonstrate that seamless cross-border data flows are a key driver of AI performance and innovation, as well as being a fundamental pillar of the global digital economy and the rise in AI-related services. And although data is generated and flows globally, it is often processed, stored and monetized in a small number of digital hubs with advanced infrastructure, which concentrates value geographically.

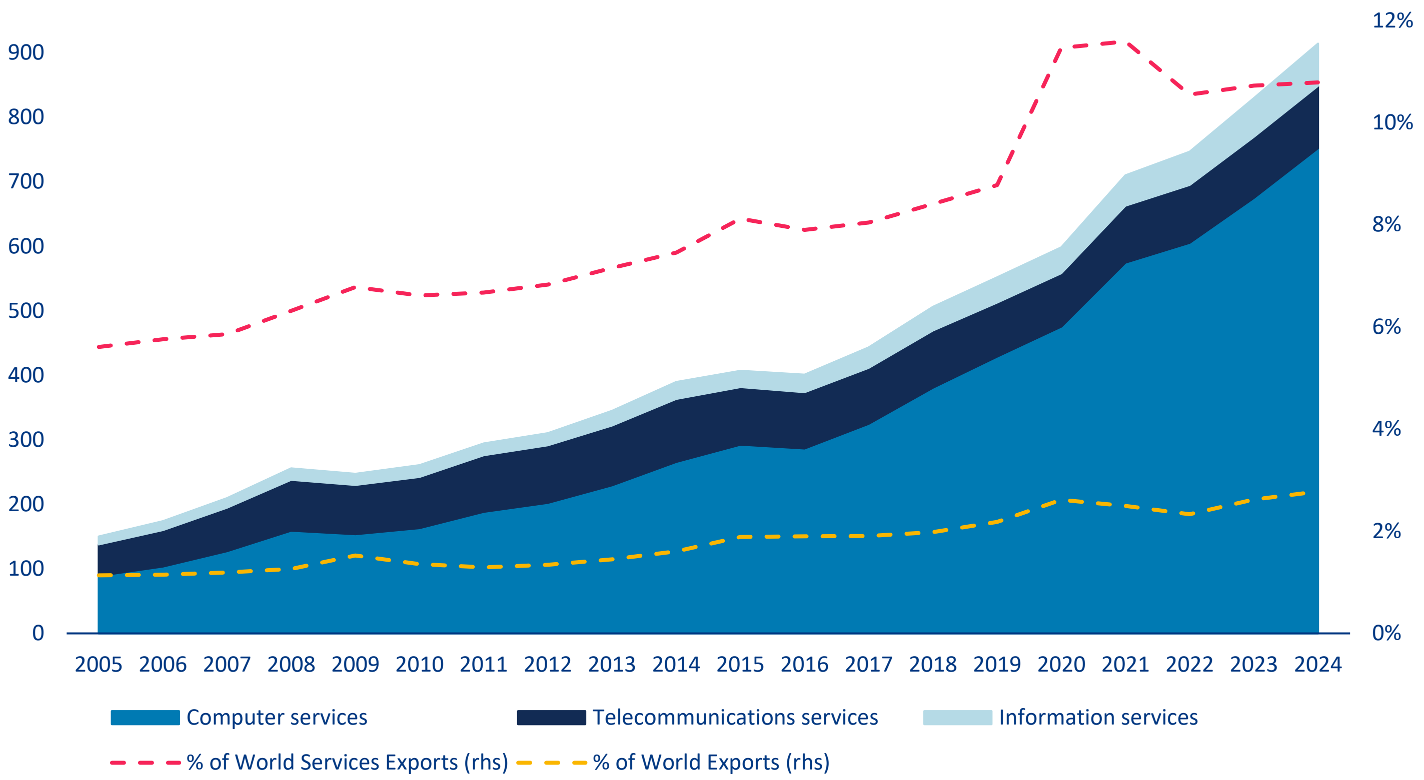
Figure 6: Cross-border global trade in ICT services (in USDbn) and digital data flows (in zettabytes)



Sources: OECD BATIS, Statista, Allianz Research. Notes: 2025* is an interpolated approximation for ICT services as no data is yet available.

The value of international trade in ICT services is approaching USD1trn, growing more rapidly than overall services (11% of global services trade, up from 8% a decade ago). This growth has been driven by rapid expansion since 2016–17, with the sector being highly concentrated in a few hubs. The sector, which is dominated by computer services such as software, consulting, infrastructure and support, reached approximately USD900bn in 2024 (around 3% of the total value of goods and services traded globally, and 11% of the value of services) (Figure 7). Ireland stands out as a key hub for US tech multinationals providing digital services to European core markets. It exported ICT services worth USD173bn in 2024, accounting for an exceptionally high 60% of its total exports. This is largely due to intra-firm flows linked to datacenters and licensing revenues. Consequently, Ireland has become the world's largest ICT services exporter, surpassing the EU (excluding Ireland) with USD142bn and the US with USD108bn of ICT exports to the rest of the world, respectively. Within the EU (for which we count both extra and intra-EU trade), major ICT exporters include Germany, the Netherlands and France. Meanwhile, India remains a major player, while China lags behind in providing digital services to the world.

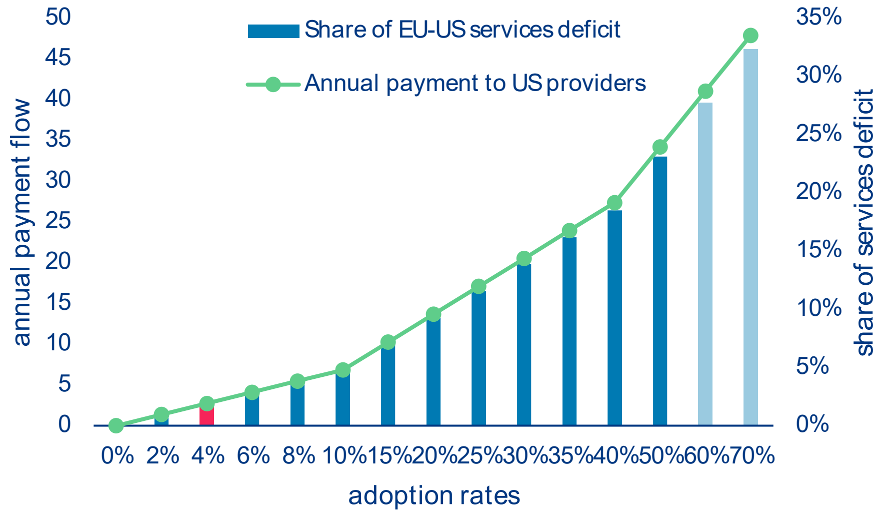
Figure 7: AI-related ICT services trade breakdown (in USD bn)



Sources: OECD, WTO, Allianz Research

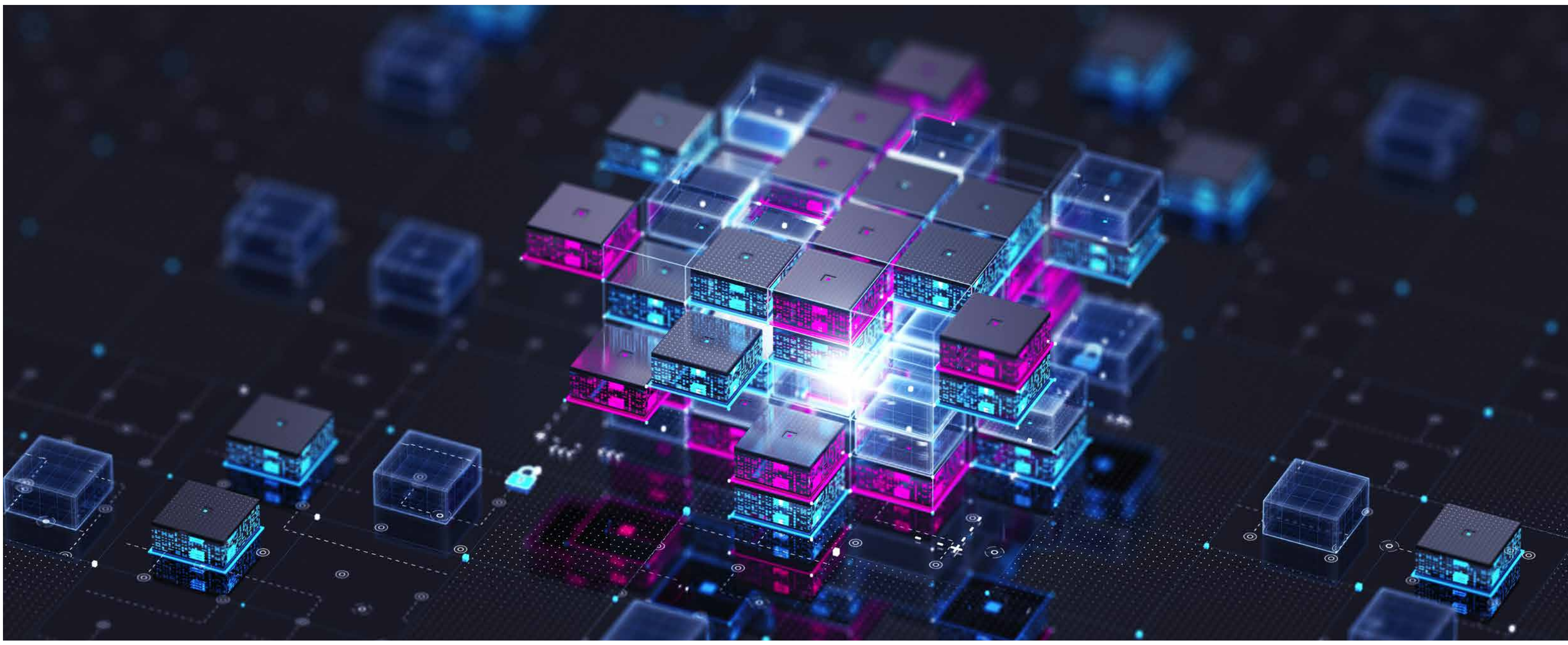
AI diffusion will be the next big driver in EU-US services imbalances due to EU dependence on AI model provision. Adopting AI in the Eurozone could significantly increase the transatlantic services imbalance through recurring subscription payments to US AI providers. Assuming an adult population of around 285mn in the Eurozone and a Claude Pro or ChatGPT Plus-style price of EUR20 per month (EUR240 per year, the current standard pricing for individual users), even modest adoption would result in significant cross-border services imports. With an estimated current penetration rate of around 4%, this equates to approximately EUR2.7bn per year in payments to US providers; already a significant portion of digital services imports, albeit still modest compared to the EUR148bn EU–US services deficit in 2024. Increasing adoption to a 10% penetration rate would raise outflows to approximately EUR6.8bn, and to around EUR17.1bn at a 25% penetration rate (Figure 8). At this point, AI subscriptions alone would be comparable to a tenth of the entire current services deficit. Under a high-adoption scenario in which 50% of adults subscribe, annual payments would total approximately EUR34bn, implying an additional impact of over one-fifth of the current EU–US services deficit, representing a significant increase in US services exports driven purely by AI diffusion.

Figure 8: Back-of-the-envelope calculation of annual payment flow from the EU to the US (in EURbn) and share of EU-US services trade deficit (in %) along AI penetration rates in the adult population



Source: Allianz Research. Notes : Assumes an adult European population of 285mn along hypothetical AI adoption rates, a subscription fee of EUR20/month (current standard pricing of ChatGPT Plus or Claude Pro for individual users) and a EU-US services trade deficit of EUR148bn.

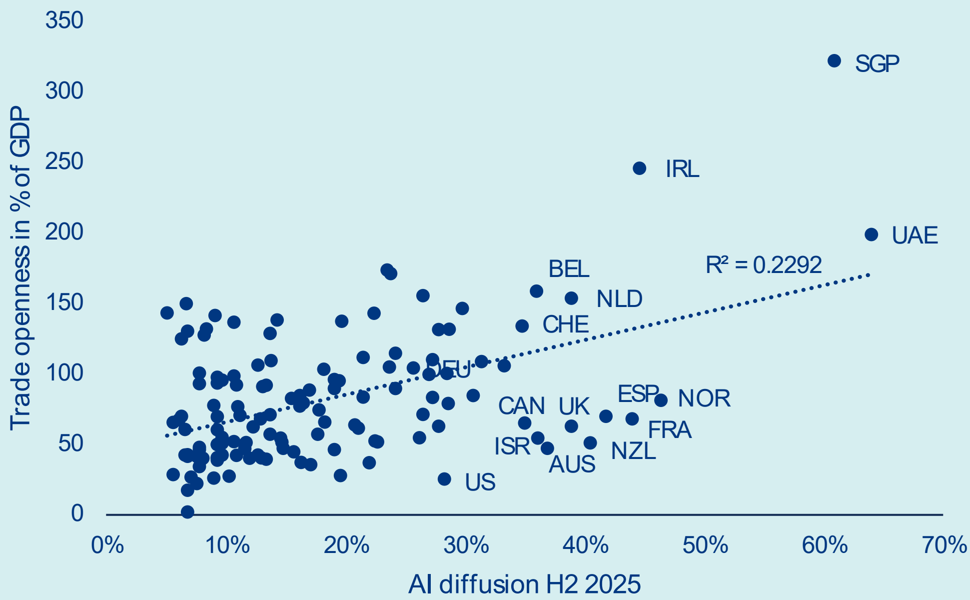
More broadly, AI acts as a scalable, recurring channel for importing services to the US economy via software subscriptions, cloud computing, APIs, IP licensing and digital advertising. These flows are intangible, highly scalable and concentrated in US hyperscalers. This matters because the EU's apparent digital surplus is already distorted by profit shifting and US tech revenues booked via Ireland. Stripping this out suggests a structural digital services deficit exceeding EUR100bn. AI accelerates this dynamic further as, unlike infrastructure-heavy digital services, subscriptions represent near-pure cross-border revenue transfers with limited domestic value creation. Furthermore, the impact is likely underestimated in the simple subscription model: enterprise AI usage (APIs, Copilot integrations and SaaS embedding) already exceeds consumer subscriptions, and productivity effects could also reduce Europe's own services exports. Therefore, unless offset by strong domestic AI platforms, European digital champions or localization of AI infrastructure and billing, AI diffusion risks becoming a persistent structural driver of widening EU–US services imbalances.



Box: Europe struggles with strategic dependencies

Despite its openness to trade, Europe faces uneven AI adoption, restricting its ability to compete with leading rivals. Trade openness explains 23% of the variation in AI diffusion across countries (Figure 9), with highly open economies such as Singapore, the United Arab Emirates and Ireland also recording the highest diffusion rates. But Europe contradicts this trend. Though its economies are highly open and integrated into global value chains, AI adoption remains uneven due to strict data privacy and regulation, limiting the continent's ability to benefit from AI-driven productivity gains. Overall, while AI can significantly boost productivity and growth, its benefits are unlikely to be evenly distributed, potentially increasing inequality between capital and labour, high- and low-skill workers, and larger versus smaller firms.

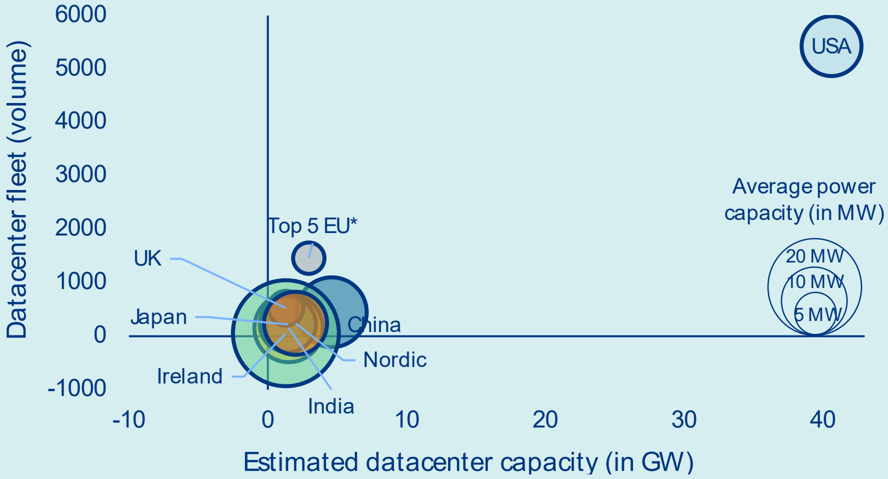
Figure 9: Brade openness (total trade as % of GDP) versus AI diffusion in H2 2025 (in %)



Sources: World Bank, Microsoft, Allianz Research

When it comes to data centers, the US is racing far ahead. With an estimated 40+ GW of installed capacity today (Figure 10), projected to surpass 100 GW by 2030, the US already has the world's largest computing infrastructure. This is underpinned by over USD600bn in capital expenditure committed by US hyperscalers alone this year. By contrast, Europe remains structurally underequipped. Its data-center capacity is heavily concentrated in five core markets (FLAP-D: Frankfurt, London, Amsterdam, Paris and Dublin), accounting for around 60% of the total European capacity. Meanwhile, emerging hubs such as the Nordic countries are only just beginning to attract significant interest, thanks to competitive energy costs and an increasing proportion of renewable energy sources. The challenge for Europe is not merely one of scale, but also of capital and political architecture. Unlike the US, where private hyperscalers flood the market with investment, or China, where the state substitutes for private capital, Europe suffers from a dual deficit: insufficient private investment and an absence of coordinated public policy. The fragmentation of Europe's regulatory landscape prevents the emergence of a clear, common framework dedicated to AI infrastructure, leaving European ambitions dangerously reliant on external resources.

Figure 10: Data-center fleet & estimated capacity (in GW) per region

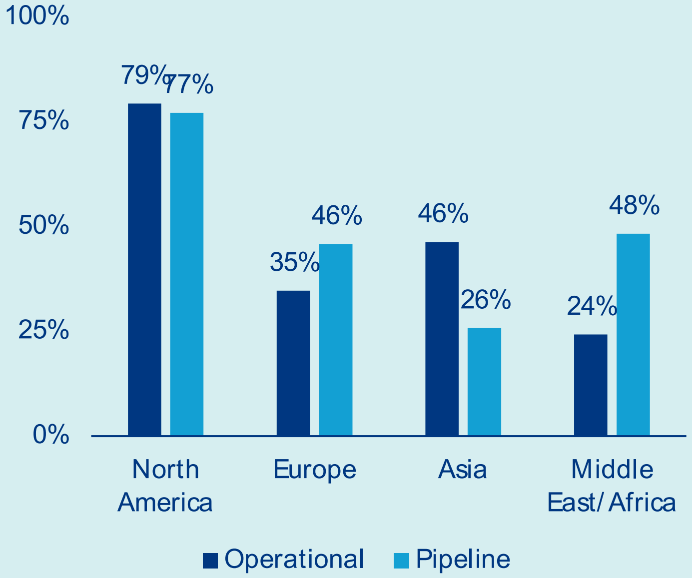


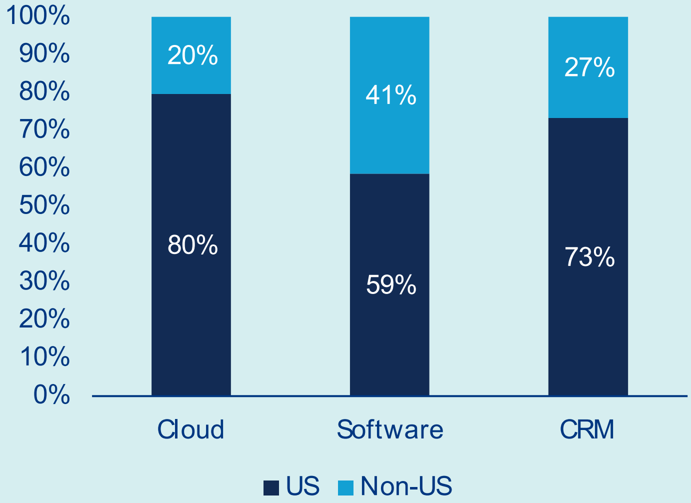
Sources: Cloudscene, Cushman & Wakefield (Datacenter H2 2025 update), Allianz Research. Notes: *Country group including Germany, France, Italy, Spain & Netherlands.

To avoid falling into a technology dependency trap, regulatory and capital constraints need to be cleared. Building data-center infrastructure in Europe is a formidable undertaking, constrained by a unique combination of structural, regulatory and financial hurdles that are unparalleled in the US or China. Limited land availability in urban areas, lengthy and complex permitting processes and increasingly stringent environmental regulations – particularly regarding water consumption and carbon footprint – can extend development timelines significantly, often stretching from planning to commissioning to over four years. Perhaps the most acute bottleneck is power connectivity: grid connection queues in key European markets have grown dramatically, with some operators reporting wait times of five years or more for adequate power access. This is a direct consequence of ageing grid infrastructure and competing demand from industrial and residential users. Financing conditions add another layer of complexity, as the capital-intensive and long-term nature of data-center projects does not align with the risk appetite of many European institutional investors. This is due to the absence of the deep and liquid private credit markets that underpin US hyperscaler investment.

The consequences of this investment gap are already evident in the ownership structure of infrastructure. As of 2025, US hyperscalers accounted for roughly 40% of total data-center operational capacity across Europe, but almost half of the pipeline (Figure 11, left) as large US tech firms are taking profits from weak private capital market in Europe and low competition from local players to consolidate their presence within a key market for them. The digital services layer tells an even starker story: US firms currently capture 80% of European cloud revenue, 59% of enterprise software revenue and 73% of CRM expenditure, leaving non-US players, including European vendors, to compete for the remaining margins Figure 11, right). This dominance is no accident. US hyperscalers' substantial financial resources enable them to absorb permitting delays, pre-fund grid upgrades and negotiate directly with national governments on energy and land access, structural advantages that no European operator can currently match on an equivalent scale. Their ample cash reserves effectively enable them to fill the void left by the lack of traction in private and public investment on the continent, as they steadily expand their physical and commercial footprint with each passing planning cycle.

Figure 11: Weight of US hyperscalers in datacenter capacity (current & future) across the globe (left) and market share of IT services and software applications revenue in Europe per firm nationality (right)





Sources: CBRE, Synergy Research, European Software & Cyber dependencies report (European Commission, Dec. 2025), Allianz Research. Notes: Data as of Q4 2025.

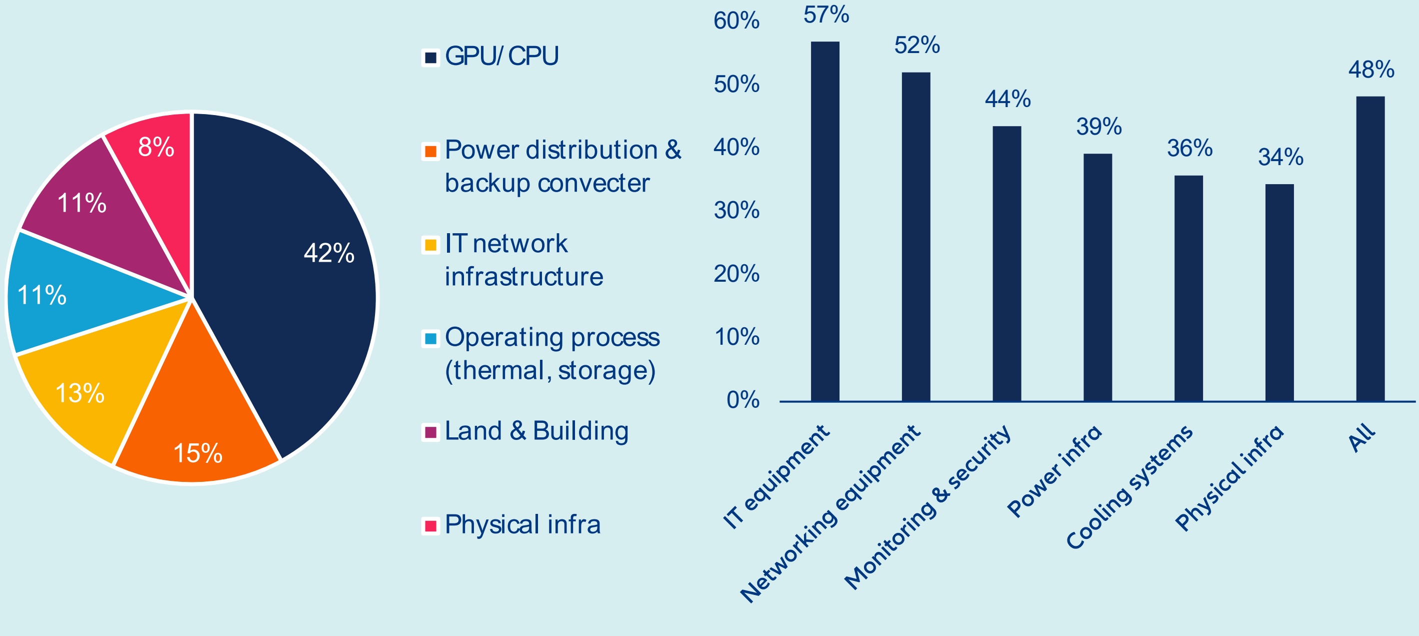
Initiatives such as Mistral AI and sovereign computing capacity projects in France and Sweden are promising, albeit modest, counterweights. Without decisive, coordinated action from Europe, including the mobilization of sovereign funds and development banks, as well as the establishment of a harmonized permitting framework, the continent risks becoming dependent on foreign infrastructure providers, losing not just market share, but also strategic autonomy over the digital backbone of its economy.

Strong industrial-linked specialization and regulatory constraints undermine European ambitions to build up a domestic semiconductor ecosystem and take part to AI race. Historically, European industrial strategy oriented domestic semiconductor production toward application-specific and industrial-grade chips, well suited to its automotive and manufacturing base but leaving the continent without meaningful capacity in the advanced nodes where AI demand is now concentrated. In the meantime, the relatively small weight of the ICT industry in Europe (less of 1% of GVA) but also the absence of a large domestic hyperscaler with ample cash reserves like in the US meant there was never sufficient demand to justify leading-edge foundry investment at scale, a gap that policy has failed to compensate for. Regulatory complexity and slow subsidy disbursement have further deterred the large infrastructure commitments that catching up would require. The cancellation last year of Intel's plan to build a foundry in Magdeburg is a good example Underlying both is a structural weakness in private capital markets: With only 6% of global AI funding flowing to European firms, the continent lacks.

Finally, Europe's data-center ambitions depend on fragile, Asia-heavy supply chains. Building AI-dedicated data-centers is, first and foremost, a hardware problem, and Europe currently owns very little of it. GPUs and CPUs alone account for around 42% of the total capital cost of a 1GW AI data center (Figure 12, left), and this expenditure largely flows outside of Europe: non-EU import ratios exceed 57% for IT equipment and 52% for networking gear (Figure 12, right). Five Asian economies (Taiwan, China, South Korea, Malaysia and Vietnam) collectively supply around two-thirds of this expenditure. Taiwan's near-monopoly on sub-3nm logic chips and South Korea's dominance in memory represent structural dependencies, not cyclical ones, and cannot be meaningfully substituted in the short term. The next tier of components – power distribution, IT network infrastructure and cooling systems, each representing 10–15%

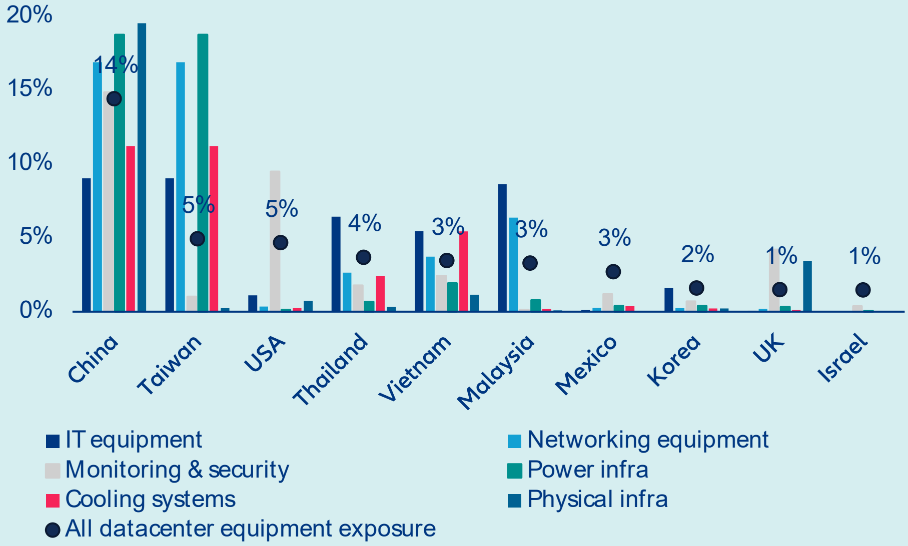
of datacenter value – shows only modestly better numbers: monitoring and security sits at 44% non-EU exposure, power infrastructure at 39%, and cooling systems at 36%. Even physical infrastructure, the least technology-intensive segment, runs at 34% external reliance. At the country level, China stands out with a total exposure share of 14% across all datacenter equipment categories (Figure 13), nearly three times the weight of both the US and Taiwan, reflecting its dominance in advanced electronics and physical and cooling inputs. Vietnam primarily supplies computers and electronic transmission devices, while Turkey is a notable partner for cooling systems. Despite its influence on the software and services stack through its hyperscalers, the US has a more modest hardware footprint concentrated in IT and networking. Overall, Europe has little domestic manufacturing depth in advanced electronics and limited battery and power storage capacity. Its supply chain is also acutely exposed to two live geopolitical risks: renewed trade tensions between Washington and Beijing (currently on pause but far from resolved) and any energy disruption in Asia stemming from Middle East instability. For a continent with sovereign AI ambitions, this is a precarious foundation.

Figure 12: Estimated cost breakdown of an AI-dedicated 1GW capacity data center* (left) and share of non-EU imports ratio for critical materials used in AI data-center infrastructure construction, 2025 (right)



Sources: Bernstein, Eurostat, Allianz Research.** GPU share based on Nvidia GB200 cost.

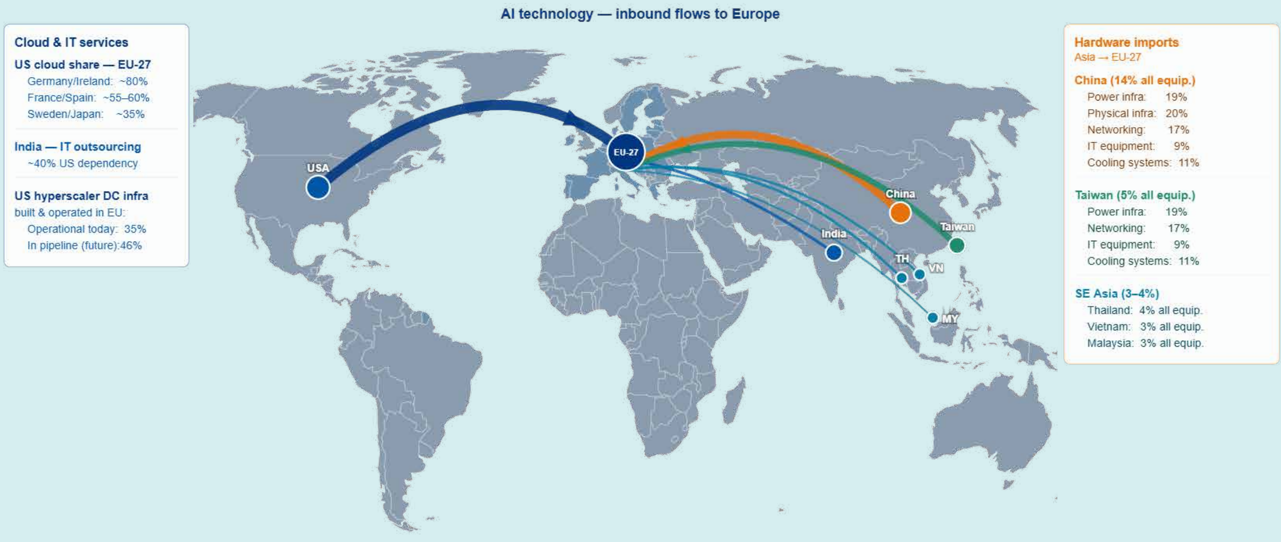
Figure 13: Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction



Sources: Eurostat, Allianz Research

This structural dependency has direct macroeconomic consequences. Europe's limited control over AI infrastructure and its reliance on imported hardware and foreign hyperscalers imply that a growing share of AI-related value added will be captured abroad, dampening domestic productivity multipliers and GDP gains from AI adoption. At the same time, higher data-center build costs, prolonged permitting cycles and constrained grid access raise the effective cost of AI deployment in Europe, slowing diffusion into the broader economy. This combination risks a dual drag: weaker productivity growth relative to the US and higher unit costs for digital and industrial transformation. Over time, it could also widen the transatlantic gap in investment intensity, reinforce Europe's structural current account outflows in digital services and increase vulnerability to external price-setting in critical digital inputs.

Figure 14: Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction

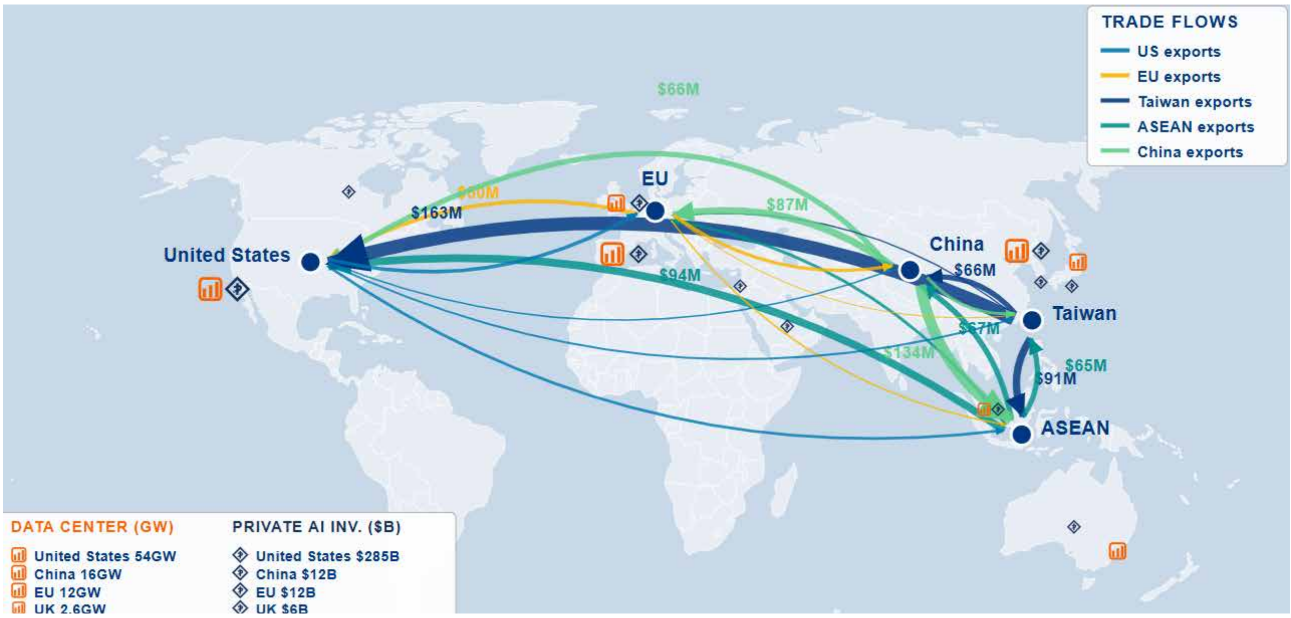


Sources: Allianz Research

The world is flat: AI reconfigures the geography of trade and dependencies

The AI supply chain is becoming more geographically distributed in assembly and packaging, but more concentrated and fragile at the technological frontier, with diversification creating the appearance of resilience while reinforcing dependency on a small number of critical nodes. The most critical parts of the AI supply chain (advanced chips, high-bandwidth memory, and the equipment to produce them) remain highly concentrated in a small set of countries, especially Taiwan, South Korea, and the Netherlands, and this concentration has intensified since 2022 due to export controls and the AI buildout. This creates systemic single points of potential failure in the global AI system. Partial diversification took place only in the middle layers of the supply chain. Server assembly, packaging and networking are shifting geographically under the influence of US policy and export controls, with gains for Vietnam, Mexico, Japan, Taiwan and parts of Eastern Europe. But this friendshoring is still limited in scale and far from replacing established hubs.

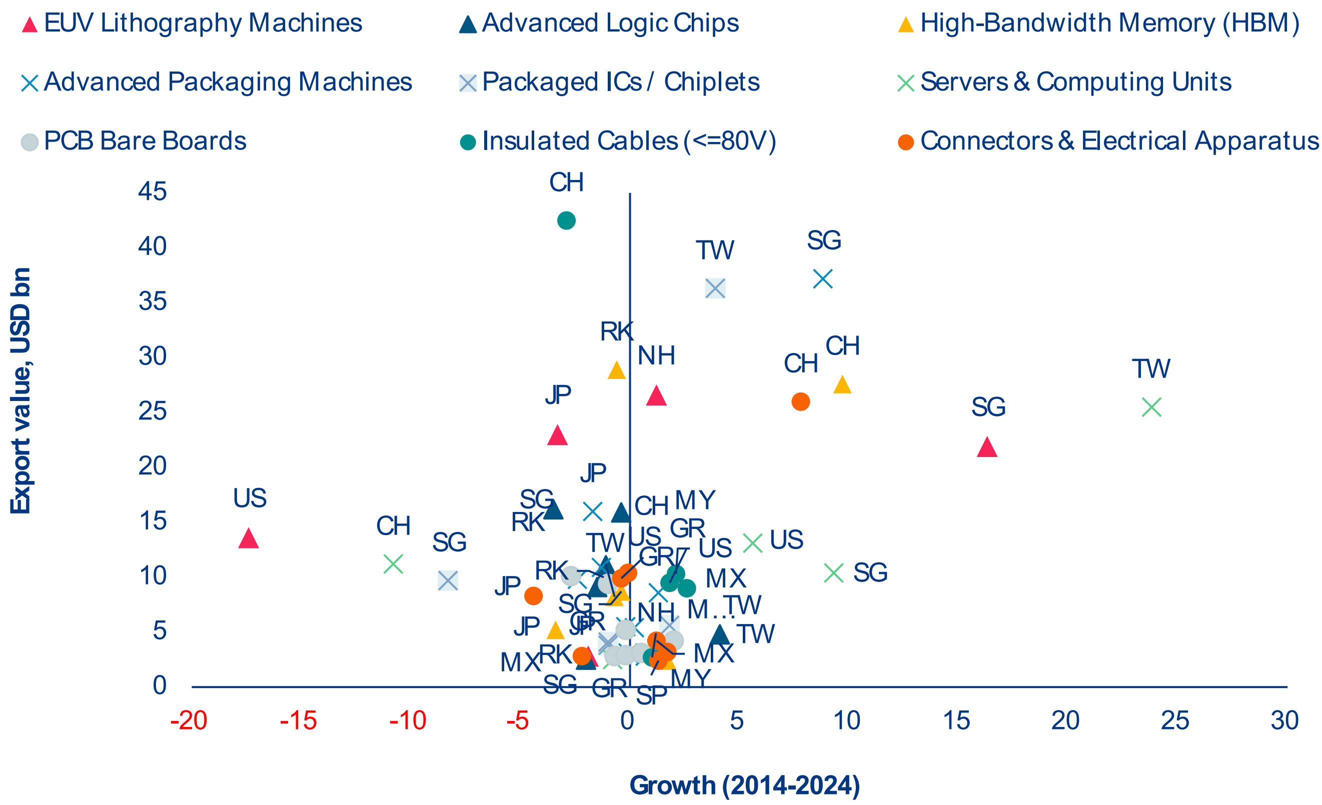
Figure 15: AI supply chain concentrated in several key geographies



Sources: Stanford HAI 2026, Synergy Research, Trade Map, Allianz Research

Despite friendshoring efforts, diversification away from China remains limited. A decade of diversification investment has produced incremental shifts in specific sub-segments but no geography has emerged as a credible large-scale alternative. Vietnam and Mexico have attracted investment and attention as potential alternative bases for electronics assembly, and there are signs of incremental share gains in specific sub-segments, but the scale of Chinese capacity in these categories means that meaningful diversification remains a multi-year undertaking for most supply chains. China continues to dominate low-value, high-volume components that are essential but underappreciated (Figure 16). Diversification away from China in these segments is slow, meaning supply chains remain exposed to geopolitical shocks despite rhetoric of decoupling. The risk here is overexposure to Chinese production in case of a new geopolitical shock or policy change in Bejing.

Figure 16: Concentration towards Asia in the higher value added parts of the AI supply chain, while China further integrates low-value, high-volume



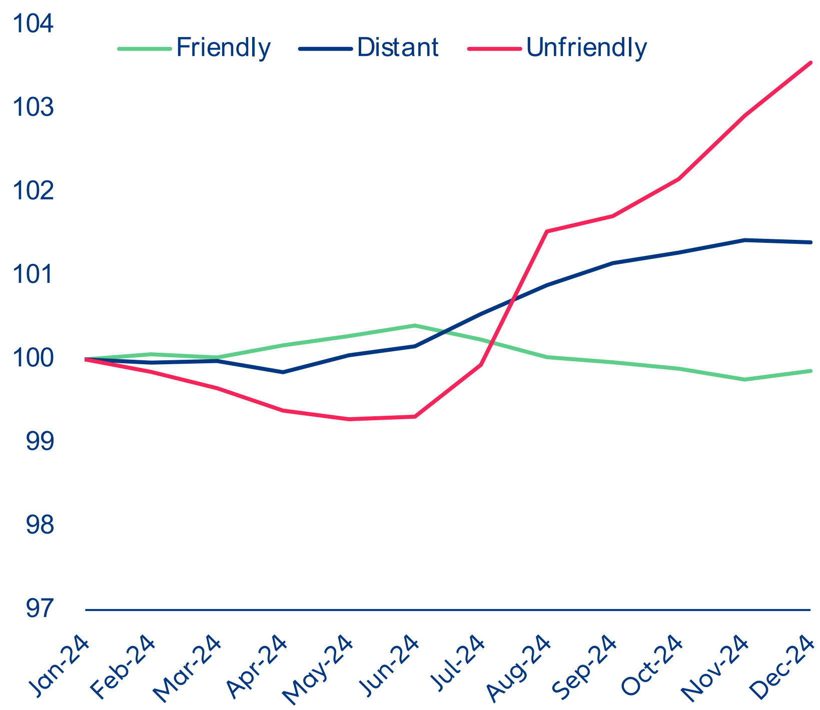
Sources: UNComtrade, Allianz Research. Note: Shapes of the markers correspond to different sections of AI-supply chain: triangle = higher value, cross = middle value, circle = lower value.

Hidden upstream vulnerabilities emerge with the current conflict in the Middle East. Shortages of key inputs for semiconductor production, notably helium, would impact the most leading-edge fabs, which could translate to losses above USD10bn a month. Qatar accounts for a third of global production of helium, an upstream input of the AI supply chain that is also almost entirely shipped through Gulf terminals. While the inflationary cost impact is relatively small on semiconductor production compared to the rest of the supply chain, the production shortage would be much more impactful, with an asymmetric impact. Semiconductor producers in South Korea are most at risk, given their exposure of imports from Qatar, and early estimates suggest 4 to 6 months of reserves. However, everything hinges on gas storage in non-producing countries. The US holds the largest known reserves and has historically been the dominant producer, but its federal helium reserve has been progressively wound down under legislation mandating its disposal. Japan, one of the world's largest helium consumers given its electronics and semiconductor manufacturing base, holds no domestic reserves and is almost entirely import-dependent. New countries could increasingly offer their helium reserves to the market, including Russia, Australia and Algeria, as well as Tanzania, which holds what may be one of the world's largest undeveloped helium deposits in the Rift Valley, though it remains years away from commercial production.

US trade policy has reinforced rather than fundamentally reshaped semiconductor supply chains. China now accounts for just 0.7% of US semiconductor imports in tariffed categories, subject to a 25% tariff, while production (or rerouting) shifted toward Taiwan, Vietnam, Malaysia, Thailand and Mexico. Despite headline tariff threats of up to 46% on Vietnam and 36% on Thailand, semiconductors remain largely exempt from new trade measures, with effective tariff rates near zero for most major suppliers and around 1% for Malaysia. As a result, the post-2018 semiconductor supply chain continues to operate under near free-trade conditions. The more ambitious reshoring effort has been the CHIPS Act which redirected USD52bn toward domestic manufacturing and research. Its investment allocation has been substantial, but the production results remain limited and represent only a small fraction of Taiwan's output.

Geopolitics is changing where semiconductors are manufactured rather than with whom they are traded. The commercial logic of the supply chain remains largely intact: Taiwan, South Korea and the broader Southeast Asian production ecosystem continue to sell to every major economy without any need of political alignment. This creates a form of structural interdependence that cuts across geopolitical blocs in ways that policy narratives tend to obscure: China remains deeply integrated into the commercial semiconductor market for everything below the frontier of advanced logic and memory, and continues to source the vast majority of its chip requirements from the same Taiwanese and Korean suppliers that serve US hyperscalers. The meaningful exceptions are narrow but strategically significant, US export controls have effectively excluded China from sub-7 nanometre logic, high-bandwidth memory for AI applications, and the lithography equipment required to produce either domestically, creating a two-tier market in which the frontier is controlled and the mainstream is not. This commercial logic is increasingly visible in the trade data. For most goods, trade has fragmented along political lines, rising between geopolitically „friendly" countries and declining between „unfriendly" ones, and until recently AI-related goods followed the same pattern. Since 2024, however, this trend has inflected, with trade in AI-related goods between „unfriendly" countries stabilising and subsequently rising, even as broad goods trade between the same countries has continued to fragment (Figure 17). This divergence suggests that, outside the narrow frontier segment subject to export controls, commercial interdependence in the AI supply chain is proving more resilient than political alignment.

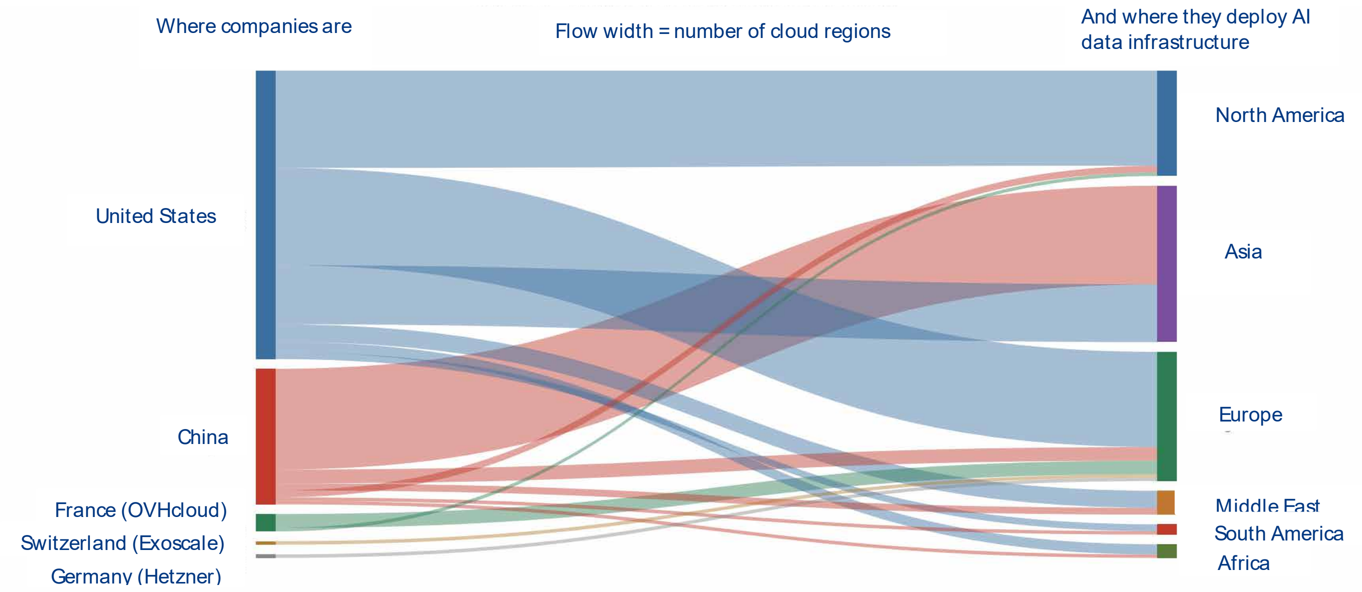
Figure 17: Growth of AI-related goods between unfriendly countries experiences the largest increase during 2024



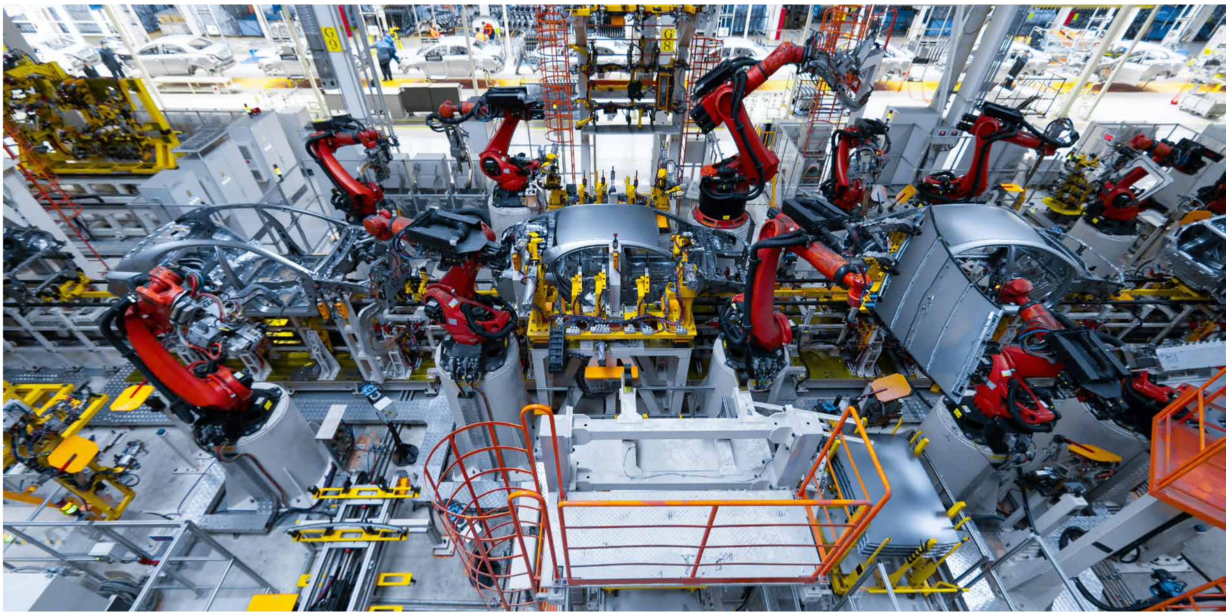
Sources: UN Comtrade, Allianz Research

AI-related production and infrastructure provision will remain concentrated. Geopolitics is reshaping semiconductor investment globally, but despite massive subsidy programs across the US, Europe, China, Japan and India, the likely outcome is not full supply security but a more geographically distributed industry that remains highly dependent on a small number of leading-edge producers for the most critical AI and defense technologies. The geographic concentration of large-scale AI-capable data centers is becoming a new source of geopolitical power, as control over compute infrastructure increasingly determines which economies can develop, export and benefit from AI services. The US dominates the global AI infrastructure landscape in the Western hemisphere, while China dominates Asia (Figure 18). Large parts of Africa and Latin America remain largely excluded from the compute capacity and so far depend half-half on the US and China with no dominance decided yet. This growing concentration risks reinforcing technological dependence, widening digital divides and shifting economic and strategic influence toward the countries that control cloud infrastructure, data flows and AI platforms.

Figure 18: Distribution of headquarters of AI companies and AI data infrastructure deployment



Sources: WTO (2025), Lehdonvirta, Wu and Hawkins (2024), Allianz Research

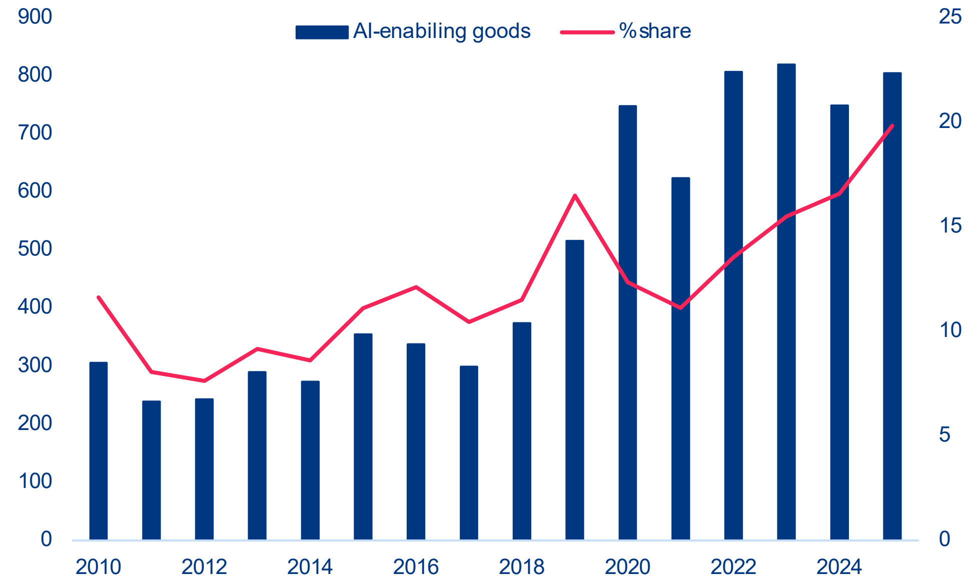


The race for AI dominance is increasingly also being fought through industrial policy

Tariffs on AI-enabling products remain low. Industrial policy and trade rules are becoming central to competition in the field of AI, influencing access to critical minerals, semiconductors, cloud infrastructure, IT services and cross-border data flows, all of which are essential for the development and deployment of AI. Open and predictable trade regimes support innovation, productivity and the integration of AI into global value chains. However, these tools are increasingly being weaponized in an increasingly geopolitical world. Despite rising tensions, tariffs on AI-enabling products remain relatively low. Worldwide, most-favored-nation (MFN) applied tariffs on AI-related goods fell from 5.6% in 2015 to 2.8% in 2025, compared with 7.8% on manufacturing goods overall. The US maintains an average tariff of 1.4%, the EU of 2.0%, the UK of 1.3%, and EFTA countries effectively zero, while China applies the highest average tariff in the group at 3.5%. Overall, AI-related goods continue to be traded under relatively open conditions, reflecting the strong interdependence of global AI supply chains and the strategic importance of maintaining access to computing infrastructure, digital technologies and knowledge flows.

Non-tariff measures (NTMs) on AI-enabling goods account for a fifth of all harmful NTMs in force. Over the last decade, NTMs rose from 355 in 2015 to 805 in 2025 (Figure 19) with their share of the total constantly increasing since the deployment of AI to the general public in 2022. These measures, including export restrictions and financial assistance, increasingly reflect efforts to protect technological leadership and secure strategic industries. China has been the top player in implementing trade restrictions with a peak 2020, while the US has more recently picked up since 2022 overtaking China only in 2025. The US and China implemented the most restrictions in 2025 with 208 new restrictions from the US and 159 from China. At the same time, the US is the most affected nation as well, followed by Germany, Malaysia, Thailand, China, the Netherlands and Japan. While the most common new measure implemented in 2025 were import tariffs, this has shifted clearly from financial assistance in foreign markets which was top of the list since 2011 and peaked in 2020 with 431 new measures, followed by tariffs and trade finance and the occasional export ban on AI-enabling goods. While such restrictions may slow technology diffusion in the short term, they also carry significant costs, are difficult to enforce and can accelerate innovation and diversification elsewhere, limiting their long-term effectiveness unless backed by strong domestic industrial capabilities and coordinated policy frameworks.

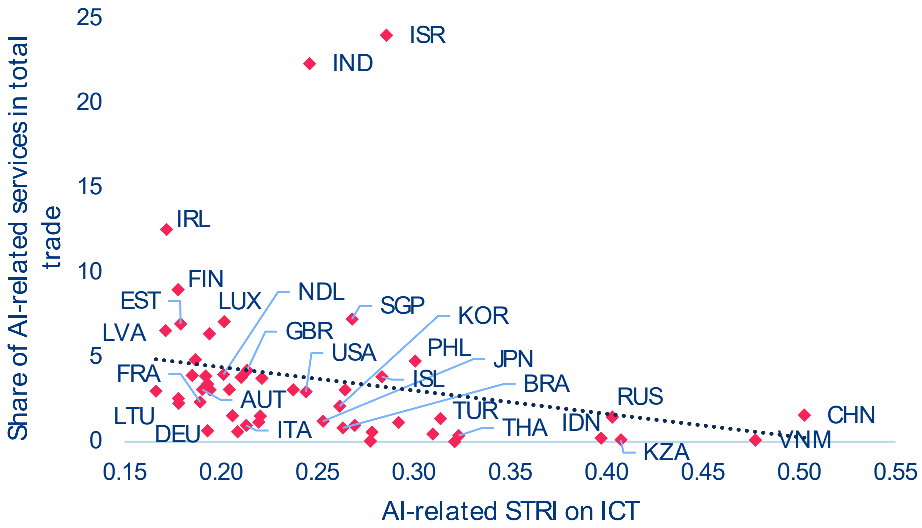
Figure 19: NTMs on AI-enabling goods, number of cumulative measures implemented (left) and percentage share (right)



Sources: GTA, Allianz Research

Trade in AI-enabling services is limited by restrictive regulations. The trade of AI-related services is increasingly being shaped by restrictions on digital services and cross-border data flows. These restrictions are becoming a key determinant of global AI market share. Although AI is making services such as finance, software and telecommunications more tradable, these sectors are still limited by foreign equity restrictions, licensing regulations, localization requirements and barriers to cross-border delivery. Economies with more restrictive ICT and digital trade regimes capture smaller shares of AI-enabled services exports. In fact, ICT trade restrictiveness explains around 5% of a country's share of AI-related services trade (Figure 20). There is significant regulatory fragmentation: the OECD Digital Services Trade Restrictiveness Index (STRI) records 78 economies requiring a local presence out of 129 countries covered and 46 imposing data localization rules in 2025. The number of countries requiring data to be processed or stored locally nearly doubled in the last decade. These barriers disproportionately affect AI-intensive sectors such as telecommunications, banking, insurance and software, where cross-border supply and commercial presence remain tightly regulated. Furthermore, fragmented data governance amplifies divergence by raising compliance costs, limiting access to cloud infrastructure and global datasets, and weakening participation in AI innovation networks, particularly for smaller economies. While some regulation is necessary for trust and security, openness to digital trade and data flows is now a key factor in achieving AI competitiveness. Economies with restrictive regimes are increasingly failing to capture AI-driven growth.

Figure 20: AI-related services share in total trade (in %) to AI-related services trade restrictiveness index

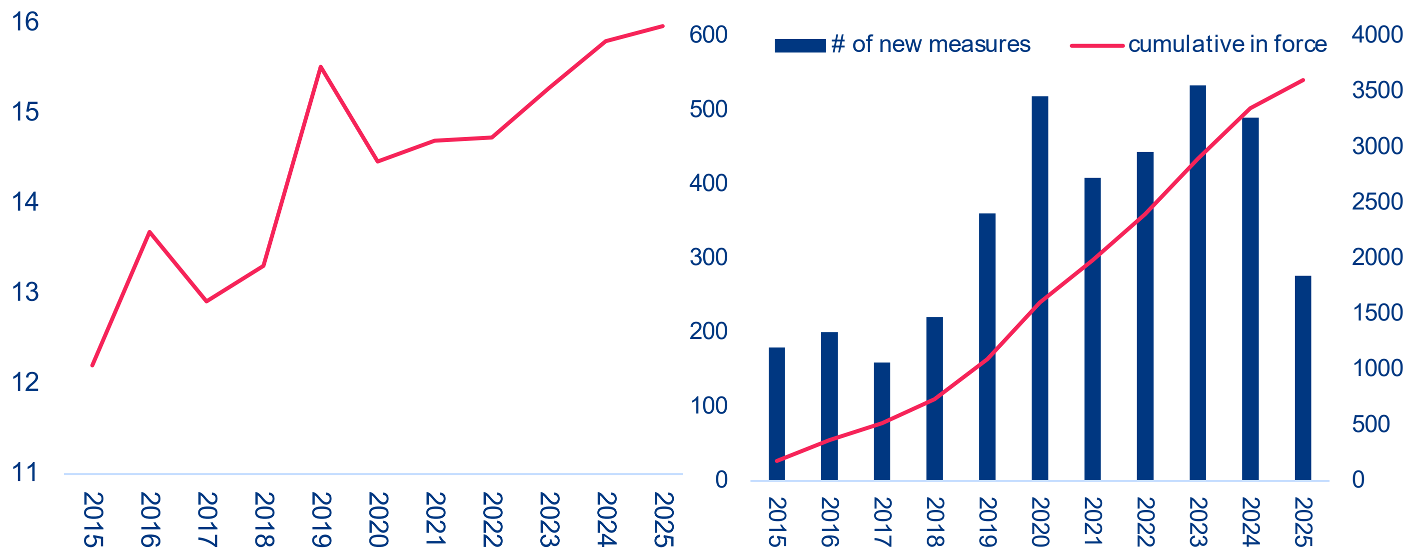


Sources: OCED TIVA, OECD STRI, Allianz Research. Notes: Sector shares are as reported in OECD TiVA VA exports for 2022. AI-related sectors are ICT services. Services restrictiveness based on 2025 data only on ICT services sectors.

The growing financial support for AI is exacerbating global technological disparities. Higher-income economies are implementing large-scale AI strategies with the backing of fiscal resources, industrial policy, and targeted support for domestic champions. In contrast, lower-income economies are hampered by fiscal and infrastructure constraints, which limit their ability to take comparable action. This is widening the structural gaps in innovation and trade capacity. The scale of AI-related subsidies is also expanding: the proportion of AI-enabling goods covered by subsidies increased from 12% in 2015 to almost 16% in 2025 (Figure 21, left), and the total number of AI-related subsidy measures now exceeds 3,600 worldwide (Figure 21, right). Although the introduction of new measures slowed in 2025, the overall stock continues to rise. China is in the lead by a large margin, with its subsidy coverage of AI-related goods trade increasing from 14.5% in 2015 to 55% in 2020 and continuing to expand nearly twofold since then. It is followed by South Korea, Malaysia, the US and Japan as the top five AI-related subsidy measure players to expand or protect their position in the AI value chain. These measures primarily target critical materials, advanced semiconductors, AI accelerators (GPUs), semiconductor manufacturing equipment, optical fibres and data-centre components. They are funded through grants, tax credits, loans, state-backed equity injections, electricity subsidies, preferential financing and public support for R&D and infrastructure.

Initiatives to develop and regulate AI are expanding rapidly, but they are also creating a fragmented policy landscape. Regulation now extends beyond algorithms to encompass the entire physical and digital infrastructure of AI systems, including data centers, semiconductor supply chains, energy-intensive computing facilities, and cross-border data flows that facilitate AI-driven trade. Governments are no longer merely shaping trade in AI-enabling inputs; they are also actively building industrial ecosystems to secure positions in AI value chains. 23% of countries have set up national AI governance bodies (32 in Europe, 10 in Asia, four in Africa, five in North America, four in Central and South America, and two in Oceania), and 28% have adopted AI regulations, guidelines or standards. These frameworks are increasingly integrating regulation, intellectual property, competition policy, skills development, labor markets and infrastructure investment in order to foster innovation while managing concentration risks. Regional approaches differ significantly. North America has traditionally relied on lighter regulatory frameworks, allowing faster innovation with fewer constraints. China's model is more state-driven, with AI governance closely tied to industrial policy and oversight. Europe occupies a middle ground: while regulation strengthens trust, transparency and adoption, key factors in scaling AI, it also introduces

Figure 21: Share of cumulative subsidies on AI-enabling goods in total subsidies, cumulative in force as % of total (left) and number of new subsidy measures in force (right)



Sources: GTA, Allianz Research

compliance and security requirements that can slow innovation cycles, even though many processes are becoming increasingly automatable. Overall, however, regulation is not currently a binding constraint in Europe. AI governance is emerging as a global trade regime in its own right, which could further reinforce the existing dominance of a select few. At the multilateral level (Table 1), frameworks such as the OECD AI Principles and the G7 Hiroshima Process promote interoperable standards and trusted data flows in order to reduce friction in the trade of AI services. At the same time, national strategies are becoming increasingly geopolitical. The US ‚Pax Silica' approach links AI directly to trade security through export controls, investment screening, and domestic incentives for the production of chips and computing power.

Meanwhile, China's Digital Silk Road initiative integrates AI infrastructure exports, such as datacenters and cloud platforms, into broader trade and connectivity strategies. Middle Eastern economies are leveraging their energy abundance and sovereign wealth to establish themselves as global compute hubs with AI-friendly regulatory environments. Other initiatives, including the UNESCO AI Ethics Framework and the UN Global Digital Compact, seek to establish shared norms for data governance, model oversight and infrastructure efficiency. Together, these overlapping regimes are increasingly determining where AI is produced, how it is governed and how AI-enabled goods and services flow globally, effectively turning AI governance into an emerging trade regime in its own right.

Table 1: Major AI-trade-related international policy initiatives

Initiative	Lead Actors	Core Focus Areas	Trade Relevant Impacts
OECD AI Principles (2023)	OECD countries	Responsible AI, transparency, data governance	Supports interoperable standards for cross border data flows and cloud services; reduces regulatory fragmentation.
G20 AI Principles (2019)	G20 economies	Human centric AI, innovation, digital economy	Encourages harmonized data governance approaches that facilitate AI enabled services trade.
UNESCO AI Ethics (2021)	UNESCO members	Ethical standards, inclusiveness, data rights	Shapes global norms for data handling, influencing access to datasets and model training resources.
G7 Hiroshima Process & AI Code of Conduct (2023)	G7	Frontier model safety, AI supply chain security	Promotes common benchmarks that support trusted AI supply chains; potential trade conditions for high risk AI systems.
Bletchley Declaration (2023)	28 countries incl. US, UK, EU, China	Safety of frontier models	Introduces shared risk management expectations, influencing compliance requirements for cross border AI deployment.
UN General Assembly AI Resolution (2024)	UN member states	Safe, inclusive global AI governance	Calls for capacity building in digital infrastructure; facilitates AI adoption in developing countries.
Council of Europe Framework Convention on AI (2023)	Council of Europe	Human rights, democratic governance of AI	Sets obligations that may become preconditions for AI trade with EU markets.
International Network of AI Safety Institutes (2024)	Multinational	Safety research, model evaluations	Could lead to standardized audits that function as “technical passports” for AI model trade.
UN AI Advisory Body Final Report (2024)	UN	Global governance mechanisms	Recommends infrastructure investment frameworks and compute governance strategies.
UN Global Digital Compact (2024)	UN	Digital cooperation, data flows, public infrastructure	Promotes open digital markets and trusted cross border data exchange.
International Scientific Report on Advanced AI Safety (2024)	Multilateral science groups	Scientific evaluation of AI risks	Provides technical benchmarks and safety thresholds that may condition AI exportability.
US Pax Silicia Strategy	United States	Chip controls, AI supply chain security, cloud regulation	Reshapes global trade in semiconductors, cloud capacity, and advanced models; creates new dependencies.
China’s Digital Silk Road	China	Digital infrastructure, cloud, AI services exports	Embeds AI capabilities into BRI corridors, strengthening China centric digital trade routes.
Middle East AI Hub Strategy	UAE, Saudi Arabia, Qatar	Compute clusters, cloud zones, energy powered AI	Turns the region into a global exporter of compute and AI services; attracts multinational AI firms via regulatory free zones.
OECD Cross Border Data Flows	OECD	Data governance, interoperability, trust frameworks	Provides stock taking of global data flow policies; supports common G7/G20 approaches for enabling AI driven digital trade.
OECD Going Digital: Trade & Data Flows	OECD	Policy landscape for international data flows	Identifies how data regulations affect international trade and AI-enabled services; informs trade negotiators.
OECD aligned research on AI & digital trade rules	OECD affiliates, legal scholars	Next gen rules for AI, digital trade, cross border data	Explores regulatory divergence, national security restrictions, and future international digital trade agreements.

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Allianz Research | 19 May 2026

Transatlantic spread – from monetary signal to fiscal mirror

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In Summary

- **Is one of the world's most important pricing signals entering a new era?** Since 2015, the 10-year spread between US Treasury yields and Eurozone German Bunds has ranged between 100bps and 270bps. Since early 2025, pushed first by Germany's fiscal pivot and then by converging rate expectations (hawkish ECB vs Fed easing bias), the 10y US–Bund spread has narrowed by roughly 90bps. If this continues, the transatlantic spread could finally break through the lower trading range that has held for more than a decade.
- **The latest bond sell-off suggests that the convergence trade has run its course: We expect a gradual reversal rather than a new tighter-for-longer regime.** Over the last few weeks, the spread has stopped narrowing and rewidened to around 140bps. In the short run (next 6–12 months), the move should remain modest (+20bps) and still driven by rate expectations as markets reprice a less hawkish ECB against a Fed that stays on hold longer than consensus expects or even proceeds to hike.
- **In the medium term (3–5 years), the transatlantic drivers will shift decisively from rate expectations to risk premia**. We see the US neutral rate rising by 10–20bps on stronger AI-driven productivity gains while the Eurozone neutral rate drifts 10bps lower, consistent with a historical 20–50bps widening of the real-rate spread. On top of that, the US Treasury term premium looks underpriced relative to Bunds: the US convenience yield has flipped from -30bps to +20bps as Treasury supply erodes the safety premium, while the Bund retains a -40bps convenience yield anchored in collateral scarcity. The ECB's faster QT has already cheapened Bund duration by 35bps versus 10bps for Treasuries and we expect that gap to close. Adding a structural +10bps from the OIS/swap-spread channel, we see the spread widening by ~75bps to around 200bps.
- **Ultimately, the transatlantic spread is evolving from a pure monetary-cycle gauge to a barometer of fiscal credibility and sovereign liquidity.** For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a mirror of fiscal sustainability and sovereign market liquidity. For duration positioning, this means active positioning in Bunds versus US Treasuries not only in terms of rate expectations but also where the term premium repricing will be most pronounced (longer maturities). In addition, the marginal buyer of Treasuries will increasingly demand a higher term premium, supporting a steeper US curve and a structurally weaker dollar against the euro. For corporate issuers, the EUR–USD funding-cost differential is expected to widen, favoring euro issuance (reverse Yankee issuance) that may support EUR investment-grade bonds over their US counterparts.

The transatlantic spread reaches the lower bound of its long-term trading range

Is transatlantic spread entering a new lower-for-longer era? The transatlantic spread, which refers to the differential between US Treasury yields and Eurozone benchmark yields (German Bunds), is among the most important pricing signals in global financial markets. It influences two of the largest fixed-income markets and moves the EURUSD, the most important currency pair in global trade and financial transactions.

Any sustained shift in the transatlantic spread carries transmission effects across global capital flows, equity valuations, corporate funding and hedging costs. Since the 1980s, transatlantic spread has experienced three distinct phases. From 1980 to 1995, covering the Volcker Era and the post oil-shock disinflation, it was characterized by a high and volatile spread, fluctuating between +500bps and -150bps for the ten-year maturity. The period from 1995 to 2015 marked a phase of moderation and convergence, during which the spread appeared capped at an upper bound of 100bps and rarely fell below -100bps. With the onset of ECB quantitative easing in 2015, which pushed long-term yields in the Eurozone persistently lower, a third phase began, with the ten-year spread ranging between 100bps and 270bps (Figure 1). Since 2025, the transatlantic spread has narrowed by ~90bps, approaching the lower bound of the third phase trading range. This raises the question of whether we are at the start of a new lower-for-longer regime or should expect a near-term reversal of the narrowing trend (Figure 2).

Figure 1: Long-term 10y transatlantic spread

Spread in %

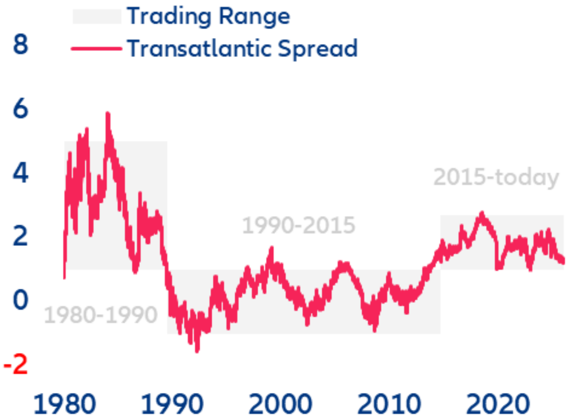
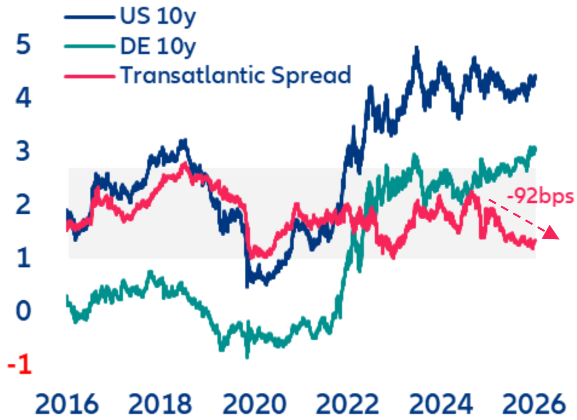


Figure 2: Transatlantic spread at lower bound

Yield and spread in %



Sources: LSEG Workspace, Allianz Research

US and Eurozone yields: from term-premium suppression to repricing rate expectations

At its core, the transatlantic spread signals divergence in the monetary policy stance as well as in the growth and inflation outlooks. But it also embodies risk premia for the uncertainty around future inflation and neutral rates in addition to premia for fiscal risk and market liquidity. Term-structure models help to refine this view by decomposing observed market yields and rates into three components:

- **Rate expectations** capture growth and inflation expectations as expressed by expected nominal short-term rates.
- **Term premium** reflects the uncertainty around the future path of short-term rates and inflation expectations. It also captures duration supply/demand pressure with fiscal policy (debt issuance) on the supply side and changes in central bank bond purchases and holdings – Quantitative Easing (QE) or Quantitative Tightening (QT) – as a major factor on the demand side.
- **OIS spread**, defined here as the difference between government bond yields and the swap rate, represents the premia for market liquidity. Its level and sign signal whether government bonds are scarce or abundant, and whether they are cheap or expensive to finance and hedge (market liquidity). The OIS spread is an indicator of imbalances in government bond supply and demand. The higher the OIS spread, the more constrained governments are in issuing additional debt without paying a liquidity premium to the market.

The weight of these components varies over time. If we look at the composition of US and German 10y yield over the last 10 years, we can clearly see these shifts. Pre-Covid, both yields were subject to prolonged suppression of term premia (negative), reflecting QE duration demand, while rate expectations reflected different monetary policy stances: tightening in the US versus stable in the Eurozone. Post-Covid, both yields surged, driven by rapidly rising rate expectations. During this time, the term premium shifted from a dampening to an amplifying factor once central banks reduced duration demand through QT while governments increased supply by fiscal expansion. Since 2025, US rates have held steady as a rising term premium was offset by falling rate expectations, reflecting the Fed's easing bias amid a late-cycle slowdown. German Bund yields, however, rose steadily. First this was driven by a higher term premium following Germany's fiscal pivot (*Sondervermögen*). In a second phase, from mid-2025 onwards, the yield increase was driven by a shift in rate expectations as markets priced upside risk to euro short-term rates on the back of an emerging economic recovery. Over this period, market liquidity played a minor role for US Treasuries in absolute terms. Structurally, however, OIS spreads switched into positive territory in 2025, marking a shift from the US government being a receiver of a liquidity premium to being a payer. For German Bund yields on the other hand, OIS spreads were a major driver – at times even equal to the term premium. Until 2022, this reflected the relative scarcity of German government debt in the euro sovereign market. Investors paid a substantial premium to hold these ultra-safe assets, contributing to structurally low Bund yields. With the fiscal pivot, this privilege has at least diminished, reflecting the increased supply of German government bonds (Figure 3 and 4).

Figure 3: Decomposition of US 10y Treasury yield
Yield and components in %

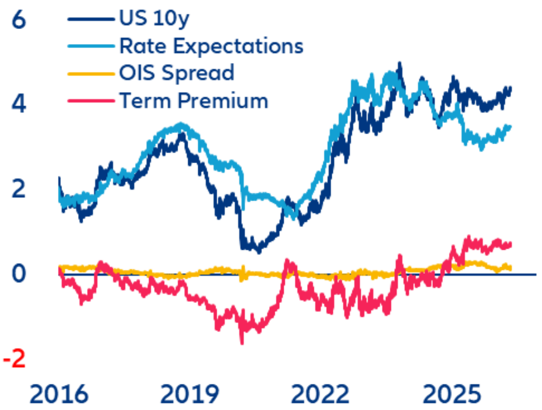
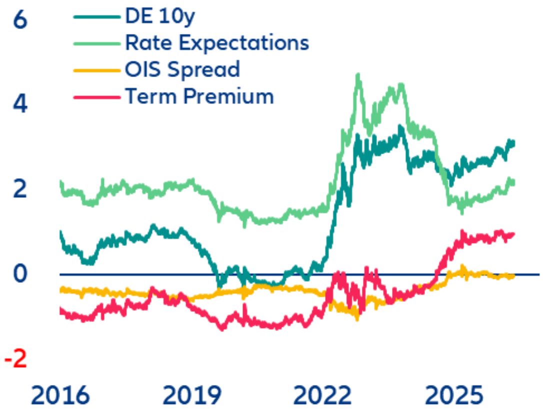


Figure 4: Decomposition of 10y German Bund yield
Yield and components in %

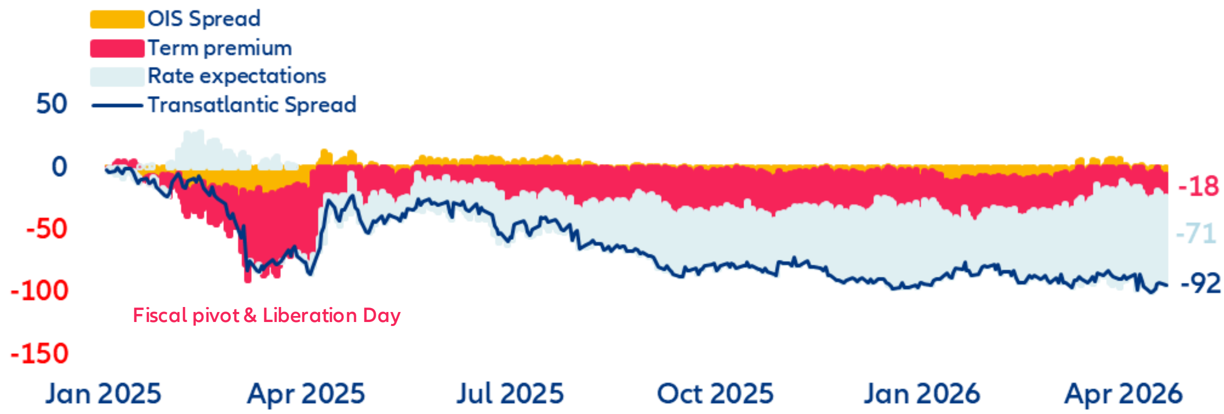


Notes: Term structure decomposition based on Abrahams et al. (2016)
Sources: LSEG Workspace, Allianz Research

Latest transatlantic spread narrowing driven by rate expectations not risk premia

The latest narrowing of the transatlantic spread (-92bps since 2025) was initially driven by a term-premium repricing after the German fiscal pivot, reinforced by the "Liberation Day" trade shock. After that, however, it increasingly became the product of converging rate expectations (reflecting the easing bias of the Fed versus an ECB on standby) and divergent growth dynamics (slowdown in US vs early recovery in Eurozone). The energy-price shock triggered by the Iran war reinforced this trend but inverted the signs of the rate expectations. Markets now expect the Fed to stay on pause while pricing at least two ECB rate hikes by end-2026 (Figure 5). If this trend continues, the transatlantic spread could finally break through the lower trading range of the last 10 years.

Figure 5: Contributions to the 10y transatlantic spread since 2025
Yield and components in bps



Notes: Term structure decomposition based on Abrahams et al. (2016)
Sources: LSEG Workspace, Allianz Research

However, we believe the current convergence trend is nearing its end and is about to reverse. In the near term, markets are pricing in 25bps of excess ECB rate hikes through year-end, in our view. This should push the transatlantic spread wider again in H2 2026 as long-term inflation expectations remain largely aligned, offering little offset. In the medium term, we see real rates as a major driver for a wider transatlantic spread. The expected real rate differential currently sits at its long-term average but we estimate the US neutral rate will rise by 10-20bps over the next five years, driven by stronger productivity growth. In contrast, we see the Eurozone neutral rate declining by 10bps. Historically, such divergence in neutral rates has been followed by a widening of the expected real rate spread between US and Eurozone of at least 20–50bps

Figure 6: Difference in neutral rates and expected rates
Real yield in %

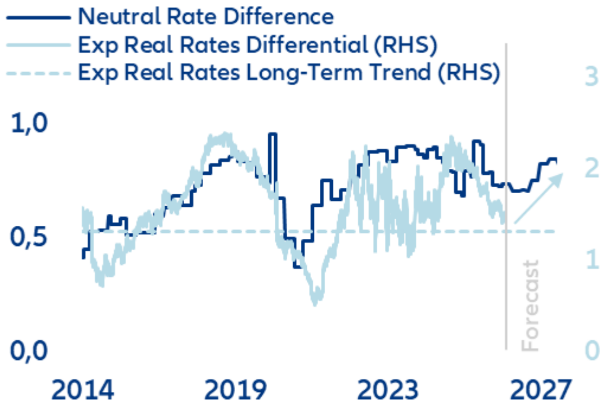
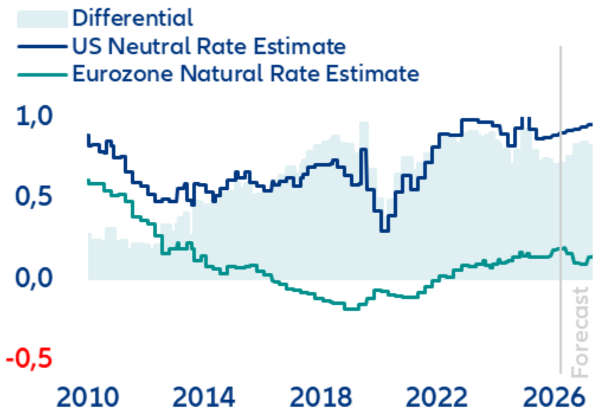


Figure 7: US vs Eurozone neutral rates
Real yield in %



Notes: US neutral rates estimate as median value of Lubik & Matthes (2015), Del Negro et al. (2020), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)
Eurozone neutral rate estimate as median of Lubik & Matthes (2015), Brand et al. (2024), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)
Sources: LSEG Workspace, Allianz Research

Diverging QT pace drives term premium

On the term premium side, we also see no further impulse for convergence in US and Eurozone sovereign yields. The term premiums for 10y Bund and 10y US Treasury tracked closely for an extended period. From mid-2024, however, the Bund term premium began rising significantly faster than its US counterpart. Today, the German term premium stands 60bps above the US level (Figure 8). We consider this gap excessive. Notably, half of the divergence is attributable to the ECB's faster pace of quantitative tightening (QT). We estimate that the run-off of ECB holdings has cheapened 10y Bund duration by 35bps, versus only 10bps in the US (Figure 9).

Figure 8: German term premium exceeds US by 60bps
in %

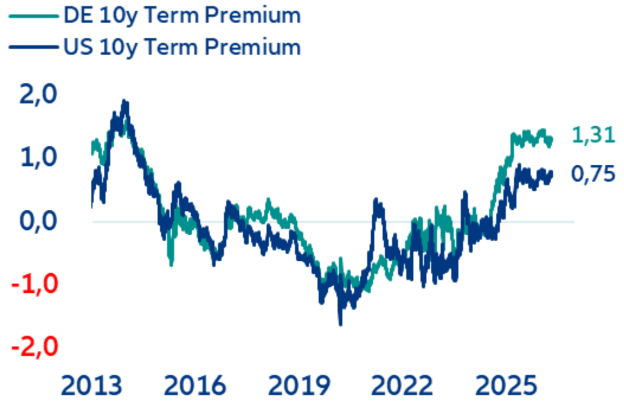
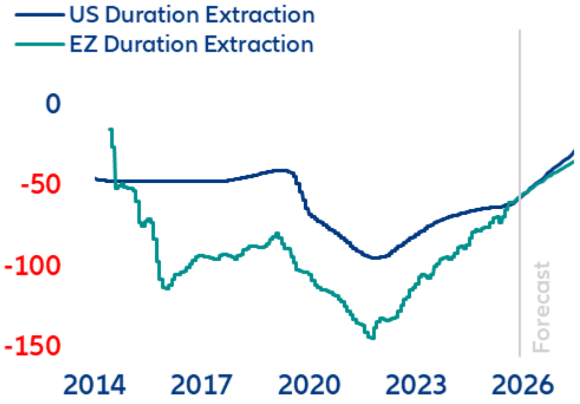


Figure 9: QE/QT effect on 10y term premium
in bps



Notes: Term structure decomposition based on Abrahams et al. (2016)
Notes: for US: Li & Wei (2013) ; Eurozone: Eser at al. (2019)
Sources: LSEG Workspace, Allianz Research

We expect this QT effect to equalize over the next years. In particular, recent Fed T-Bill purchases do not cause a renewed yield-dampening effect. Confined to very short-dated securities, these purchases do not drain duration from the market, therefore the upside pressure on US duration pricing from of Fed portfolio run-off remains intact. The elevated German term premium relative to the US also appears stretched given the US's higher risk profile. Strong expansion of Treasury supply (fiscal deficit) has eroded the US convenience yield – the premium investors pay for an asset's combined safety and liquidity properties. US Treasuries still carry a significant liquidity privilege based on the USD-based collateral (repo) ecosystem. But the safety component has deteriorated to a point where the US convenience yield flipped from a dampening effect of -30bps at end-2024 to +20bps most recently. The Bund convenience yield has also declined with rising issuance (fiscal pivot) but remains at -40bps, which is 60bps below US Treasuries (Figure 10). The scale of the US Treasury risk premium is also visible when comparing the term premium of the US 10y Treasury bond with the one implied in the USD 10y swap rate. The two tracked almost identically until "Liberation Day", after which markets have priced an additional +50bps premium for the sovereign (Figure 11). In our view, this is not yet sufficiently priced relative to the Bund. On this basis, we see clear upward pressure on the US Treasury term premium relative to the Bund.

Figure 10: Convenience yield US vs Germany, 10Y maturity
in %

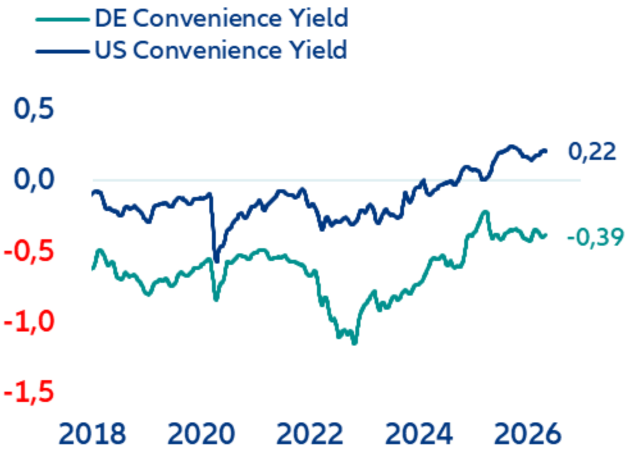


Figure 11: US 10Y Treasury risk premium vs swap
in %

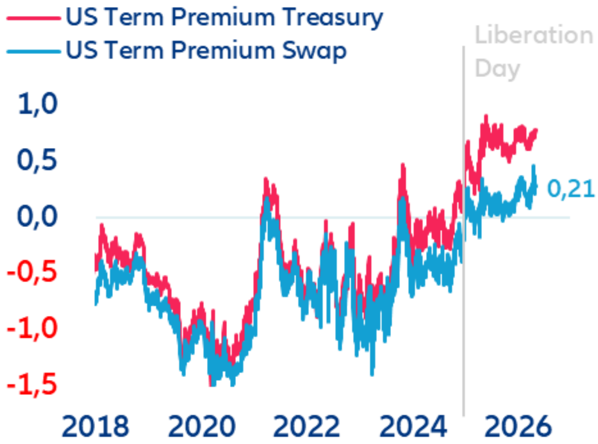


Fig.10 Notes: to illustrate the effect on the total yield, the sign of the convenience yield is inverted. A negative value represents a yield dampening effect, a positive value represents an upside pressure on the total yield. Convenience yield is calculated as the average of synthetic proxies mentioned in Lustig et al. (2024) and the intersovereign convenience yield in Du et al. (2018) Fig 11. Notes: Term structure decomposition based on Abrahams et al. (2016)

Sources: LSEG Workspace, Allianz Research

The transatlantic liquidity divide will drive OIS spreads

The transatlantic OIS spread differential remains structurally biased toward widening in Europe's favor, driven by fundamentally asymmetric liquidity and supply dynamics. In the US, persistently high net Treasury issuance (USD2-3trn/year), structurally constrains dealer balance sheets and weaker foreign demand keeps US swap spreads biased wider. Near-term regulatory relief (Supplementary Leverage Ratio adjustment) could provide a temporary tightening impulse, but the underlying trend remains intact. In the Eurozone, the picture is the mirror image. Despite the step-up in Bund supply (EUR170-180bn additional issuance), superior bank absorption capacity, ample excess reserves and persistent high-quality collateral scarcity in repo markets support a structural tightening bias on euro swap spreads. The specialness[1] of German Bunds is still driven by collateral scarcity rather than credit. This is the repo market signal for of the higher Bund convenience yield. It reinforces this dynamic, as does solid foreign demand on the back of euro strength and a more favorable hedging profile for non-US investors. The net result is a widening US–EU liquidity differential, with euro swap spreads tightening relative to their US counterparts over the medium term. We see potential of around +10bps effect on the transatlantic spread.

[1] Specialness being the premium of a strongly desired collateral which allows cheaper borrowing in repo transactions.

We expect the 10Y spread to widen by approximately +20bps in the short run. In the medium term (next 3 to 5 years) we see a widening of +75bps to around 200bps, reversing the bulk of its compression since 2025 (Table 1). However, we expect a clear shift in the drivers of that move. The transatlantic spread is transitioning from a predominantly rate-expectations-driven signal reflecting monetary policy divergence to one equally shaped by risk premia repricing.

Table 1: Transatlantic spread components

Component	Sub-component	Since 2025	Short-term Outlook	Mid-Term Outlook
Rates Expectations	Real Rates Expectations	-67bps	+15bps	+30bps
	Inflation Expectations	-5bps	+5bps	+10bps
Term Premium	QT Effect	-27bps		
	Risk premium	+12bps		+25bps
OIS Spread	OIS Spread	-5bps		+10bps
TOTAL		-92bps	+20bps	+75bps

Sources: LSEG Workspace, Allianz Research

We find the US Treasury term premium underpriced relative to the Bund, given the scale of US fiscal deficits, eroding convenience yield and tighter liquidity. By contrast, the Eurozone benefits from a supportive convenience yield and more favorable liquidity conditions. This expected change in the composition of the transatlantic spread shows how it is evolving from a pure monetary cycle signal into a barometer of sovereign fiscal credibility. Treasury supply dynamics will increasingly set the floor for US long-end yields as the US is forced to offer elevated real rates to retain foreign capital inflows. Europe, meanwhile, stands to benefit from a relative repricing: even as Bund supply rises, the Eurozone's stronger structural liquidity position and lower fiscal risk premium provide a durable anchor. For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a gauge of fiscal sustainability and sovereign market liquidity. A cross-curve duration positioning between Bunds and US Treasuries should therefore not only reflect rates expectations but take into account which curve segment will see the most pronounced term premium repricing. A structurally higher term premium in the US should also add steepening pressure on the US curve and act as a weakening force on the USD versus EUR. On credit and issuance, we expect the EUR–USD funding cost differential to widen, favoring euro issuance for global borrowers (reverse Yankee supply). This could have a tightening effect on EUR investment-grade spreads relative to their USD counterparts. In short, the transatlantic spread is moving from a "*Fed versus ECB*" story to a "*Treasury versus Bund*" story, and portfolios built for the former need to be repositioned for the latter.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
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Allianz Research | 12 May 2026

Behind the gate: The promises and perils of private markets democratization

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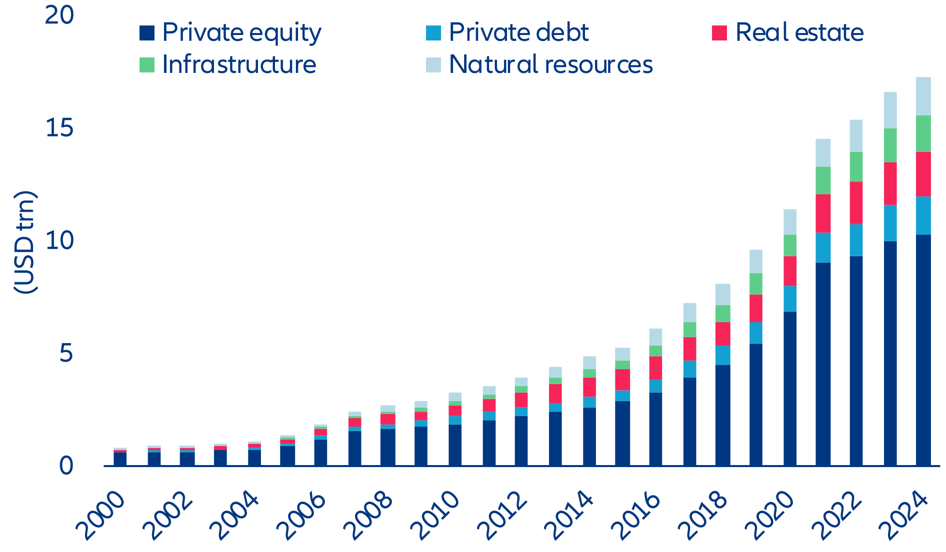
In Summary

- **Private markets are opening up to retail, and the product has to be rebuilt to fit.** Global private markets have grown more than twentyfold since 2000 to over USD17trn, propelled by institutional adoption of the Yale endowment model that tilted long-duration capital aggressively into illiquid assets. The next leg of growth runs through wealth channels and mass markets, where investors lack the governance, illiquidity tolerance and manager-selection capability. Semi-liquid evergreens have become the workhorse for wealth, while the mass market is reached primarily via DC plans, regulated retail funds, public-private hybrids, and insurance wrappers, structures in which a fiduciary, insurer, regulatory template or the product design itself makes the call rather than the end investor.
- **Three forces have converged to push democratization: savers need enhanced returns and diversification, governments need to mobilize private savings for public priorities and asset managers need new pools.** Diversified private-market exposure can improve portfolio efficiency, lifting expected returns from 6.2% to 7.9% while keeping stressed downside risk (CVaR) around -3% versus -5% for traditional liquid portfolios. Governments see a policy lever to address a USD400trn retirement gap by 2050, growing inequality as value creation stays private and infrastructure needs that public budgets cannot fund. For asset managers, the move is increasingly necessary: institutional fundraising is slowing as the traditional capital pool matures and large LPs reach their target allocations.
- **The pitch is real, but a closer look reveals a more nuanced picture.** First, the return story is compelling at face value: buyout funds have delivered 12-14% annually against 7-9% for public equities. But strip out leverage and most of the private-equity advantage disappears, leaving manager selection as the real driver of outperformance. Second, retail investors face a compounding triple disadvantage compared to institutional products: additional fee layers that erode the illiquidity premium, potentially weaker underlying assets, and no ability to select managers in an asset class where dispersion runs wide. Third, the semi-liquid promise is not illusory, but it comes with conditions; it works in normal market conditions but exposes investors to gates precisely when they want their capital back. Finally, retail money brings reactive behavior into an asset class increasingly tied to banks and life insurers, reintroducing the maturity transformation that closed-end structures had engineered away.
- **The early-2026 slowdown is a stress test, but structural drivers remain intact.** Expect product recalibration toward larger liquid buffers and tighter gating, consolidation toward mega-GPs with the scale and distribution to run retail wrappers and a new round of regulatory scrutiny on disclosure, leverage and liquidity buffers, adding cost but reinforcing product quality.
- **For investors, the dispersion between well- and poorly-built retail vehicles will widen.** Four markers separate institutional-grade retail product from repackaged versions: liquidity sized to the underlying asset, clean separation of retail and institutional pools, independent valuation governance and real manager-selection capability behind the wrapper. For institutional investors, the illiquidity premium will compress and deal-allocation conflicts will emerge as GPs spread a finite pipeline across vehicles with different liquidity profiles, making vigilance on cross-pool transfers essential.

A new investor base, a new product shelf

Private markets were built around institutions and family offices. The 1980s leveraged-buyout boom, fueled by the advent of junk bond markets, lit the fuse; institutional money turned the resulting industry into a colossus. Global private-capital assets under management (AUM) have grown from roughly USD800bn in the early 2000s to over USD17trn by 2024 (Figure 1), more than twentyfold in two decades.

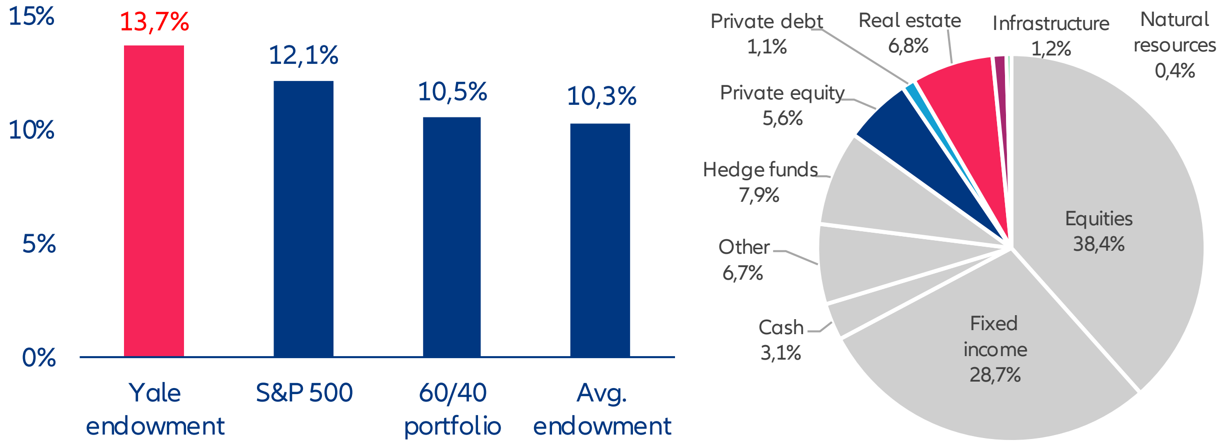
Figure 1: Global private capital AUM has grown more than twentyfold since 2000



Sources: Preqin, Allianz Research
Note: Secondaries and funds of funds are excluded to avoid double-counting.

The pivotal driver behind the transformation of private markets into a mainstream institutional allocation is Yale's endowment model. When David Swensen took over the university's investment office in 1985, endowments ran the standard 60/40 mix. Over the next three decades he tilted the portfolio aggressively into private assets, arguing that long-duration capital should earn an illiquidity premium that liquid investors could not. By 2021, Yale held roughly 70% in alternatives, nearly ten times the pre-Swensen level, and the portfolio had returned 13.7% a year over Swensen's tenure against 10.5% for 60/40 and 10.3% for the average endowment (Figure 2). Yale's success rippled outward. By the 2010s, the biggest US university endowments held 40-60% in private markets on average. Sovereign wealth funds, public pension funds and insurers followed. Today, global institutional investors allocate around 10% to private assets (Figure 2), with endowments and public pensions allocating more than a third.

Figure 2: The Yale endowment has outperformed traditional benchmarks under Swensen's management (left) and global institutional investors are now allocating around 10% in private assets (right)

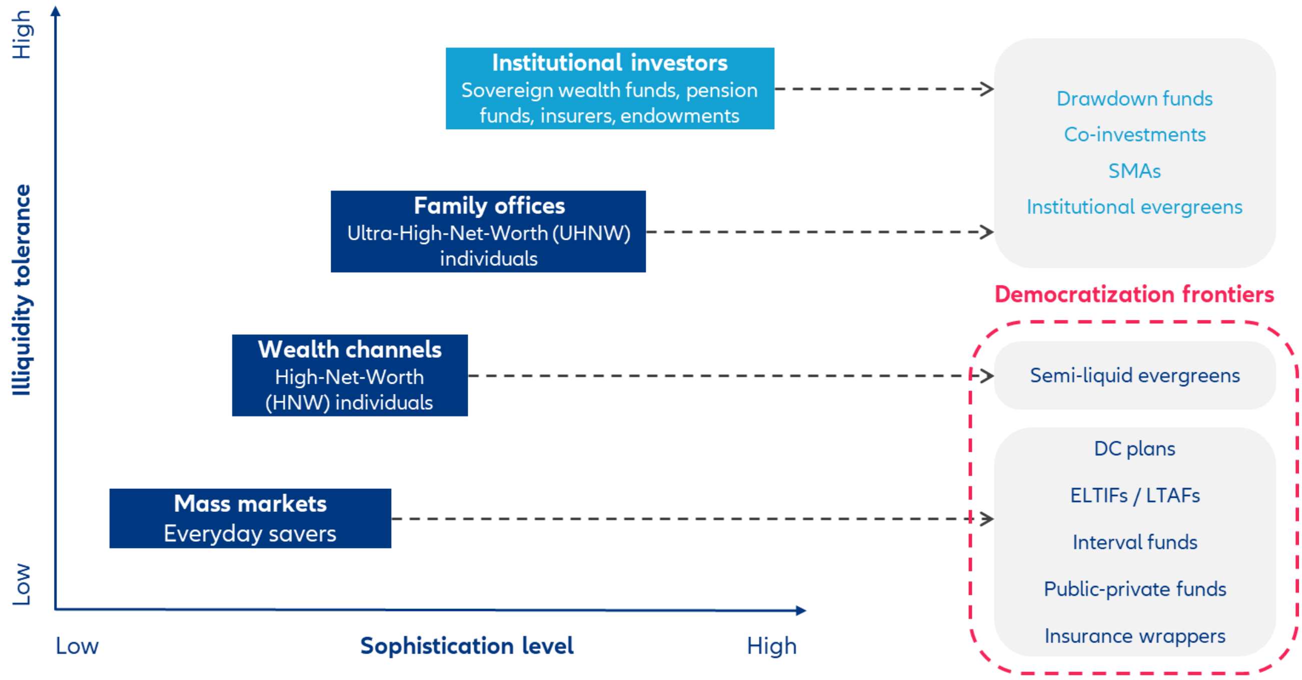


Sources: Bloomberg, The Economist, Preqin, Allianz Research

The fit is natural. Most institutional capital sits in drawdown funds run by general partners (GPs) who source, structure and manage the underlying deals. The 10-year-plus horizon suits pension funds and life insurers with liabilities measured in decades; in-house teams, investment committees and boards do the diligence that opaque assets demand and scale buys both fee leverage and access to the best managers. Beyond drawdown funds, institutions increasingly mix in co-investments (alongside the GP, often at low or zero fees), separately managed accounts (SMAs, typically USD100mn-plus, fully customized) and open-ended evergreens (operational simplicity and continuous liquidity, no J-curve). Together these tools let institutions calibrate exposure with a precision the conventional fund template cannot match. **Family offices, the private treasuries of the ultra-wealthy (a USD 30mn threshold, though in practice more is needed to access large GPs), follow the same model in miniature.** They vary in sophistication but use the same vehicles to reach the same managers, sometimes on slightly worse terms.

Now, the action has moved to the wealth and mass-market channels, where investors' needs are entirely different. "Democratization" means giving private-market access to those locked out of the institutional product suite and that requires more than lower minimums. It calls for fresh product design, with new trade-offs between liquidity, fees, transparency and asset quality (Figure 3).

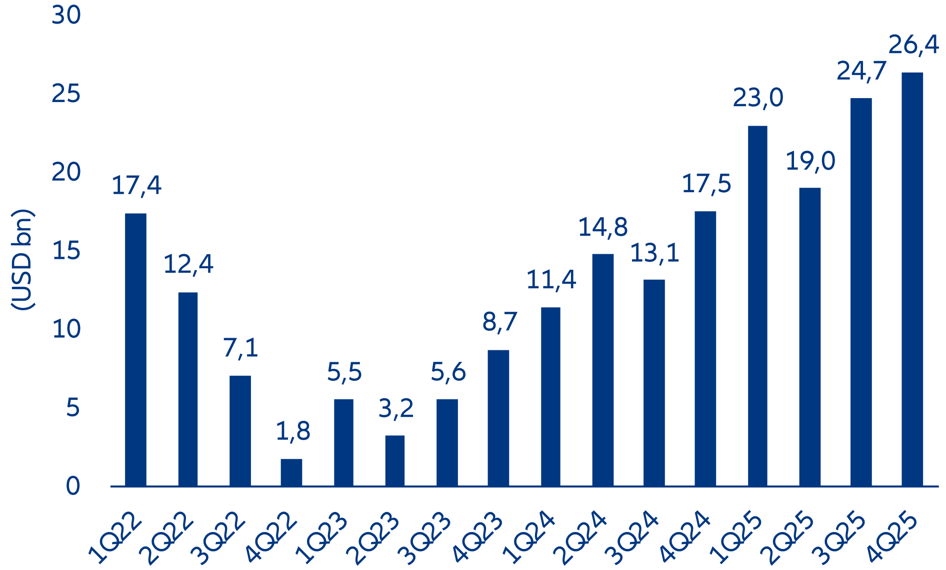
Figure 3: Investor segmentation and product structures across the wealth pyramid



Source: Allianz Research

The wealth channel covers the high- and very-high-net-worth segment, growing fast in both numbers and assets. Entry starts at USD1mn of investable assets; the more active VHNW tier kicks in at USD5mn. US households with USD5mn to USD20mn now control 40% of addressable wealth, up from 18% in 2013, turning them from once-niche cohort to a market force. They are too small to deal with GPs directly and reach private markets through private banks, wealth managers and financial advisers. This is the most commercially mature corner of democratization, and one of the fastest-growing. Inflows to the major listed alternative-asset managers have surged over the past three years, topping USD25bn in the fourth quarter of 2025 alone (Figure 4). US financial advisers now allocate USD1.9trn to less-than-fully-liquid strategies, but the average adviser still has only 2.3% of client portfolios in alternatives, and roughly half use none at all. The runway is long.

Figure 4: Net inflows to wealth products offered by major listed alternative managers have surged since early 2023



Sources: JP Morgan, Allianz Research

The mass market is a much bigger prize, and a wholly different machine. Its investors have less investment expertise than institutions, shorter effective horizons and no practical capacity to select managers themselves, which is why the channel is built around fiduciaries, insurers and regulated structures that do the selection for them. The US mass affluent and middle-market segments, 46.9m households in all, hold USD25trn of financial assets. The retirement system holds another USD49trn, of which USD14trn sits in employer-sponsored DC plans. In Europe, household savings are mostly parked in low-yielding bank deposits, around EUR10trn across the EU; the next-largest pool is unit-linked insurance, notably France's assurance-vie, Italy's PIR and Germany's equivalents.

Three structural differences set wealth-channel investors apart from institutions. They lack the governance infrastructure, they are less tolerant to illiquidity and they reach private markets through intermediaries whose operating models demand standardized features that fit into model portfolios. The result is a generation of products engineered for accessibility, with performance pursued within those constraints rather than at any cost. Semi-liquid evergreens have become the workhorse: quarterly subscription and redemption windows, no capital calls, simplified fees and standardized tax reporting. Each feature addresses a specific constraint. Well-engineered evergreens, with adequate liquidity buffers and disciplined valuation, can deliver a meaningful share of the institutional return profile to investors who previously had no access at all.

Reaching the mass market requires a different paradigm. The end investor no longer chooses the product directly; instead, the structure has to satisfy a fiduciary or regulator that it is suitable for an average participant. The result is a new range of vehicles, each tailored to a specific regulatory category or savings product rather than evaluated on its own merits. New structures are launching in every major jurisdiction; the main channels are defined contribution (DC) plans, ELTIFs and LTAFs, interval funds, public-private hybrids and insurance wrappers.

The deepest pools sit inside existing retirement and savings systems. US DC plans, accessed mainly through target-date funds with embedded private-market sleeves, are the largest mass-market pool in the world: USD14trn of employer-sponsored DC assets, including roughly USD10trn in 401(k)s. European insurance wrappers play the same role in the savings sector, with the insurer handling manager selection, valuation oversight and liquidity management for the policyholder. Both work precisely because they do not demand investment expertise of the end investor: the fiduciary or insurer takes on the discipline that retail investors lack and would struggle to acquire. The private exposure is bundled inside a familiar structure and the design ensures that no single policyholder's liquidity needs can destabilize the underlying assets. The retail experience is unchanged at the surface; private-market access, properly governed, sits one layer below. For long-horizon savers, this is one of the few realistic routes to the diversification and return premium of private markets without bearing institutional-style governance demands themselves.

Regulated retail vehicles take a more direct path, swapping regulators for fiduciaries. ELTIFs in the EU, LTAFs in the UK and interval funds in the US are standalone retail products an investor or adviser can pick actively, subject to rules on eligible assets, concentration, mandatory repurchases and disclosure. The regulatory framework replaces the institutional governance these products would otherwise lack, which is a large part of their appeal: the regulator has effectively vetted the structure. The cost is product flexibility and the channel has tilted strongly toward private debt and lower-volatility strategies that fit cleanly inside the templates.

Hybrid public-private funds are the newest design, aimed squarely at the liquidity problem. DC plans absorb redemption pressure through fiduciary scale; regulated retail vehicles use prescribed gates; hybrids embed a 30-60% public-asset buffer into the product itself. The public sleeve provides exit capacity; the private sleeve provides the alternative exposure that justifies the fund. The trade-off is return: a 40% private allocation inside a fund that is itself only a slice of a retail portfolio leaves effective private exposure of just a few percent at the investor level. Hybrids are a design solution to mass-market liquidity rather than a route to maximized private-market exposure. Their merit is extending some private-market characteristics to investors for whom pure private vehicles are not workable, accepting a return trade-off in exchange for genuinely daily-liquidity behavior.

The four channels share a single design principle, with different machinery. None assumes the end investor actively chooses the product. The decision rests with a fiduciary (DC plans), an insurer (insurance wrappers), a regulatory template (ELTIFs, LTAFs, interval funds) or the structural design itself (hybrids). This shift in who decides ultimately drives the outcome for investors, and means the routes to mass-market capital are likely to fragment by jurisdiction rather than converge on a single global model.

Why private markets are going mainstream now

For retail investors, the case rests on diversification and better risk-adjusted returns. The evidence is plain: private assets diversify; traditional ones increasingly do not. Since 2022, correlations between public equities and bonds have risen sharply, eroding the protective property of the standard 60/40 portfolio. Modern portfolio theory says that adding assets with low or negative correlations shifts the efficient frontier up and to the left; the correlation matrix in Table 1 backs this up. The numbers are forward-looking, not historical, and reflect today's regime. The arrows mark the largest moves, concentrated among traditional assets that have grown more closely correlated under stagflationary pressure.

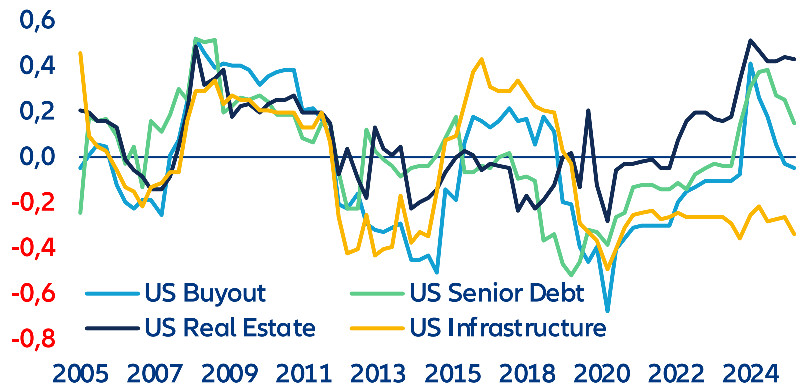
Table 1: Cross asset correlation matrix

	Euro Corporates	Euro High Yield	Euro Covered Bonds	USD Corporates	Equity BAA Aggregate	US Private Equity (Buyout)	US Private Debt (senior)	US Real Estate	US Infrastructure	Euro Government Bonds	Euro Agency bonds	Emerging Markets Bonds	Gold
Euro Corporates	1,00	0,82	0,86	0,77	0,63	0,20	0,31	0,24	0,17	0,80	0,83	0,87	0,47
Euro High Yield		1,00	0,53	0,54	0,82	0,24	0,33	0,20	0,30	0,42	0,48 ▲	0,83	0,43
Euro Covered Bonds			1,00	0,71	0,37	0,21	0,31	0,19	0,20	0,93	0,94	0,70	0,49
USD Corporates				1,00	0,43	0,17	0,23	0,30	0,12	0,72	0,73	0,76	0,53
Equity BAA Aggregate					1,00	0,14	0,26	0,09	0,24	0,28 ▲	0,33	0,74	0,46
US Private Equity (Buyout)						1,00	0,81	0,42 ▼	0,71	0,09	0,10	0,20	0,42
US Private Debt (senior)							1,00	0,43	0,62	0,22	0,25	0,31	0,37
US Real Estate								1,00	0,17 ▼	0,23	0,21	0,17	0,04
US Infrastructure									1,00	0,01	0,03	0,27	0,58 ▲
Euro Government Bonds										1,00	0,98	0,62	0,35
Euro Agency bonds											1,00	0,67	0,38
Emerging Markets Bonds												1,00	0,63
Gold													1,00

Source: LSEG Datastream, Allianz Research

Private assets are consistently uncorrelated with public ones, and barely correlated with each other. As public-market correlations climb, infrastructure stands out as the biggest beneficiary; pair correlations with euro government, agency and corporate bonds are all low. The more striking feature of the matrix is the heterogeneity within the private universe itself: real estate and infrastructure are not substitutes for private equity. The rolling four-year analysis in Figure 6 backs this up. From 2005 to 2026, all four American private asset classes, PE buyout, senior debt, real estate and infrastructure, have averaged correlations close to zero with a 60/40 public portfolio, none above 0.5. The dispersion has widened recently, with infrastructure the most decoupled, even as overall market correlations have climbed.

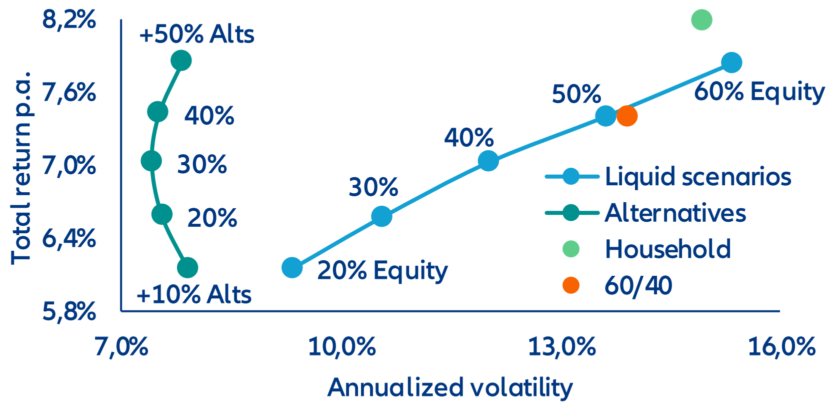
Figure 6: Cross-asset rolling correlations



Source: Pitchbook, Allianz Research

The most compelling quantitative argument for democratization sits in the efficient frontier itself. Figure 7 plots annualized volatility against total return for portfolios ranging from a pure traditional mix of 13 asset classes to ones with progressively larger alternative allocations (10-50%). As the alternatives weight rises, the frontier shifts unmistakably outward: the green alternatives frontier dominates the orange liquid frontier at every point, lower volatility and higher returns with no exceptions. Theory says low-correlation assets should improve the trade-off; the empirical gap is striking. Private assets earn an illiquidity premium without the concentration risk that comes from leaning on public equity, provided investors can stomach the illiquidity and keep enough liquid ballast for market stress. The classic 60/40 (orange dot) and the typical European household (green dot) sit on noticeably higher volatility, since bonds and equities have failed to diversify each other since 2022's stagflation. Owning a home does not fix this; a diversified slice of private markets does. The same pattern shows up in Conditional Value at Risk, which captures downside losses in bad markets and actually fits the non-normal distribution households better than volatility does.

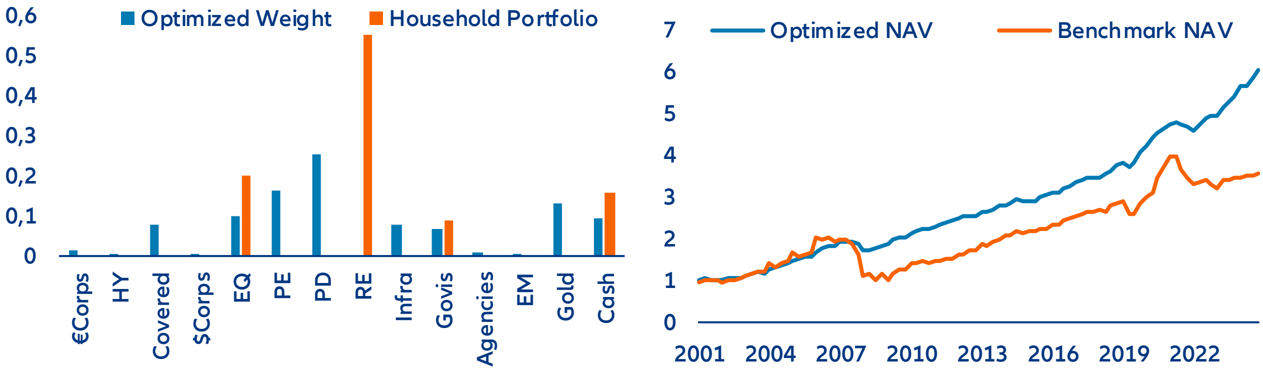
Figure 7: Efficient frontier liquid - illiquid assets



Source: Pitchbook, Allianz Research

An optimizer sharpens the picture. Figures 8 and 9 turn the theoretical frontier into a realized-performance comparison. The optimized portfolio (blue), allocating across alternatives and traditional assets, posts a Sharpe ratio of 2.85, against 0.64 for the typical household benchmark (orange). The optimizer solves one trade-off: maximize risk-adjusted return while penalizing both single-asset concentration and large tail losses (CVaR). It uses market-cap weights, forward-looking assumptions for returns, volatility and correlations, and alternative-history scenarios rather than historical averages, producing an allocation that holds up across regimes. The result a balanced portfolio, half in private markets, complemented by low-risk bonds, gold and cash. Smaller private allocations still make sense, since the diversification benefit persists across weightings. The right share for any investor depends on their tolerance for risk.

Figure 8: Portfolio allocation in % (left) and portfolio performance (right)



Source: Pitchbook, Allianz Research

For governments, democratization is more than an access question. It is a policy lever for retirement adequacy, distributional fairness and the funding of long-term economic priorities. With public finances constrained and traditional capital sources struggling to keep up, private markets are being repositioned as a way to mobilize household savings.

The most pressing motive is the retirement-adequacy crisis. The shift from defined-benefit to defined-contribution pensions has moved investment risk from employers and governments to individuals, leaving DC participants with portfolios dominated by public equities and bonds. DB plans have long held 15-30% in alternatives and benefited; DC participants have not, and the gap has compounded into a serious shortfall. The global retirement-savings gap is projected to reach USD400trn by 2050, up from USD70trn in 2023. With public-pension systems under strain, widening the DC investment universe is a way to narrow the gap without lifting public spending.

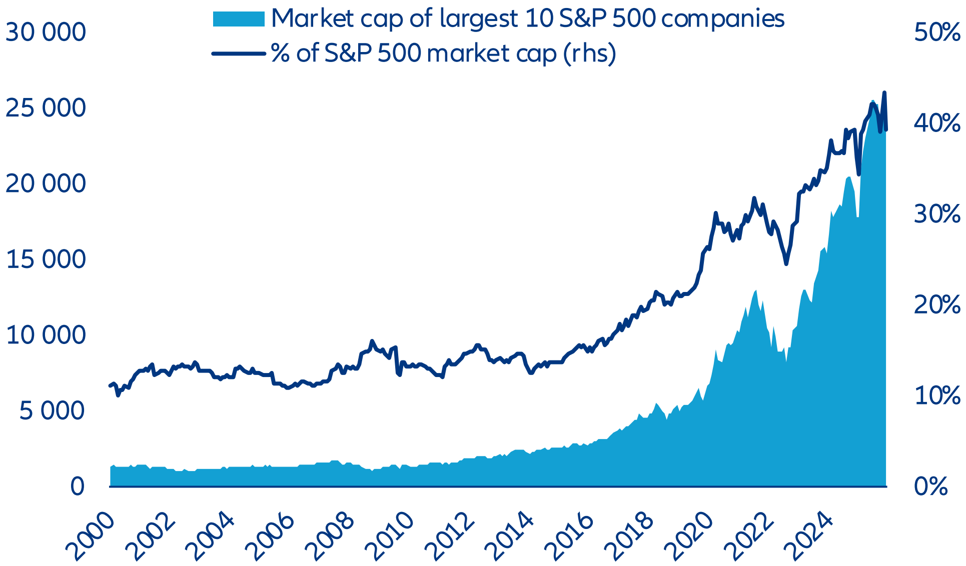
The second motive is inequality. As private markets stay walled off, a growing share of value creation slips out of reach of ordinary savers. The number of US-listed firms has roughly halved since the late 1990s, while the number with fewer than 500 employees has grown by nearly a fifth (Figure 10). Companies stay private for longer, so much of their growth, and the wealth that comes with it, is captured by sophisticated investors with broader access. Ordinary savers, stuck in public equities, watch from the sidelines.

Figure 9: US listed companies have nearly halved while mid-market firms have grown by one-fifth over the past three decades

Sources: US Census Bureau, World Bank, Allianz Research

Public-market concentration is a third motive. The top 10 S&P 500 firms now make up 40% of the index, and US equity benchmarks are more concentrated than at any point since 2000 (Figure 11). Retail investors who hold passive index funds are heavily exposed to a narrow set of large-cap tech names; private assets reach sectors and stages that public indices underrepresent. The case for retail access is not just about returns. It is also about reducing the concentration risk that the passive revolution has, unintentionally, amplified.

Figure 10: S&P 500 concentration has reached its highest level since 2000



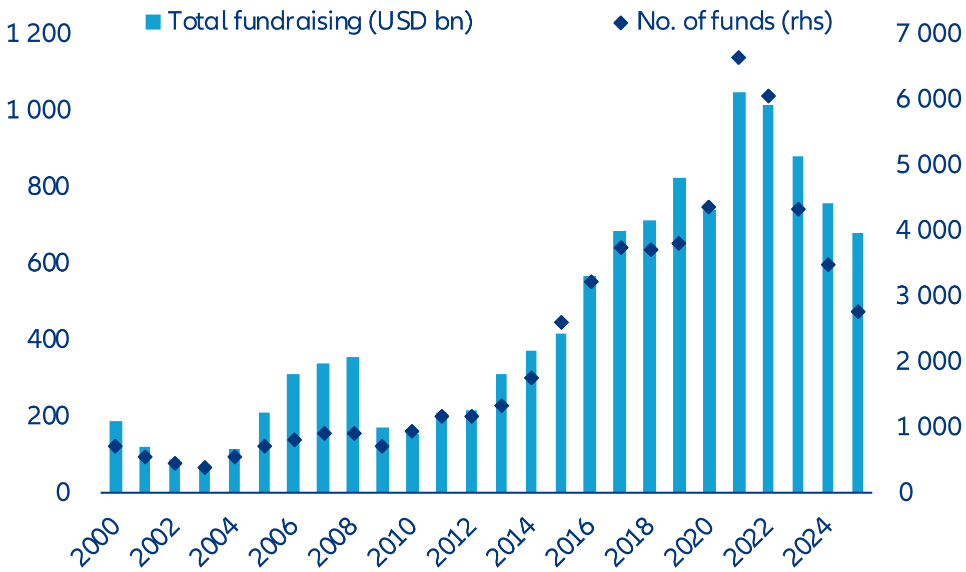
Sources: Macrobond, Allianz Research

Private capital is also being marshaled to fill gaps public budgets cannot. The EU's investment needs for infrastructure, the energy transition and digital transformation are projected at roughly 5% of GDP a year, well beyond what fiscally constrained governments can fund. Europe's EUR10trn or so of household bank deposits is largely unproductive capital. Channeling those savings into infrastructure, growth equity and energy projects through ELTIFs or LTAFs would do two things at once: lift household returns over the long term and deliver real-economy capital that the public purse cannot.

Jurisdictional competition is shaping both the design and the timing of these reforms. US, the EU, the UK, Australia and Singapore are all moving at once, none willing to see domestic savings drift offshore in search of alternatives. Britain's decision to make LTAFs ISA-eligible from 2026 was driven by competition with Luxembourg and Dublin as fund-domicile hubs; France's Green Industry Act was designed to keep retail savings inside the French insurance sector. The race is not just to channel retail capital into private markets, but to do so on one's own terms before a rival captures the flow.

For asset managers, the move into wealth and retail is no longer optional. After decades of rising inflows, institutional fundraising has hit a wall. In 2025, closed-end private-equity fundraising fell to a five-year low (Figure 12), as subdued exits left more than 16,000 portfolio companies held for over four years and LP distributions slowed to a trickle. The 2022 public-market drawdowns produced a denominator effect that left many institutions over their allocation targets, prompting a pause in fresh commitments. Beyond the cycle, the institutions that began private-markets programs in the 2000s have now reached, or nearly reached, their targets. With the traditional capital pool maturing, the industry has pivoted to investor groups that, until recently, had limited access.

Figure 11: PE fundraising volume and fund count have fallen sharply from their 2021 peaks



Sources: Preqin, Allianz Research

Box 1. Policy in motion: bringing private markets into 401(k)s

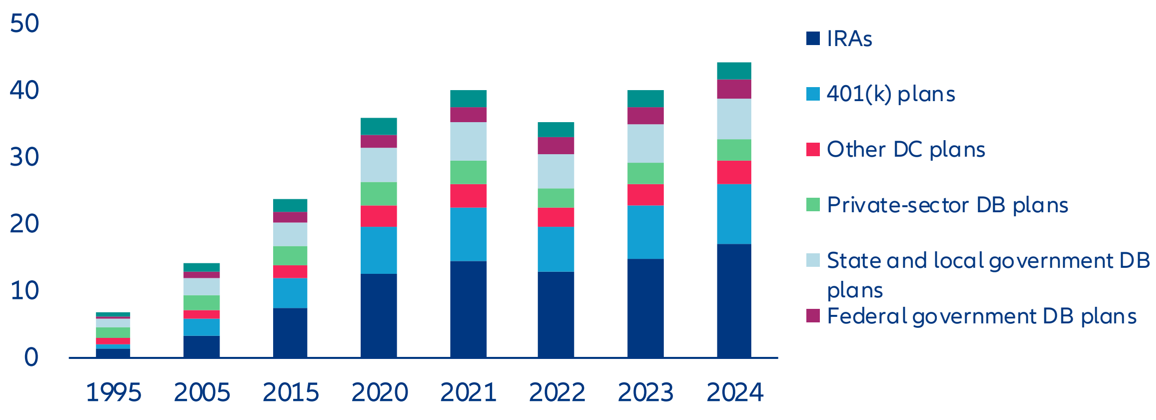
For most of their existence, American 401(k) and other defined-contribution (DC) plans have excluded private-market assets. ERISA fiduciary constraints, litigation risk and operational frictions (illiquidity, valuation complexity, fee opacity) made them poor fits. The SEC's accredited-investor rules further limited direct retail participation, reinforcing the structural divide between DC and defined-benefit (DB) plans. The first significant move came in June 2020, when the DOL issued an Information Letter allowing private equity inside diversified DC options such as target-date and balanced funds, provided it was managed prudently. A December 2021 Supplemental Statement then signaled a pause, noting that most plan sponsors lacked the expertise to evaluate such products properly.

By 2025 the policy had shifted. On 12 August 2025, the DOL formally rescinded its 2021 guidance discouraging 401(k) alternatives. The move aligns with the White House Executive Order on "Democratizing Access to Alternative Assets for 401(k) Investors" and clarifies that exposure to private-market assets may be permitted within diversified options under ERISA. The DOL is now signaling a neutral, principles-based approach: Fiduciaries must judge investment options on their merits and suitability, a clear departure from the earlier practice of singling out specific asset classes for scrutiny.

Congress has lent legislative support. The SECURE Act 2.0 of 2022 facilitated alternative access indirectly, by letting 403(b) plans use Collective Investment Trusts (CITs). CITs are often the preferred vehicle for private-market exposure in employer plans thanks to lower costs and more customization than mutual funds. Together these moves create a coherent policy environment in which private-market exposure can expand within DC plans under existing fiduciary and securities-law frameworks. An alternative investment option that sits inside a well-diversified fund managed by qualified professionals, and that the plan sponsor has documented thoroughly, can sit comfortably within ERISA's duty of prudence. Further sub-regulatory guidance from the DOL is expected, and the SEC may add guidance on valuation and disclosure as more products come to market.

The base effects are substantial. As of Q1 2025, the American DC system held roughly USD12.2trn in assets, compared with USD3.8trn in private and public DB plans. Alternatives make up around 30% of DB allocations but just 0.1% of DC portfolios, so even a modest reallocation would be material in absolute terms. A baseline scenario of 3-5% DC penetration by 2030 would deliver cumulative inflows of USD300bn-400bn; an upside of 10% penetration by 2035 would mean USD1.2trn-1.5trn in new allocations. Adoption is expected to start with large-plan default options that wrap diversified PE and private credit inside multi-asset structures, then diffuse gradually into mid-sized plans as operational frameworks standardize.

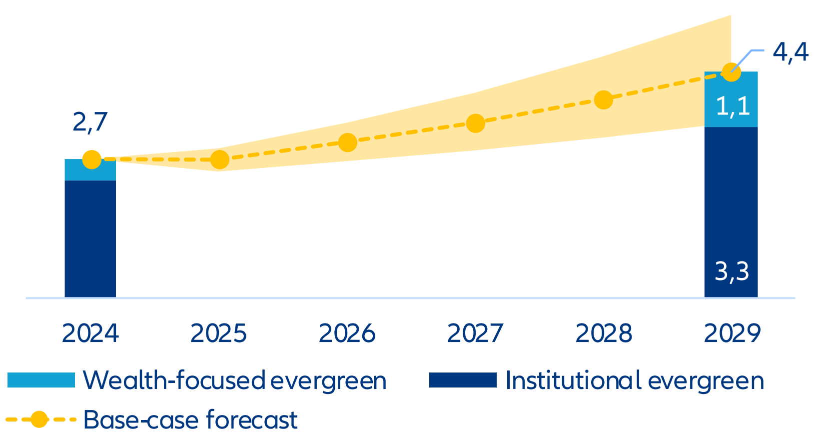
Figure 12: US retirement market assets (USD tn)



Sources: Investment Company Institute, The US Retirement Market, Fourth Quarter 2024, Allianz Research

Product evolution and opportunity set. Capital will flow along the path of least operational resistance. CITs are gaining ground because of their flexibility on liquidity and fees, smooth integration into plan recordkeeping and exemption from certain retail-disclosure requirements under the Investment Company Act. Semi-liquid '40-Act interval and tender-offer funds serve as secondary conduits for smaller, retail-aligned platforms, though their quarterly-liquidity features and higher cost structures may limit early scale. Across both channels, managers are re-engineering fund economics toward evergreen structures with quarterly or semi-annual rebalancing to accommodate DC flows. Pitchbook estimates that roughly $2.7 tn is now managed in various indefinite-life formats globally, projecting growth to $4.4 tn by the end of 2029 (Figure 13). Recordkeepers, custodians and administrators are investing in capabilities to support partial redemptions, performance-fee accruals and standardized daily valuation reporting.

Figure 13: Pitchbook estimate for Evergreen fund AUM forecast (USD tn)



Source(s): Pitchbook, Allianz Research. Geography: Global. Note: Note: Forecasts were generated on April 14, 2025. "Institutional evergreen" includes insurance AUM from Blackstone, KKR, Blue Owl Capital, The Carlyle Group, Ares Management, Apollo Global Management, and Brookfield.

Within diversified default options such as target-date funds, private equity is expected to dominate the growth allocation for younger cohorts, mainly through fund-of-funds or secondaries to achieve vintage diversification and reduce dispersion risk. Private credit is well placed to stabilize income for pre-retirement cohorts, focused on senior secured lending, infrastructure debt and core real-estate credit. Real assets (core real estate, infrastructure equity) will play a secondary role as inflation hedges and diversifiers, supported by established appraisal-based valuation practices. Allocations will stay modest relative to liquid assets, typically capped at 10-15% of the default fund, to maintain daily-liquidity compliance.

The primary concerns at this stage are fiduciary risk, liquidity and valuation governance. Fiduciary exposure under ERISA remains real: sponsors must justify private-market inclusion as prudent on a net-of-fee basis, with documented due diligence and monitoring. The mismatch between daily participant flows and illiquid underlying holdings calls for hard caps, liquid sleeves and robust queue procedures, and under stress redemption gates can still be tested. A related vulnerability is the need for valuation fairness across transacting participants, which requires consistent third-party appraisal cycles, interim model-based pricing and audit transparency. Operational dependencies, recordkeeping, performance-fee calculation and participant communications introduce a layer of execution risk that must be addressed before a large-scale rollout.

For investors, the policy shift creates a durable, multi-year reallocation theme, not a short-term trade. Early strategies will favor diversified private equity and senior private credit that fit established liquidity and valuation criteria, with real assets in a supporting role. As operational systems and legal precedents mature, DC allocations to private markets could move from pilot programs to a structural component of retirement portfolios. Early participation should bring stable, long-duration capital to managers; investors should expect risk premia in yield-oriented segments to converge gradually as the DC channel scales.

Do returns justify the hype?

Whether private assets truly outperform is less straightforward than the headline numbers suggest. At first glance, the case is compelling. Over the past two decades, buyout funds have delivered net annualized returns of roughly 12-14%, against 7-9% for broad public benchmarks like the Russell 3000[1]. But this obscures significant variation. Private debt has yielded steady mid- to high-single-digit returns; venture capital and real estate have been more cyclical, with periods of underperformance against public benchmarks.

The critical adjustment is leverage, and it transforms the comparison. Buyout funds typically run 50-70% loan-to-value at the initial stage to amplify equity returns; public benchmarks are unlevered. Once public markets are adjusted to comparable risk, the excess return collapses. Simulations from 2000-2024 show that a 20% buyout allocation generates only about 64bps of annual excess return over an unlevered benchmark, and roughly nothing once the benchmark is levered to match. Most of the apparent alpha turns out to be financial engineering, not investment skill[2].

Table 2: Private markets vs public markets (leveraged comparison, 2000–2024)

Asset Class	Portfolio Return	Benchmark (Unlevered)	Excess Return	Benchmark (Leveraged)	Excess Return (Leveraged)
Private Equity (Buyout)	7.54%	6.90%	0.64%	7.58%	-0.03%
Private Debt	7.37%	6.80%	0.57%	7.09%	0.29%
Venture Capital	5.71%	6.92%	-1.21%	,	Negative
Real Estate	6.89%	7.18%	-0.29%	,	Negative

Source: PitchBook simulations

The picture is one of conditional opportunities, not structural outperformance. Private markets can generate excess returns, but those returns are neither uniform nor guaranteed. Outcomes diverge sharply by asset class. Private debt emerges as the most consistent contributor, holding on to positive excess returns even on a leverage-adjusted basis. The buyout edge largely disappears; venture capital and real estate typically underperform. Results hinge on cycle, entry point and vintage, and the post-crisis years show notable regime sensitivity. The notion of a stable, persistent illiquidity premium is overstated; the return advantage is more fragile and context-dependent than the brochures suggest.

[1] Preqin, Burgiss, Cambridge Associates long-term private equity performance vs public benchmarks.
[2] PitchBook (2025), Are Private Markets Worth It?

Manager selection is the whole game. Private markets are defined less by an illiquidity premium than by a dispersion premium. Even modest selection skill, a moderately higher probability of landing in top-quartile funds, can lift total-portfolio return by 20-60bps a year, which is roughly 100-300bps of alpha inside the private allocation itself. The effect is largest in buyout and venture capital, where dispersion is widest, and smallest in private debt, where outcomes cluster. The risks of getting it wrong fall on direct and indirect investors alike, particularly those allocating at scale. Private markets reward access, discipline and underwriting capability far more than they reward passive exposure.

Table 3: Impact of manager selection skill (Average annualized alpha by strategy)

Asset Class	Underperformer Avoidance	Moderate Skill	High Skill
Private Equity (Buyout)	0.41%	0.35%	0.64%
Venture Capital	0.38%	0.32%	0.61%
Private Debt	0.25%	0.21%	0.38%
Real Estate	0.32%	0.28%	0.47%

Source: PitchBook simulations

Do retail investors get the B-product?

Three structural factors place retail vehicles on weaker terms than institutional ones. Higher fees eat into the return premium; retail investors risk being handed inferior underlying assets and in an asset class where the gap between best and worst is unusually wide, retail investors cannot meaningfully pick managers themselves. The three disadvantages compound, dragging down net returns.

Retail access comes with stacked fees that eat into the return premium. Institutional drawdown funds typically charge 1.5-2% on committed capital plus 20% carry over an 8% hurdle, and the largest LPs negotiate down within those bands, with within-fund management-fee dispersion of about 90bps. Wealth and retail vehicles carry additional cost layers, reflecting the work of getting private-market product to a broader audience: placement fees at the distribution platform, advisory fees from the adviser and a feeder-fund management fee on top of the underlying fund's economics (Table 4). Add it up and the gap between retail and institutional access often exceeds 200bps a year, enough, over a decade, to consume a meaningful share of the very illiquidity premium that justified the asset class to institutions in the first place.

Table 4: Fee layers in wealth channels

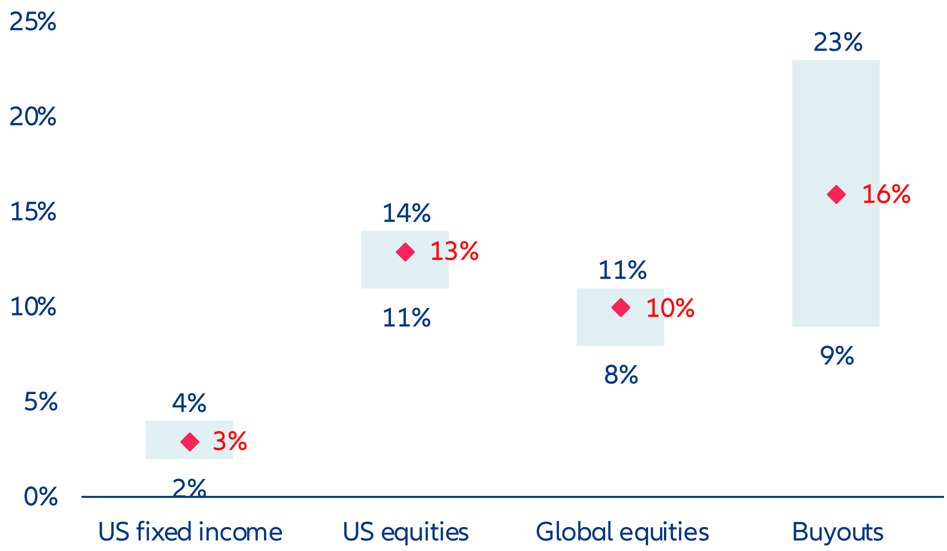
Fee type	Recipient	Purpose	Typical range	Notes
Servicing fees	Adviser (sometimes shared with brokerage)	Annual compensation for account management and client servicing	0.25% - 0.85% per year of client investment	Brokerage houses sometimes take a cut
Placement fees	Brokerage	One-time fee for offering the fund on the brokerage platform	Up to 0.5% of initial investment	Some funds cap combined servicing + placement at 0.85%
Brokerage commissions	Brokerage	Front-end commission charged at the point of sale	~2% on average (up to 3.5%)	Advisers have discretion to waive
Independent adviser fees	Independent adviser	Flat advisory fee on overall client assets	Variable (typically % of AUM)	Generally do not charge commissions or significant placement fees

Sources: FT, Allianz Research
Notes: These fees sit on top of the underlying private capital fund's own management fee (typically 1.5%-2% of AUM) and performance fee (typically 20% of profits above an 8% hurdle)

Retail and institutional vehicles also tend to diverge in what they hold. Retail evergreens are naturally suited to perpetual-hold, lower-volatility assets that support NAV-based redemptions, while buy-to-exit strategies that need patient capital sit more comfortably in institutional drawdown funds. This is a structural feature of the architecture. The liquidity profile of the wrapper shapes what can sit inside it. The implication is that two vehicles carrying the same brand can deliver materially different return and risk profiles, and advisers and platforms will need to make that distinction legible. The rise of GP-led secondaries has created a channel for moving hard-to-exit assets between funds. In these transactions, the GP is seller, buyer and valuation agent at once, an arrangement that puts the integrity of the price under strain.

The third disadvantage is the inability to choose. Manager selection is arguably the most important driver of private-market returns; the gap between top- and bottom-quartile private-equity managers is roughly 1,400bps, against around 300bps in public equity. For institutions, that dispersion is an opportunity. For retail investors, it is a risk. Mass-market participants in DC plans, insurance wrappers and platform-default products do not pick managers at all; intermediaries do, on the basis of their own incentives, including fee-sharing and distribution relationships, which can favor available managers over the most suitable ones (Figure 13).

Figure 14: Performance dispersion is especially wide for buyout funds



Sources: KKR, eVestment Alliance, Preqin, Allianz Research
Notes: Bars show median net IRRs; whiskers show first- and third-quartile thresholds. 15-year period through 31 December 2024. US equities include large- and small-cap indexes

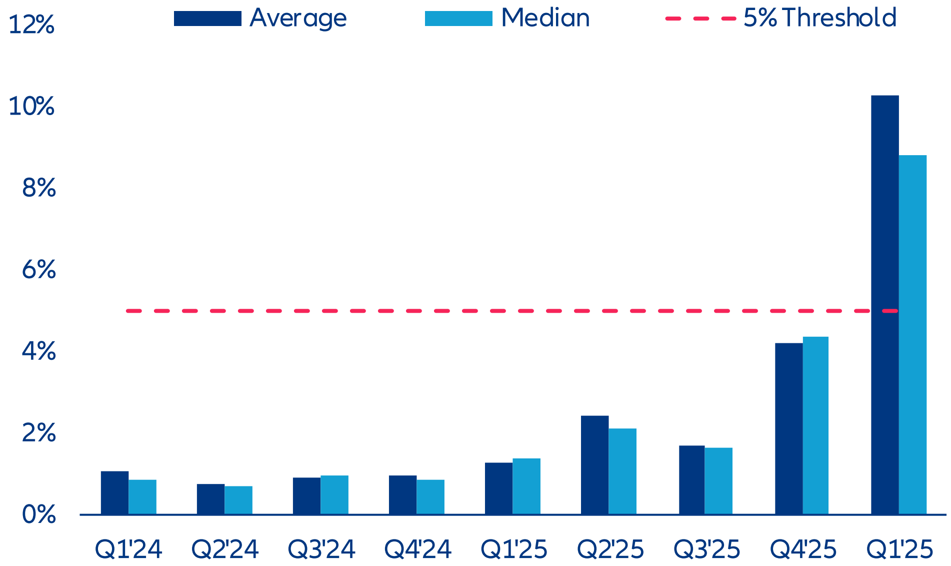
The three disadvantages mentioned above are interconnected and have a compounding effect. In order to achieve the same returns as institutions, retail investors must be willing to pay higher fees. Inferior underlying assets can negatively impact the gross return received. The inability to select top-quartile managers anchors the median retail outcome closer to the bottom of the dispersion than the top. When the effects compound, the median retail experience in an average private-market product can deliver a meaningfully lower net-of-fee return than an institutional investor in the same asset class, even though both are nominally invested in private markets. The gap is a function of product design, and well-built retail vehicles can close a portion of it.

The democratization narrative, therefore, comes with an important qualifier. Retail investors are gaining genuine access to assets that were previously out of reach, but the conditions of that access can compress the return premium that justifies the asset class in the first place. Whether the move from public to private markets pays off on a risk-adjusted basis depends on several factors: the specific product, the quality of the manager, the fee structure and the ability of the investor or adviser to distinguish well-engineered retail vehicles from those that travel less well outside the institutional setting they were designed for.

Is the "semi-liquid" promise an illusion?

An industry built on locking capital up is now selling products that promise to let it out. Retail investors face less predictable liquidity needs than institutions, and semi-liquid products try to bridge the gap with quarterly redemption windows. The bridge is narrow. Withdrawals are permitted only on set dates and after a holding period. If aggregate requests breach a preset threshold, typically 5% of NAV, the manager can invoke gates and honor only part of the request, leaving investors locked in for longer than they expected (Figure 15).

Figure 15: Redemption pressure is building for non-listed BDCs



Sources: SEC fillings, Allianz Research
Notes: the data represent the mean and median redemption request rates for the eight largest non-listed BDCs

Disclosure has not always kept up with product complexity. Semi-liquid evergreens are typically marketed as a route into previously exclusive strategies, with the added convenience of periodic redemptions. Gates and conditional liquidity terms appear in the fund documentation, but get less prominence in marketing materials than the headline features. The gap between how the product behaves in calm markets and how it behaves under stress therefore surfaces only when redemption pressure rises, often to investors who had not fully absorbed the small print.

The underlying problem is that demand for liquidity peaks precisely when supply is lowest. Semi-liquid mechanisms work smoothly in normal markets. In periods of market stress, liquidity tightness or broad portfolio rebalancing, just when investors most want their money back, managers find it hardest to provide. Gates prevent the bank-run dynamic that would force fire sales and worse outcomes for everyone, but they trap investors at the worst possible time. Retail investors who expect access to their capital may discover that access depends on the same market conditions that drove them to seek it.

Once redemption pressure builds, the dynamics turn self-reinforcing. Initial requests draw headlines, which prompt more redemptions from anxious investors. Gating works as a brake, but the act of gating itself can be read as a distress signal, accelerating the next wave. If gates persist for several quarters, managers come under pressure to sell holdings or draw on subscription lines, producing losses for those who stayed and validating the concerns that drove the run. The non-traded BDC turmoil has played out exactly that way: redemptions rose, sponsors invoked caps, the caps generated more requests and gating spread across the category.

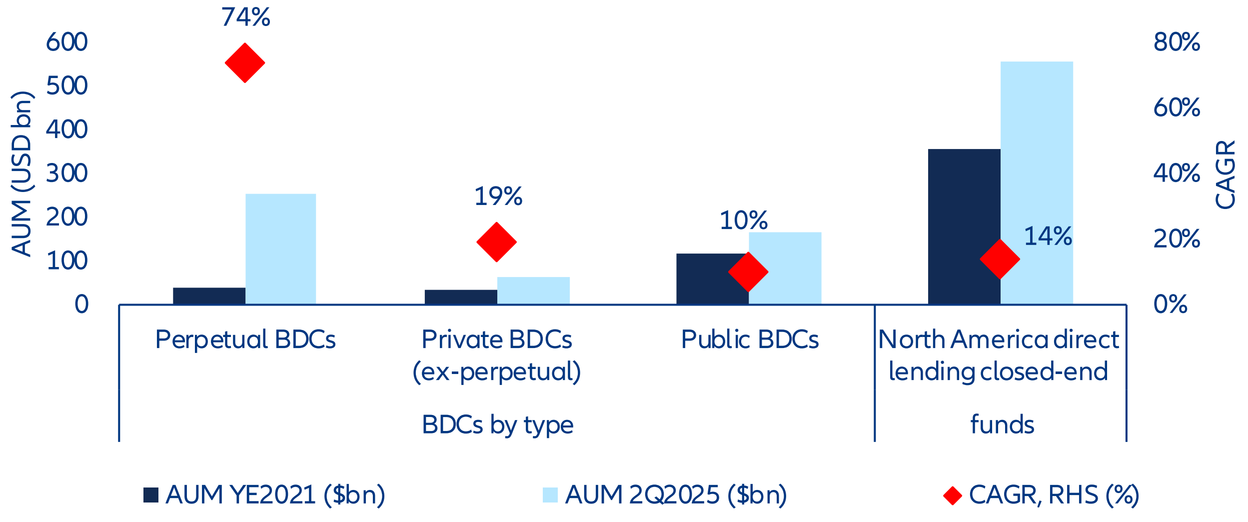
The semi-liquid promise is not an illusion; it is conditional. In normal markets, gates rarely trigger and redemptions clear smoothly. The mechanics work as intended. The case for democratization, however, depends on these structures holding up over time. The unfolding BDC episode is the first real-world stress test at scale, and a cautionary tale for the wider effort.

Box 2. The private BDC reckoning. A proof of concept for retail private credit or the proof of limits?

The Business Development Company (BDC) sector grew from USD190 bn in 2021 to over USD500bn by mid-2025, fueled by post-crisis bank retrenchment, floating-rate income and the push to bring private credit to retail and wealth-channel investors (Figure 16). The first quarter of 2026 provided the first real test of that expansion.

The result was not the broad credit crisis some had feared. It was something more telling. Wrappers designed for slow-moving institutional capital buckled when retail investors hit their limits, lagged marks and ran into limited liquidity in vehicles whose reported portfolios looked sound. This is not evidence of a failure in private credit; it is a reminder that product design, valuation architecture and investor base now matter alongside underlying credit quality. Any democratization vehicle built on periodic liquidity, manager-determined valuations and illiquid assets will carry some version of the same risk.

Figure 16. Perpetual BDCs have captured impressive AUM growth



Source Cliffwater Direct Lending Index, Preqin. As of Q2 2025

Perpetual non-traded BDCs pose a persistent design challenge. In Q1 2026, Blue Owl's technology vehicle received redemption requests totaling 40.7% of shares against a 5% quarterly cap, even though reported non-accruals were only 0.6%. The gap matters because quarterly liquidity sits on top of manager-marked, illiquid loans, giving investors an incentive to redeem before future marks absorb losses. The implication is durable: even if this queue clears, the wrapper stays vulnerable whenever confidence in reported NAV weakens.

Investors should know that reported income and reported credit quality can diverge for longer than commonly assumed. In Q4 2025, the share of private-credit loans classified as bad PIK reached 6.4%, against headline non-accrual rates of around 2%; Lincoln estimated a shadow default rate near 6%. The mechanism is straightforward: non-cash income preserves earnings optics while borrower stress is deferred, not resolved. As a result, dividend coverage, NAV stability and the perceived quality of the management team can look stronger than they really are until the recognition event arrives.

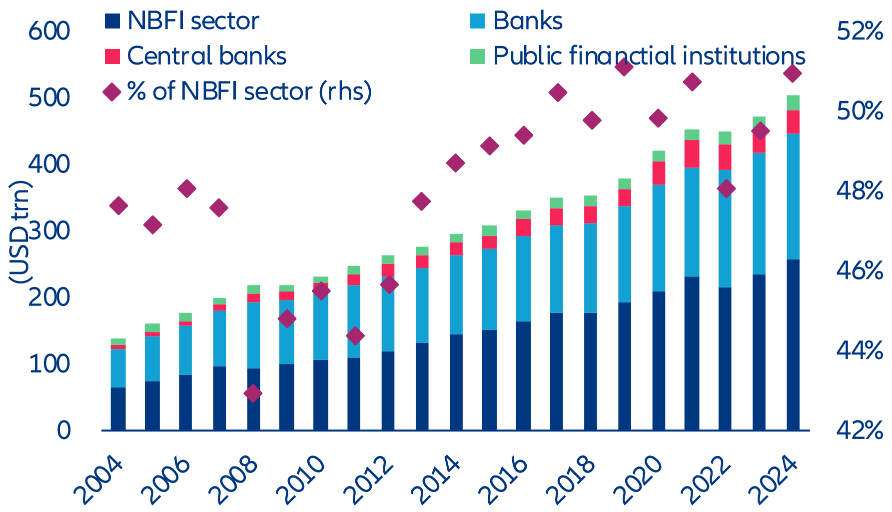
The issue is not just credit losses; it is delayed price discovery. BlackRock TCP Capital's roughly 19% NAV cut in late 2025 showed that loans previously carried near par could be repriced sharply once underlying weakness was acknowledged. This creates a feedback loop: stale marks dampen observed volatility but raise the incentive to exit before the revaluation lands. Future stress episodes are therefore likely to show up first as redemption pressure and only later as formal credit deterioration.

The 2026 episode points to structural vulnerability, not merely idiosyncratic manager error. According to Fitch, non-traded BDCs entered the period with lower leverage than their traded peers, yet sector-wide redemption requests for the largest vehicles averaged 12.1% in Q1 2026. The pattern suggests the pressure came from distribution and product design as much as from weak balance sheets. The implication for democratization is clear: expanding retail private credit will require a redesign of the framework, not just sharper marketing or stricter manager selection.

A new vector for systemic risk?

Even before retail money enters, private markets already inject several systemic risks into the financial system. Start with banks. By end-2024, the non-bank financial intermediation (NBFI) sector held 51% of global financial assets (Figure 17), and the linkages have tightened: banks' credit exposure to private credit funds alone is estimated at USD270bn-500bn. In times of stress, private funds tend to draw on their committed credit lines just as banks are absorbing market volatility and rising loan losses. The drawdowns force previously off-balance-sheet exposures onto bank balance sheets, lifting capital requirements at the worst possible moment.

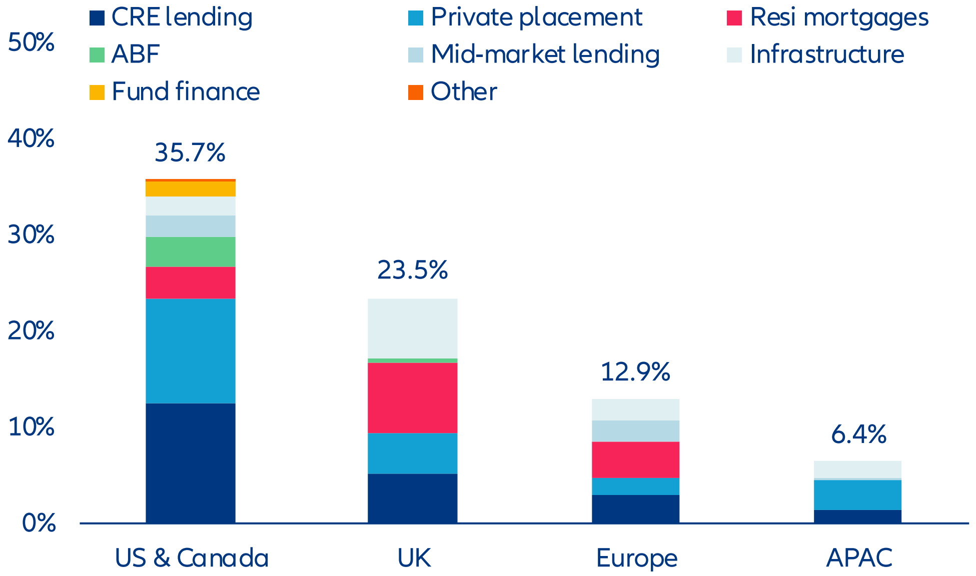
Figure 17: The NBFI sector has grown to 51% of global financial assets, tightening linkages with the banking system



Sources: FSB, Allianz Research

Within the life sector, regulators have flagged a more specific concern: the rise of private-equity-backed insurers. Private capital now controls roughly USD700bn of US life-insurance assets, and the market share of PE-backed insurers has climbed from 1% in 2012 to 13% by 2023. They have driven a broader shift toward private and structured credit, which now makes up 36% of North American life insurers' balance sheets (Figure 18). The IMF and the IAIS note that this subset of the industry tends to run thinner liquidity buffers and larger illiquid-credit positions than traditional, prudentially regulated life insurers. Stress at a PE-backed insurer can therefore feed pro-cyclical asset sales, as the 2023 collapse of Italy's Eurovita showed. The episode is a reminder that the case for retail private-market exposure rests on the quality of the wrapper as much as the quality of the asset, and that not all life-insurance balance sheets are built the same way.

Figure 18: Private credit accounts for 36% of North American life insurers' balance sheets



Sources: Moody's, Allianz Research

Democratization broadens these channels by tilting them toward a more reactive investor base. The institutional money that powered three decades of growth came with long horizons, contractual lock-ups and governance structures that resisted short-term liquidity pressure. Retail capital tends to chase peaks and redeem under stress, often on the same signals that drive broader volatility: news headlines, equity drawdowns, employment shocks. As retail's share of private-market AUM grows, each existing channel widens and the marginal investor becomes more skittish. The asset class may shift from absorbing market stress to amplifying it.

Democratization also introduces new risks. The mismatch between quarterly redemptions and decade-long assets revives run dynamics that the asset class had engineered away. Traditional institutional funds run on closed-end capital and multi-year commitments, which contain rapid redemption pressure: if a few funds suffer big losses, capital stays locked and sophisticated investors absorb the impact without a broader confidence shock. That mechanism weakens sharply once the investor base turns retail. Quarterly redemption rights on fundamentally illiquid assets reintroduce the maturity transformation that closed-end structures had removed. Gating can in principle prevent a bank-run dynamic, but the channels funds use to raise liquidity, credit lines and new loan issuance, are directly tied to banks and broader credit markets. If redemption pressure persists for several quarters, forced sales at distressed prices cannot be ruled out.

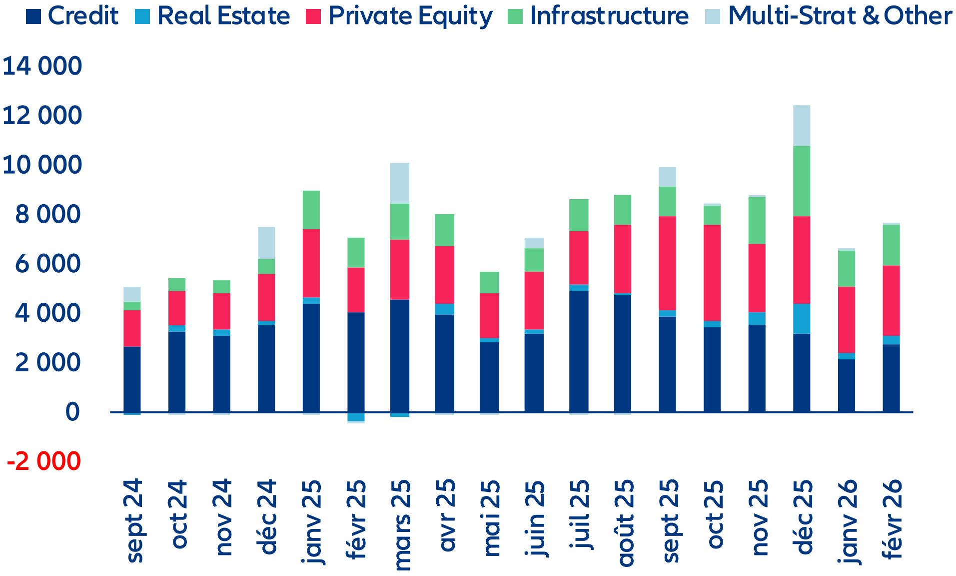
Valuation contagion is amplified when retail money is involved. Private assets are appraised by management, not by markets, and NAV marks can lag reality by several quarters. When a publicly traded vehicle holding similar private-credit loans trades at a persistent discount to its stated NAV, the market is signaling that it does not trust the valuation. That skepticism rarely stays local: it can prompt LPs to rethink commitments, lenders to tighten collateral terms and providers to pull subscription lines. Retail investors, who may not always tell well-marked vehicles from poorly-marked ones, tend to redeem indiscriminately when confidence breaks. Incentives in retail-focused funds can lean toward smoother, more optimistic NAV paths, since stable marks support both fee continuity and the credibility of the redemption mechanism. Where governance does not push back, this can build feedback loops of optimism that unwind sharply when tested.

Concentration among a handful of major managers can turn firm-level stress into a market-wide contagion. Six firms now control nearly half the retail-alternatives market, and the largest individual vehicles top USD100bn in assets. The GPs that have been most aggressive in raising retail capital also run multiple semi-liquid wrappers across product categories. Stress in one large manager's product can therefore spread quickly through correlated vehicles run by the same firm, and through the wider retail-alternatives universe, where so much AUM sits with so few entities.

What happens next?

Democratization has hit its first substantial obstacle. Inflows to wealth products at the major listed alternative managers have fallen sharply from their December 2025 peak (Figure 19). BDC sales were down roughly 40% year-on-year in early 2026, and redemption caps were triggered across major non-traded vehicles. The narrative has gone from "growth drivers" to "discipline and selectivity" in a single year.

Figure 19: Inflows to wealth products have retreated from their December 2025 peak



Sources: JP Morgan, Allianz Research

This looks like a stress test, not a fundamental shift. The BDC episode exposed real vulnerabilities in semi-liquid product design, and the fundraising slowdown reflects multi-year exit constraints and investor over-allocation. The policy scaffolding put in place over the past three years – the DOL safe-harbor rule, ELTIF 2.0 and LTAF ISA eligibility – is built and unlikely to be dismantled as the pressures that drove it – retirement-adequacy gaps, infrastructure funding needs and the hunt for broader investment opportunity – remain unresolved.

The next phase will be recalibration. Product designs are evolving in three visible directions. First, larger liquid-asset buffers inside semi-liquid wrappers, with some new launches reported to hold 25-40% in public securities to support redemptions. Second, tighter gating, longer notice periods, smaller per-quarter caps and clearer disclosure of restrictions at the point of sale. Third, a migration toward asset classes better suited to retail liquidity profiles.

The most likely structural outcome is consolidation. Mega-GPs are well placed to ride the coming wave of retail capital; smaller managers without distribution scale, brand recognition or the operational capacity to run semi-liquid wrappers will struggle to compete for retail flows. The projected parity of retail and institutional fundraising by 2030 is therefore likely to coincide with a sharp concentration of AUM among a few large managers, deepening the concentration risk noted above.

The regulatory environment has been benign so far, but the BDC turbulence will probably trigger a new round of scrutiny. Adding liquidity has resurfaced concerns that closed-end structures had previously sidestepped. Lock-ins removed deposit-like risk by stopping investors from pulling capital on demand, which limited the channels through which fund stress could spread. Open-ended structures bring back the duration mismatch and tilt the investor base toward less sophisticated participants more likely to react to market volatility. As private-markets funds push into territory once held by bank lending, regulators face a familiar dilemma: the same maturity transformation that justified the post-2008 reforms is being recreated outside the bank-regulatory perimeter, and now sold to retail investors.

Regulators themselves are divided. The SEC has explicitly rejected the idea that private credit is a systemic threat, prioritizing capital formation. The IMF, the Bank of England and the IAIS take a different view, flagging interconnections, valuation opacity and less sophisticated investors. The disagreement is essentially about how much weight to place on tail risks that have not yet materialized.

Four developments will shape the next 12-24 months. The DOL's safe-harbor rule for 401(k) alternatives, in public comment through mid-2026, will decide whether plan sponsors get the legal certainty to offer alternatives broadly in DC plans. Anderson v. Intel, before the US Supreme Court in the autumn 2026 term, will decide whether ERISA fiduciary-breach claims require "meaningful benchmarks", with direct implications for plan-sponsor litigation risk. ELTIF 2.0 will get its first proper stress test as semi-liquid launches multiply and distribution widens across the EU. And the BDC gating events of early 2026 will set the tone for retail investor protection for years, depending on whether the SEC or FINRA pursues formal enforcement on point-of-sale disclosure.

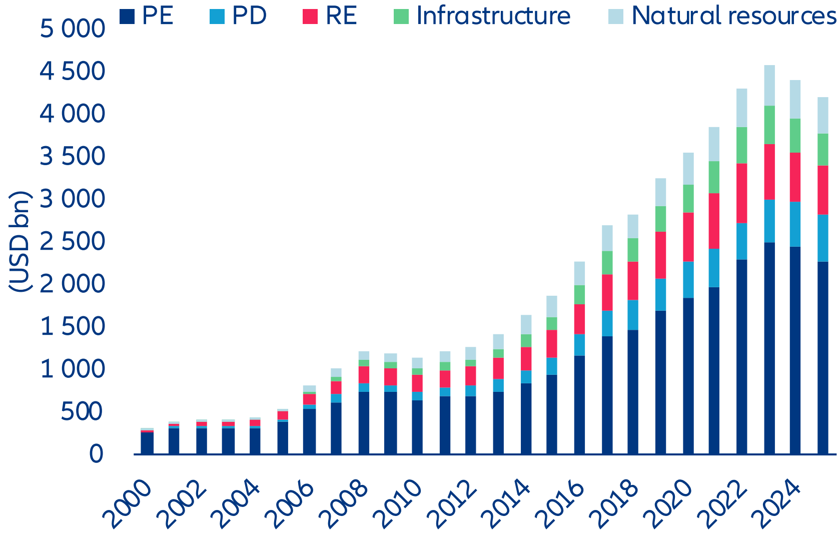
Whatever its form, the regulatory response will add cost and complexity. The BDC episode has already pushed the SEC to tighten oversight of how BDCs value illiquid holdings in falling markets. New disclosure rules, leverage limits, restrictions on retail-targeted marketing and mandatory liquidity buffers are all on the table in the major jurisdictions. Each addresses a real concern; each adds operational burden on GPs, costs that will be passed on to LPs and will erode the net-of-fee returns that justified the asset class in the first place. The deeper worry is that regulation will arrive after retail losses, eroding four decades of structural advantage.

What does this mean for investors?

Not all retail private-market vehicles are built the same way, and the distance between well-engineered and poorly-engineered product is likely to widen as the channel scales. Four markers separate the two. The first is liquidity architecture: adequate public-asset buffers, conservative gating thresholds and redemption windows calibrated to the underlying asset, rather than to the distribution channel. The second is clean separation between retail and institutional pools, so that allocation conflicts and cross-subsidies do not arise. The third is valuation governance: independent pricing, regular external review and transparency on NAV methodology. The fourth is manager selection capability, both at the GP level and, where the wrapper allows it, at the underlying-asset level. Investors and advisers will increasingly need to look past the brand and the headline strategy to these design features. For the industry, the opportunity is to build retail product that is genuinely institutional in its discipline, rather than institutional only in its marketing.

If retail capital arrives at scale, the illiquidity premium is likely to compress. Broader participation will erode the conditions that historically underpinned private-market outperformance, with more capital chasing an opportunity set that is already increasingly competitive. The pressure is acute now: managers are sitting on more than USD4trn of dry powder (Figure 20) to deploy in an environment of constrained exits and intense competition for quality deals. Institutions have historically captured the illiquidity premium through patient capital. That advantage is diluted as retail capital enters at scale.

Figure 20: Managers are sitting on USD4trn of dry powder



Sources: Preqin, Allianz Research

More fundamentally, the rise of retail products creates a structural conflict of interest in deal allocation. GPs must now spread a finite pipeline of deals across products with very different investor bases and liquidity profiles. That complicates life for institutional LPs, who lose visibility on whether their interests are being protected, whether assets are being transferred between vehicles on unfavorable terms, and whether their original strategies are being reshaped to accommodate retail liquidity needs.

If GPs prioritize institutional vehicles, institutional returns are not directly compromised, but new risks emerge. Retail products receive systematically secondary deal flow while institutional LPs continue to see the most attractive transactions. A more troubling possibility is that retail vehicles end up serving as liquidity outlets for the institutional side, especially in tough exit environments, with hard-to-sell assets channeled into retail evergreens. In these trades the GP is seller, buyer and valuation agent at once, an arrangement whose integrity depends on a valuation discipline that is hard to sustain when the same firm is on both sides.

If GPs allocate deals on broadly equivalent terms across institutional and retail products, the critical question is whether retail and institutional capital is cleanly separated. In pooled vehicles, retail capital flows directly into vehicles with the same exposure as institutional LPs, which exposes institutional capital to the risk of more reactive retail behavior. In deployment, retail inflows tend to peak when valuations are high and competition for deals is fiercest, pushing GPs to deploy into transactions that would otherwise pace through. In the holding period, retail redemption pressure during market stress can force premature sales at distressed prices, with losses that institutional capital would have absorbed and recovered through patience. In distribution, the same pressure can disrupt the orderly exit timing institutional LPs rely on, with assets sold into weaker markets to meet retail liquidity needs rather than held until the optimal exit window. The institutional return premium earned from patient, locked-up capital is diluted in this scenario.

The corollary is that the firms best positioned in the next phase will be those that combine institutional underwriting with retail-aware product design: large balance sheets that can carry liquidity buffers without diluting returns, integrated insurance and asset-management capabilities that internalize valuation and governance discipline and distribution that reaches savers without adding unnecessary cost layers. For long-horizon retail investors, the structures that meet these tests are not a watered-down version of institutional product. They are a route to diversification and return outcomes that public markets, increasingly concentrated and increasingly correlated, can no longer deliver on their own.

These assessments are, as always, subject to the disclaimer provided below.

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Allianz Trade

Thinking fast, building slow: The energy cost of the US AI boom

12 May 2026

Allianz Research

Content

Page 3-4
Executive Summary

Page 5-10
Can power markets handle the AI surge?

Page 11-12
Beyond the baseline: Agentic AI, adoption velocity and the rebound effect

Page 13-17
The hidden bill?: AI's impact on power prices

Page 18-19
Preparing AI for energy and energy for AI: policy recommendations

Executive Summary

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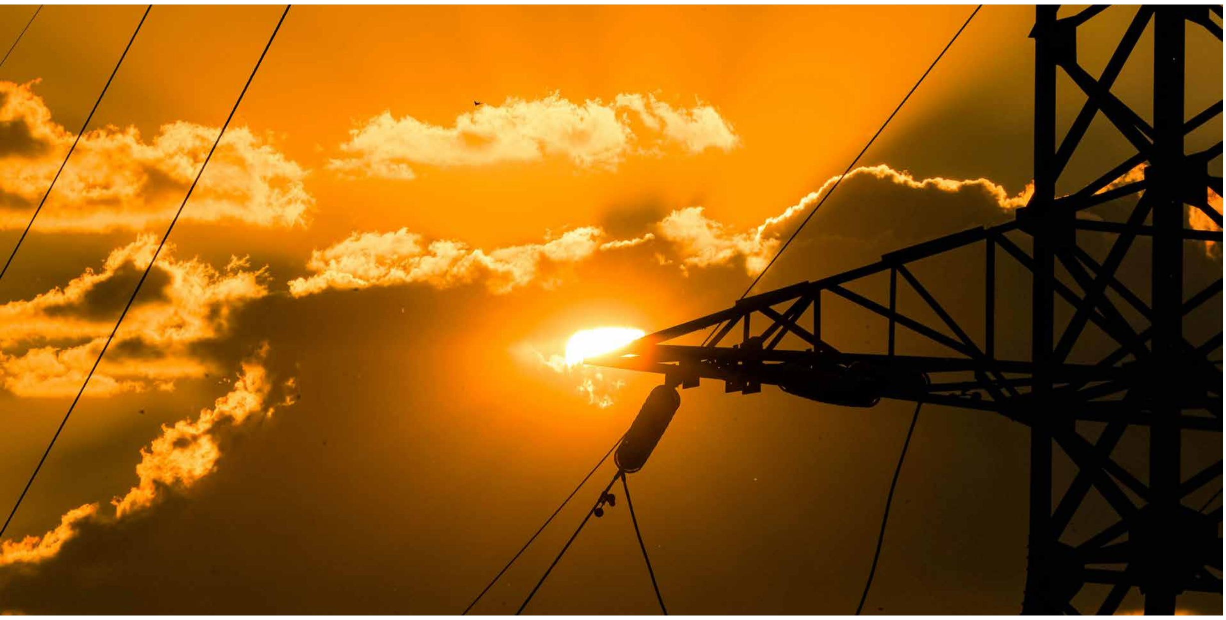
- **Artificial intelligence is about to impose the largest sustained demand shock on US electricity infrastructure in decades.** By 2030, data-center power consumption is expected to nearly double, lifting the sector's share of total US electricity demand from roughly 5% to around 9%. Although planned generation additions look sufficient on paper, data centers may absorb nearly half of projected new capacity, leaving thin margins if electric-vehicle adoption or industrial electrification accelerate faster than expected. Yet demand itself is evolving faster than forecasts can capture: generative AI reached 53% population-level adoption in just three years, agentic systems consume far more energy per interaction than conventional workloads and a 280-fold decline in inference costs since 2022 is driving a rebound effect that efficiency gains alone cannot offset.

- **The critical bottleneck, however, is not generation capacity but the grid itself.** Data centers can be built in under two years, but grid connections can take up to seven years in congested markets such as Northern Virginia. Nationwide interconnection requests now total 1.84 terawatts, exceeding total installed US generating capacity. Supply-chain shortages have pushed lead times for critical grid equipment to several years, while projected demand would require building roughly 8,000 km of high-voltage transmission lines annually – around ten times the current pace. The strain is already showing: In early 2026, the Department of Energy invoked emergency powers to shift data centers onto backup generation during peak demand periods. Rising public opposition and legislative scrutiny add a further layer of uncertainty that conventional supply forecasts have yet to fully capture. The strain is already visible in project pipelines, with half of the 12 GW of US data-center capacity planned for 2026 being delayed or cancelled.

- **Aggregate electricity prices have not yet fully reflected these pressures, but a growing „data-center premium" is emerging.** States hosting the highest concentration of data-center activity have so far seen price inflation below the national average, reflecting favorable grid conditions, economies of scale and the lagged structure of utility rate-setting. Beneath the surface, however, the impact has become increasingly visible since 2023. US residential customers are already paying USD1.4bn more per year on their electricity bills as a direct result of data-center demand, with just five utilities serving 4.4mn households in Northern Virginia, the Pacific Northwest and Arizona accounting for more than 40% of that total. The sales-weighted average price effect across all utilities sits at just 0.6%, but for the most exposed utilities roughly 7.8pps of a 24.5% cumulative price increase between 2020 and 2024 are directly attributable to data-center demand, adding 0.19pp to headline inflation over four years through the direct electricity channel alone. These markets historically benefited from below-average electricity costs, a gap that has already narrowed from 5% to 3.7% since 2020 and is set to close further. Data-center investment grew 32% in 2025 and is set to rise a further 75% in 2026 alone, pointing to an additional electricity price effect of close to 14pps for the most exposed utilities over 2025–2026, nearly doubling the cumulative four-year effect in just two years. Meanwhile investor-owned utilities filed USD18bn in rate-increase requests in 2025, the highest since the mid-1980s, with costs largely falling on existing rate-payers rather than the facilities driving them.

- **Preparing energy infrastructure for AI and addressing community concerns is as urgent as building the AI infrastructure itself.** The immediate priority is interconnection reform: binding timelines, penalties for speculative queue filings and priority treatment for shovel-ready projects with firm power commitments would help relieve the most acute bottlenecks. Cost allocation is equally important. Unless data centers bear a proportionate share of the infrastructure costs they create, public opposition and permitting delays will intensify further. Mandatory energy-use disclosure, incentives to redirect investment away from saturated regions, stronger efficiency standards, demand-flexibility mechanisms and stricter additionality requirements for power-purchase agreements would fill the most critical gaps in a policy framework that remains largely inadequate to the scale of the challenge – and lay the foundation for AI ambitions that the grid can actually support.

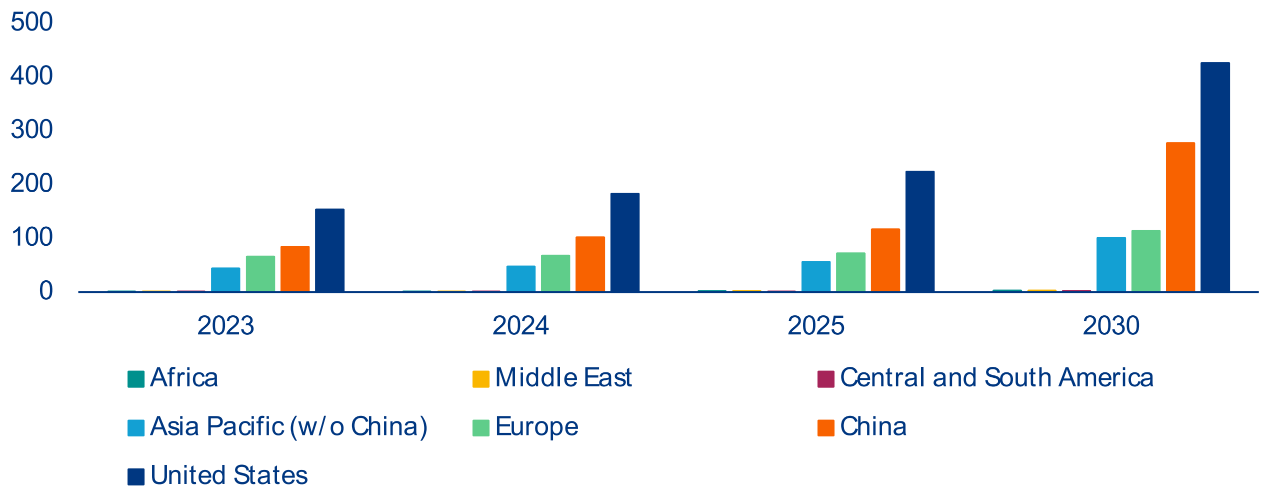


Can power markets handle the AI surge?

The AI revolution is reshaping global economies, starting with power markets. In the IEA's baseline scenario, AI-fueled growth in data-center power demand is expected to increase between 58% and 137% across major world regions between 2025 and 2030, supported by annual investment that now rivals the scale of the entire global oil and gas industry (Figure 1).[1] The shift is particularly pronounced in the US, where data-center power demand is projected to reach 426 TWh by 2030, nearly double the 2025 level. Based on long-term EIA projections, our base case estimates suggest this figure could double again by 2050, reaching roughly 810TWh. While the precise share attributable to AI remains difficult to isolate, bottom-up modeling suggests AI inference already accounts for 12–14% of total consumption, with total AI data-center capacity reaching 29.6 GW by end-2025.[2] The implications for power systems are already visible in the US, where the data center share of total electricity consumption could jump from around 5% today to between 7% and 12% by 2028, with our estimate, based on current supply pipelines and baseline demand assumptions, pointing to a share of around 9%. [3]

[1] Key Questions on Energy and AI (IEA)
[2] The 2026 AI Index Report | Stanford HAI
[3] 2024 United States Data Center Energy Usage Report | Energy Technologies Area

Figure 1: Data-center power demand growth by region (in TWh)



Source: IEA baseline projection

This significant demand growth is fueling a massive expansion in generation capacity. According to the Energy Information Administration's (EIA) Short-Term Energy Outlook, power generation in the US is expected to grow by 441 TWh between 2023 and 2027, an increase of 10.5%.[4] This would mean the five years since the breakthrough of generative AI see a larger increase than the preceding two decades, over which power generation grew by only 7.7%. 76% of the increase in power generation is expected to be driven by wind and solar (21.9% wind and 54.1% solar) with the remaining 23% supplied by gas (15.3%), nuclear (5.2%) and hydropower (3.0%). Coal remains on its post-2007 declining trend, though the pace has slowed markedly from -4% annual reductions before the AI era to just -0.5% today.

On paper, the capacity additions currently included in the EIA's pipeline of planned and under-construction projects appear broadly sufficient to accommodate rising data-center demand, with data centers expected to absorb around 47.1% of projected new supply.[5] That apparent adequacy, however, rests on the fragile assumption that competing sources of electricity demand remain relatively contained. In reality, the grid is coming under pressure from multiple directions simultaneously. Beyond data centers, electrified transport alone could materially tighten the balance. Under the IEA's baseline scenario, US electricity consumption from electric vehicles is projected to rise from 25.2TWh in 2024 to 91.4TWh by 2030. Near-term signals are mixed, but the risks remain tilted to the upside: Secondary-market EV sales continue to rise despite the expiry of the federal tax credit, while gasoline prices elevated by the escalation in the Middle East are reinforcing the economic case for adoption. Historically, sustained fuel-price pressure has been one of the strongest catalysts for accelerating EV uptake.[6] Industrial electrification adds a further layer of uncertainty: Manufacturing reshoring and the electrification of industrial processes point towards structurally higher power demand that remains difficult to forecast with precision. Taken together, these competing demand vectors suggest that the supply cushion that seems comfortable could narrow far more quickly than current projections imply.

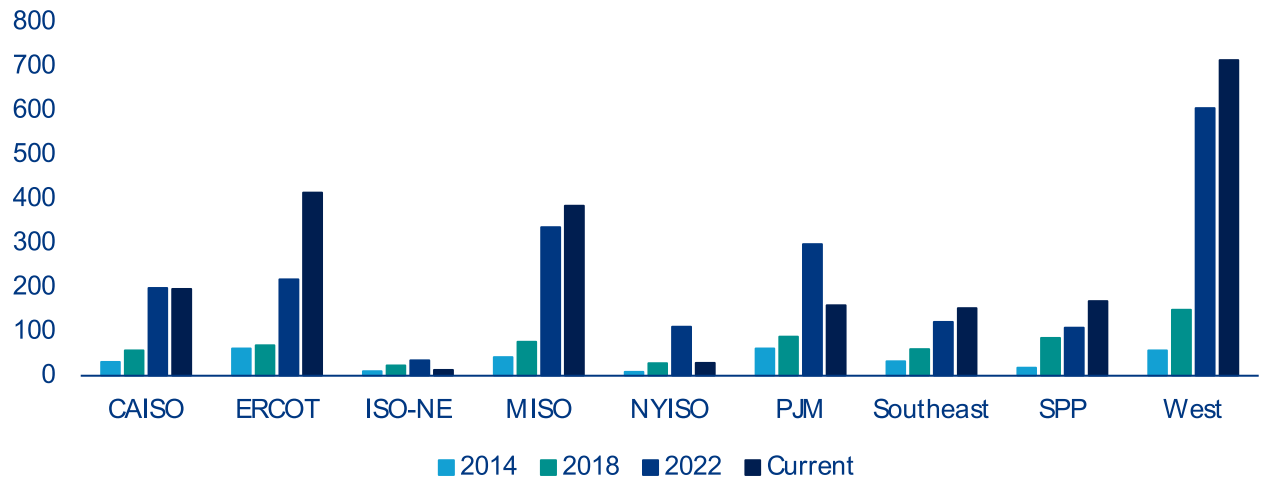
The bigger issue lies on the supply side and is rooted in US power grids, which struggle to keep pace with data-center interconnection demands. While physical construction typically takes under 24 months, grid-connection queues can more than double total

[4] IEA STEO, April 2026 update
[5] Based on EIA-860M (February 2026 update)
[6] Global EV Outlook 2025 - IEA

project timelines, with hotspots like Northern Virginia reporting waiting times of up to seven years.[7] This bottleneck reflects a deeper structural problem as both new generation capacity and new demand face the same overloaded interconnection queues (Figure 2). In Texas (ERCOT region), generation and storage projects queuing for grid connection nearly quadrupled between 2020 and 2024, while nationally active interconnection requests now stand at 1.84 TW, exceeding total US installed generating capacity. That figure overstates the genuine pipeline as only around 20% of requests ultimately reach commercial operation, with the remainder abandoned due to speculative filings, rising costs and mounting delays. Resolving these constraints requires grid expansion on a scale the US has not attempted in decades: In their 2024 National Transmission Planning study, the DOE estimates that in a lowest-cost scenario the total transmission system would increase by 2.1-2.6 times until 2050 and up to 3.3 times in a high demand case. This would imply building around 8000km of high voltage transmission lines per year, a pace that appears far beyond current construction rates, which have averaged well below 1,000 km annually in recent years.[8]

Getting there is complicated by constraints that span the entire supply chain. The grid equipment needed (transformers, switchgear and cables) faces lead times of up to four years alongside a 30% supply shortfall.[9] Building new transmission lines compounds the problem further, taking four to eight years in advanced economies, a timeline fundamentally incompatible with the pace of data-center deployment. Beyond power infrastructure, the AI build-out is also running into semiconductor supply constraints: According to the IEA, HBM chip shortages have emerged as a binding bottleneck over the past six months, with insufficient supply of this specialized memory threatening to slow AI data-center deployment independently of grid availability.[10] A construction workforce shortage of around 439,000 skilled workers and increasingly complex permitting processes compound the problem further. As a result, almost half of the approximately 12 GW of US data center capacity planned for 2026 is expected to be delayed or cancelled, with only around 5 GW currently under active construction.[11]

Figure 2: US power capacity in queue for grid connection by electricity market region (in GW)



Source: LBNL, interconnection.fyi

[7] Epoch AI and Energy and AI
[8] DOE and Grid Strategies
[9] Wood Mackenzie
[10] Key questions on energy and AI - IEA
[11] Slightline Climate Data and Bloomberg

Meanwhile grid stress from data centers has moved from projection to reality, helping explain the prolonged integration assessments that slow their expansion. Unlike traditional commercial or residential load, data centers represent large, concentrated blocks of power demand that can disconnect or transfer to backup power in a coordinated manner during grid disturbances. Compounding this, AI-optimized data centers can exhibit sharper and more frequent load swings than conventional facilities: As workloads shift between high-intensity processing and lower-utilization periods, power draw can fluctuate rapidly, placing greater demands on frequency regulation and reserve markets. When a disturbance triggers a simultaneous disconnect, as occurred in Virginia in 2024, where 60 facilities representing 1.5 GW dropped off the grid at once, the sudden loss of load forces operators to curtail generation equally fast to avoid a frequency imbalance that can cascade into widespread blackouts.[12] The reverse risk is equally real: When data centers draw at full capacity during peak periods, they can overwhelm local transmission capacity, forcing power to reroute through congested lines and driving up congestion costs across the broader grid. This strain is increasingly forcing active intervention, as seen in January 2026 when the DOE invoked emergency powers to authorize PJM and ERCOT to shift data centers onto backup generators to prevent residential blackouts during a severe winter storm. Both failure modes reflect the same underlying mismatch: a grid designed for distributed, flexible demand is being asked to absorb loads that are geographically concentrated and operationally rigid.

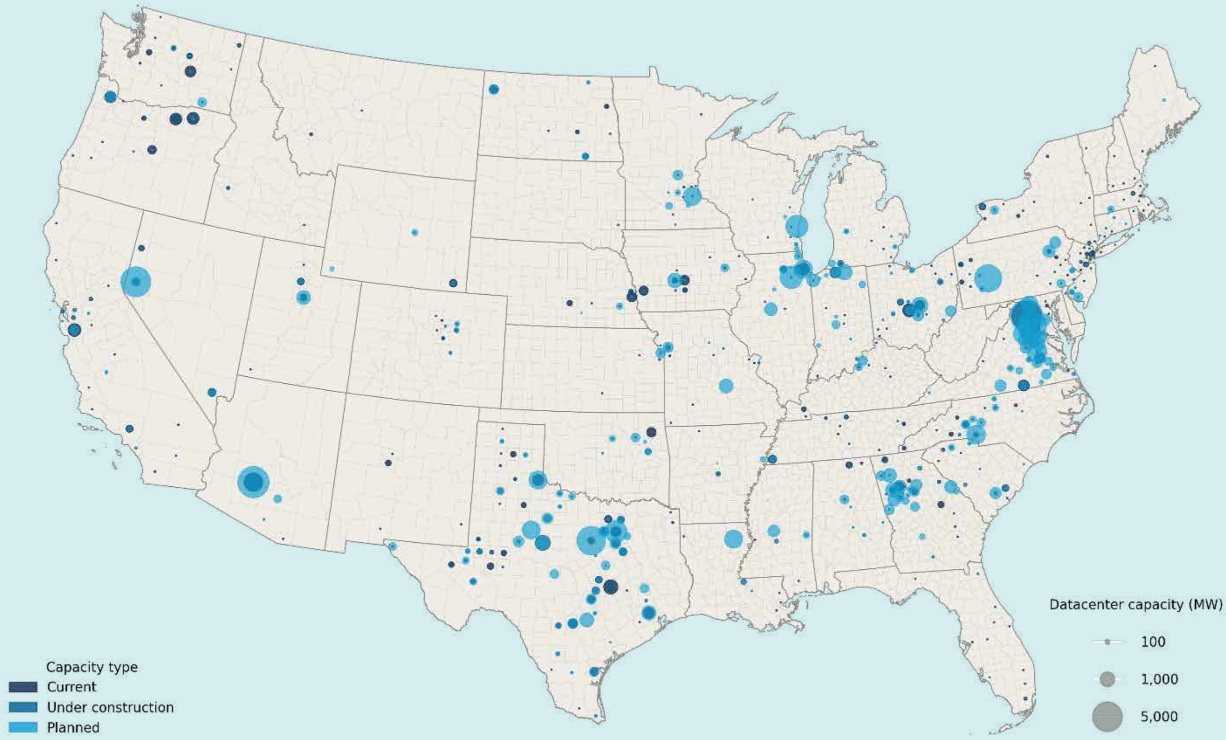
These physical bottlenecks are now increasingly intersecting with political and social constraints. Organized opposition to data-center development has emerged as an additional factor shaping the outlook. In 2025, an estimated USD156bn worth of data-center projects were blocked or delayed by local opposition, moratoriums and litigation in the US, with project cancellations more than quadrupling from six in 2024 to 25 in 2025.[13] The opposition is largely rooted in concerns about electricity costs as the infrastructure required to accommodate data-center load growth is typically funded through broader rate-payer tariffs rather than charged directly to the facilities driving demand. In PJM alone, the grid's independent market monitor estimates that data-center load growth added USD16.6bn in capacity costs across the 2025/26 and 2026/27 delivery years, spread across 67mn consumers.[14] Legislative responses are following, with moratorium bills introduced in at least 11 states in 2026, with some like Maine considering statewide construction pauses.[15] While unlikely to halt the AI infrastructure buildout, rising opposition introduces permitting delays and regulatory uncertainty that compound existing supply-side constraints and underscores a broader question about who ultimately bears the cost of AI infrastructure.

[12] DCD
[13] Data Center Watch and Construction Dive
[14] Comment of the Independent Market Monitor for PJM
[15] Axios

Where are data centers built and what determines their location?

Data center location reflects a specific set of infrastructure, cost and regulatory requirements that have historically concentrated capacity in a small number of markets. Today, Texas and Virginia alone account for 42% of total US data center capacity, though the nature of that concentration differs markedly between the two. Virginia, and Northern Virginia's Loudoun County in particular, dominates current installed capacity and represents the largest data-center cluster in the world. Texas, by contrast, carries the largest pipeline of planned and under-construction capacity, reflecting its abundance of land, a deregulated electricity market, and aggressive state-level incentives (Figure 3).

Figure 3: US data center capacity by county (in MW)



Source: National Laboratory of the Rockies.

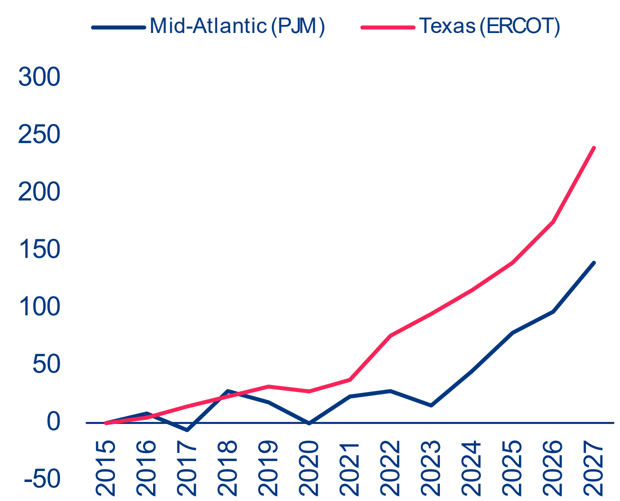
Four factors dominate siting decisions. Power availability and cost are the most fundamental as cheap and reliable electricity underpins the economics of large-scale compute. Fiber connectivity to major internet exchange points determines latency and operational viability. Land availability and permitting speed determine how quickly large campuses can be developed. Tax incentives, particularly data-center exemptions offered by Virginia, Texas and several other states, have actively directed investment toward specific markets.

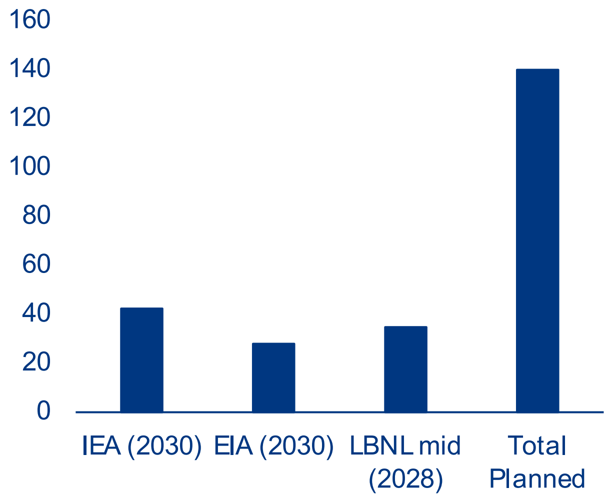
The consequence of this concentration is both operational and systemic risk. When grid capacity tightens in saturated markets, the AI infrastructure buildout slows not just regionally but globally, given the role these clusters play in serving international demand. A regional grid failure in these clusters would ripple globally, translating into higher prices for AI and cloud services, slower model deployment and capacity-driven rate limiting for end users. Overclustered markets also face compounding vulnerabilities across power supply, water availability, cooling infrastructure and skilled labor, making geographic diversification of future capacity an increasingly urgent priority.

While aggregate power-supply growth appears sufficient on paper, the regional picture is more nuanced. In the two most exposed markets, ERCOT (Electric Reliability Council of Texas) and PJM (PJM Interconnection), power generation growth has accelerated sharply since 2022, jumping from 10 TWh p.a. to 22 TWh p.a. in Texas and from 4 TWh p.a. to 31 TWh p.a. in PJM (Figure 4 a). This momentum is expected to continue, with the EIA's Short-Term Energy Outlook projecting an additional 99 TWh of supply in Texas and around 60 TWh in PJM by end-2027. On the demand side, Texas and Virginia each carry approximately 30 GW of data center capacity in their pipelines, which, if fully implemented, would imply an increase in annual electricity demand of around 140 TWh per region. In practice, however, given typical build-out timelines and project attrition, a more realistic estimate puts incremental data center demand at 30–60 TWh per region by 2030 (Figure 4 b). At those levels, both markets can realistically absorb expected AI-driven load growth, provided their supply pipelines materialize.

Delivering on that supply potential is, however, not without obstacles. In ERCOT, the supply pipeline is theoretically comfortable, but interconnection queues have grown sharply in recent years, raising the risk that capacity additions lag behind the pace of data center buildout. In PJM, the challenge is more structural. Generation growth has been slower and surrounding regions offer limited import relief, given mostly flat supply growth expectations. Data centers are also not the only source of incremental demand, with electrification of transport and industry adding further pressure on available grid capacity. With 105 GW currently queued in PJM's interconnection queue, processing delays alone could push a significant share of planned capacity beyond the 2030 horizon. Should capacity additions in either market fail to keep pace with demand, persistently tight grid conditions would translate into additional upward pressure on regional electricity prices.

Figure 4: Estimated power generation growth by grid region vs 2015 (lhs, TWh) and estimated data center power demand in Texas and Virginia (rhs, TWh)





Source: Allianz research based on EIA (STEO), IEA, LBNL and National Laboratory of the Rockies; Note: Planned power demand was estimated based on state level NLR capacity data for facilities planned or under construction. IEA, EIA and LBNL country level estimates were downscaled to regions based on planned capacity shares.

Beyond the baseline: Agentic AI, adoption velocity and the rebound effect

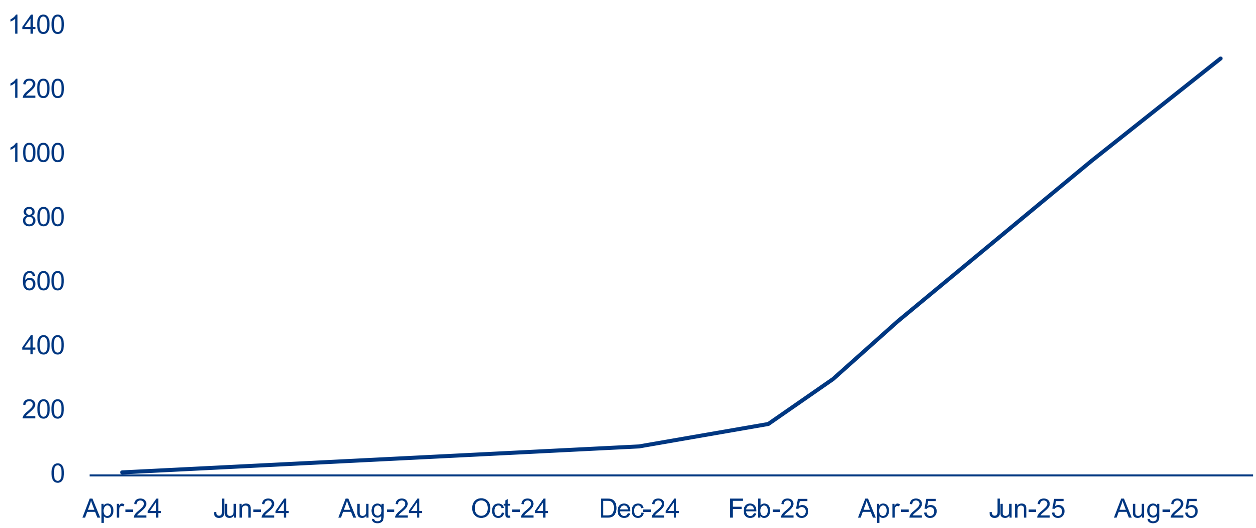
The future trajectory of AI's impact on power markets is shaped by two opposing forces. Rapid efficiency improvements are materially reducing the energy cost per token, while surging adoption, growing usage complexity and the emergence of agentic workflows are pushing demand in the opposite direction. Which force dominates will determine whether current supply projections prove adequate or significantly understated.

The energy efficiency of AI systems has improved dramatically across all layers of the technology stack... Compute performance per watt has increased by around 34% per year since 2008, with successive GPU generations delivering compounding gains.[16] At the model level, architectural innovations such as Mixture-of-Experts designs, quantization, speculative decoding and model distillation have each contributed further reductions in energy per task, in some cases cutting compute requirements by several multiples without meaningful loss in output quality. On the infrastructure side, hyperscale and AI-specialized data centers have become significantly more efficient at converting electricity into useful computation, with overhead losses from cooling and power distribution falling from around 40–50% of total facility energy in older facilities to just 10–20% in modern ones.[17] According to the IEA, widespread adoption of existing AI applications across industry, buildings and transport could deliver aggregate energy savings of over 13 EJ by 2035, equivalent to around 3% of global final energy consumption.[18] Realizing this potential, however, is far from guaranteed, as widespread adoption remains contingent on overcoming persistent barriers including fragmented data, limited digital skills and inadequate regulatory incentives.

[16] Epoch AI
[17] LBNL
[18] Key Questions on Energy and AI - IEA

...yet demand is expanding at least as fast, driven by rapid adoption, growing model capability and a structural shift in how AI is used. Today, generative AI has reached 53% population-level adoption within three years of its mass-market introduction, faster than the personal computer or the internet, with 88% of companies reporting AI use in 2025 according to Stanford's AI Index.[19] The models powering this growth have scaled at an equally striking pace, with the compute required to train frontier AI models growing at roughly 4–5x per year since 2020 and GPT-4 estimated at around 1.8trn parameters, roughly 10 times larger than GPT-3 released just three years prior.[20] Usage intensity has followed the same trajectory. According to OpenRouter's platform data, average prompt length has grown roughly fourfold since early 2024 as users shifted from simple queries to context-heavy, multi-step workflows, with reasoning models growing from a negligible share to over 50% of token usage by late 2025.[21] This growth is reflected in token consumption figures across major AI companies. OpenAI's API alone grew from 300mn tokens per minute in 2023 to over 6bn by late 2025, a 20-fold increase in under two years, while Google's Gemini token consumption increased approximately sevenfold between February and September 2025 (Figure 5).[21]

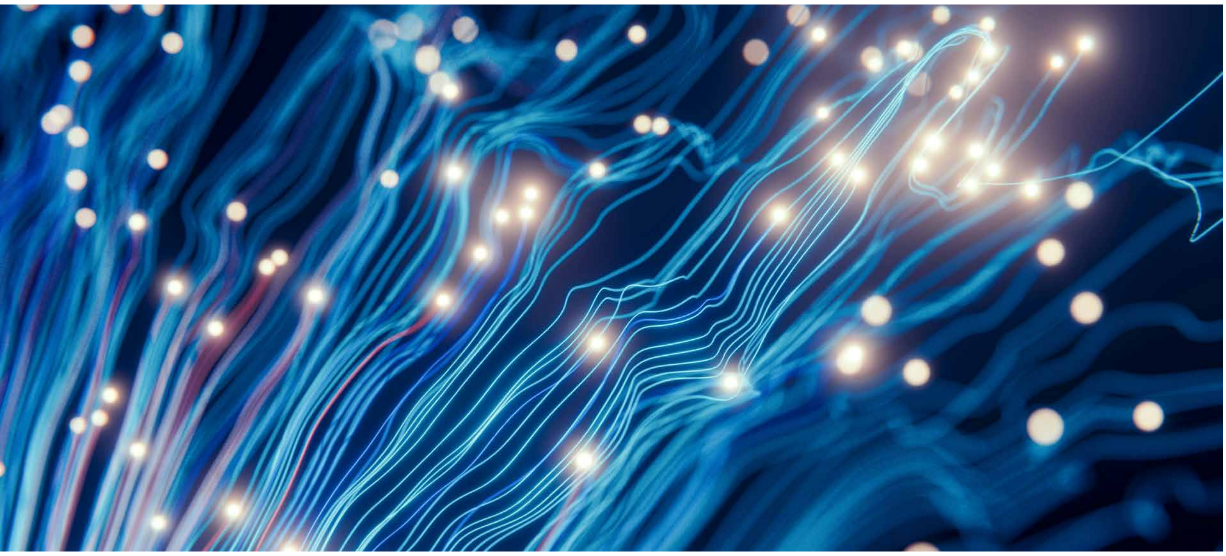
Figure 5: Google Gemini monthly token consumption (trillion tokens)



Sources: Google DeepMind; Alphabet Q2 2025 earnings.

The primary driver behind this acceleration is a fundamental shift in how AI is being used. Unlike conventional chat interactions, agentic systems plan and execute multi-step tasks autonomously, dramatically increasing token consumption per interaction. The energy implications are substantial, with an agentic task with reasoning consuming around 50Wh compared with roughly 0.3Wh for a standard text query, roughly 150 times more energy per interaction. Another reinforcing factor is the sharp decline in inference costs, which fell more than 280-fold between late 2022 and late 2024.[23] As lower prices stimulate broader and more intensive use, efficiency gains at the model and hardware level risk being offset by a classic rebound effect, whereby falling costs per unit of consumption drive total consumption higher. Taken together, these trends suggest that AI-driven electricity demand is likely to continue growing faster than efficiency improvements can offset in the near term, reinforcing the supply-side pressures on regional power markets.

[19] The 2026 AI Index Report - Stanford HAI
[20] Epoch AI
[21] OpenRouter
[22] Gemini's token growth partly reflects its integration into Google Search, which significantly expanded query volumes beyond standalone AI product usage.
[23] The 2025 AI Index Report - Stanford HAI



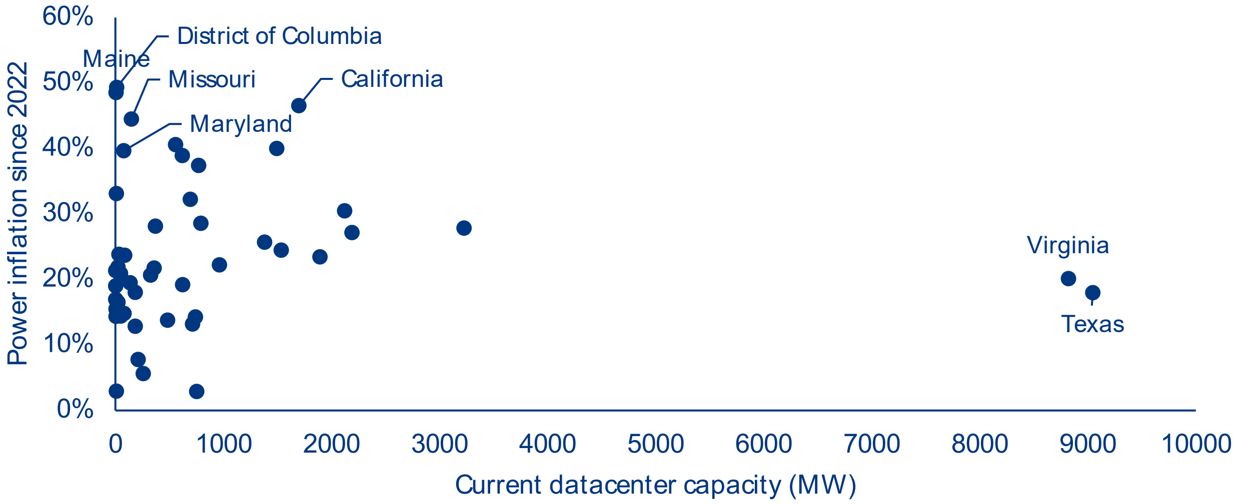
The hidden bill?: AI's impact on power prices

Data-center demand affects electricity prices through several compounding channels. Large concentrated loads drive up capacity market auction prices as grid operators must procure additional firm generation to meet anticipated peak demand, with costs socialized across all ratepayers. Grid-connection infrastructure represents a second channel as transformers, switchgear and dedicated transmission lines are typically recovered through broader utility tariffs rather than charged to the facilities driving demand. These pressures land on grids already strained by decades of underinvestment, amplifying the cost impact of each new interconnection. A third channel operates through generation mix: Data centers require firm, around-the-clock power that renewables alone cannot provide, pushing utilities toward more expensive gas and nuclear capacity. Finally, the renewable energy that data centers rely on for sustainability commitments is often located far from load centers, creating congestion costs that ripple across the broader grid. Network charges have so far remained relatively stable, but are structurally bound to rise in the regions where AI infrastructure is most concentrated. Because each channel operates on a different timescale and affects different consumer groups, the overall price impact is neither immediate nor evenly distributed across consumers and regions.

So far, aggregate state-level data tell a surprisingly benign story. Despite hosting more than 40% of total US data-center capacity, Texas and Virginia have seen power price inflation of only 18% and 20% respectively since 2022, below the 24% US average (Figure 6). This is consistent with research from Lawrence Berkeley National Laboratory (2026), which finds that state-level load growth from 2019 to 2025 was generally associated with lower all-sector average retail electricity prices as rising fixed infrastructure costs are spread across a larger base of consumers and kilowatt-hours sold.24 A further moderating factor is that data centers have historically clustered in regions with favorable grid conditions and low base energy costs, insulating those markets from the price pressures that concentrated demand might otherwise create. Regulatory lag compounds this dynamic: Utility rate cases can take well over a year from filing to approval, and capital investments are recovered over asset lifetimes rather than immediately, meaning that retail prices today largely reflect costs incurred years earlier. That lag is, however, beginning to unwind: Investor-owned utilities filed USD18bn in rate increase requests in 2025, the highest level since the mid-1980s, with regulators approving 64% of requests.

[24] Energy Markets & Planning

Figure 6: Data-center capacity and retail electricity price inflation since 2022, by state



Source: Allianz Research based on EIA and NLR.

State-level data mask considerable heterogeneity across individual utilities. A utility directly serving a major data-center cluster faces fundamentally different demand dynamics than a rural cooperative in the same state, yet both are folded into the same state-level price average. To move beyond the aggregate picture, we estimate the effect of data-center demand growth on residential electricity prices at the utility level, exploiting the fact that utilities entered the AI investment boom with very different pre-existing exposure to data center capacity.[25] The intuition is straightforward. Utilities that happened to serve counties with significant data center infrastructure were more directly in the path of the subsequent investment wave than those that did not. By interacting each utility's data center footprint with the national surge in construction investment – a design known as a Bartik shift-share instrument – we can estimate the price effect of data center demand while accounting for time-invariant utility characteristics and aggregate macroeconomic trends. The analysis covers approximately 1,200 US utilities over 2020–2024, drawing on EIA utility data matched to county-level data center capacity data from the National Laboratory of the Rockies (NLR) and national construction investment figures from the Census Bureau.

The results suggest that price pressures are more visible at the utility level than aggregate data imply, with three findings standing out. First, the price effect is concentrated entirely in the post-2023 period, precisely when AI-driven investment accelerated sharply, while the pre-2023 period shows no significant relationship between data center exposure and price changes (Table 1, column 2). This timing is difficult to explain by anything other than the AI demand shock itself, and provides confidence that we are capturing a genuine effect rather than a coincidental correlation with pre-existing trends. Second, the demand-side results validate the approach: utilities with higher data center exposure show markedly stronger commercial electricity consumption growth post-2023, consistent with hyperscale facilities connecting to utility grids and driving up load (Table 1, column 3). The effect is broad-based across utilities rather than driven by any single outlier. Third, in terms of magnitude, a utility with 4% of national data center capacity in its service territory, broadly representative of the most exposed markets outside the largest hubs, is associated with a cumulative residential price increase of approximately 3% over 2020–2024, based on the USD22bn growth in national data-center construction investment observed over that period. Modest in isolation, this effect is best understood

[25] Data center capacity shares reflect 2025 installed capacity, the large majority of which was planned and permitted before the generative AI boom took hold in 2023, making them a reasonable proxy for pre-existing exposure rather than a response to the shock we are trying to measure.

as an early signal rather than a final estimate – given the regulatory lag discussed above, much of the cost pressure already absorbed by utilities is still working its way through to consumers.

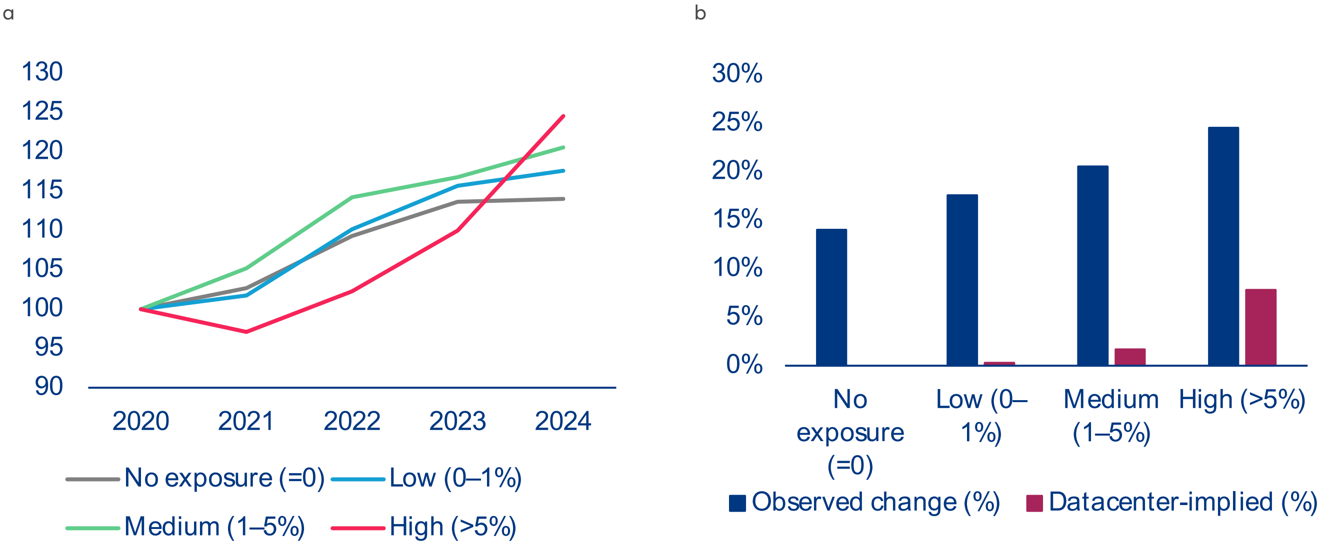
Table 1: Estimated effect of data-center demand on electricity prices

	Dep. var: log(Residential Price)		Dep. var: log(Commercial Sales/Customer)
	(1) Baseline	(2) Pre/Post Split	(3) Post-2023
Bartik IV	**0.0342*** *(0.018)*	—	—
Bartik IV × Post-2023	—	**0.0321**** *(0.0156)*	**0.2416***** *(0.0556)*
Bartik IV × Pre-2023	—	0.0145 *(0.021)*	—
Observations	6,586	6,586	6,388
Within R^2	0.0098	0.0115	0.033

Source: Allianz Research; Note: All specifications include utility fixed effects and year fixed effects. Standard errors are clustered at the utility level and reported in parentheses. The Bartik instrument is $B_{ut} = z_u \times G_t$, where z_u is utility u's predetermined share of national data-center capacity and G_t is annual US data center construction investment (US Census Bureau). Specifications (1) and (2) use log residential prices as the dependent variable; (1) uses the full-period IV while (2) splits it at 2023 to test for pre-trends. Specification (3) uses log commercial sales per customer as the dependent variable with the post-2023 IV, providing demand-side validation of the instrument. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Data-center demand is leaving a measurable imprint on electricity prices, with effects concentrated in the most exposed markets. Figure 7a) shows that while all utility groups experienced meaningful price increases over 2020–2024, the pattern is far from uniform. The no-exposure group saw the smallest increase at 14%, while the high-exposure group ended the period 24.5% above its 2020 level, a gap that widened noticeably after 2023 as AI-driven investment accelerated. Of that 24.5% rise, approximately 7.8pps are directly attributable to data-center demand based on our regression estimates (Figure 7b)), with the remainder reflecting the broader energy cost pressures that affected all utilities over the period, including fuel-price volatility, rising infrastructure investment costs and general inflation. The difference between the two groups of around 10pps therefore reflects both the data-center premium and other structural differences between high and low-exposure utility markets.

Figure 7: a) Price index by exposure group (2020=100) and b) Observed vs data center-implied price change 2020-2024 (in %)



Source: Allianz Research based on EIA and NLR; Note: Exposure groups defined by utility share of national data center capacity: no exposure (n=718), low 0–1% (n=442), medium 1–5% (n=73), high >5% (n=5). Data center-implied effect = beta × avg. group capacity share × DC investment growth (USD21.91bn)

Across the US, residential customers are already paying around USD1.4bn more per year on their electricity bills as a direct result of data-center demand. Just five utilities, serving 4.4mn households – roughly 4% of residential customers – in Northern Virginia, the Pacific Northwest and Arizona, account for more than 40% of the total, with their customers facing an average extra annual bill of USD139, attributable to data-center growth. A wider group of 73 medium-exposure utilities pays around USD28 per customer per year, while the vast majority of US utilities are not affected. Strikingly, the most affected markets are not pricier than the national average to begin with. Their historical 5% cost advantage has already narrowed to 3.7% by end-2024, and the gap is set to close further as investment accelerates, eroding a buffer that has so far masked the true scale of the data-center premium.

Table 2: Estimated financial impact of data center demand on residential electricity prices by utility exposure group in 2024

Metric	No exposure	Low (0–1%)	Medium (1–5%)	High (>5%)
Utility characteristics				
Number of utilities	718	442	73	5
Avg. exposure share (%)	0	0.17	2.3	10.47
Market size				
Avg. residential price (USD/kWh)	0.143	0.161	0.159	0.147
Total sales (TWh)	203.2	724.5	247.4	52.5
Total customers (mn)	17.3	71.42	24.08	4.37
Estimated price impact				
Estimated price effect (%)	0	0.13	1.72	7.84
Extra bill per customer (USD/yr)	0	2.1	28.05	138.65
Extra bill total (USD mn)	0	150	675	605
Inflation impact				
Direct CPI contribution (pp)	0	0.0032	0.0427	0.1945

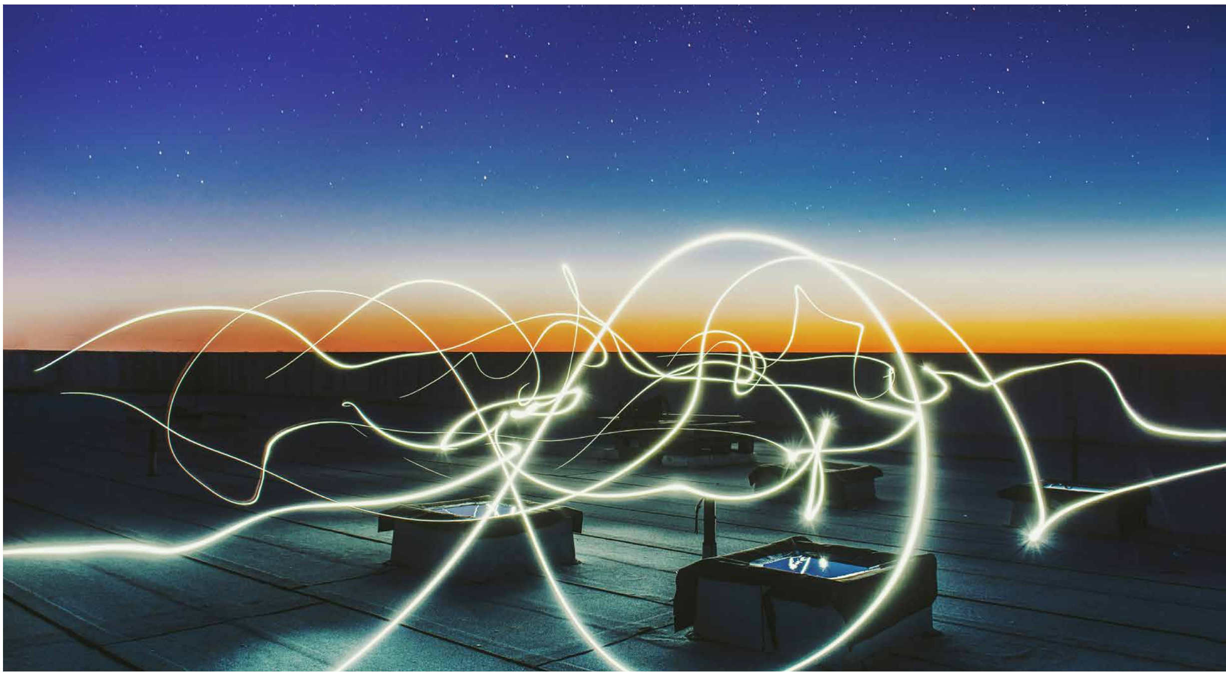
Source: Allianz Research based on EIA and NLR

Translated into aggregate consumer prices, our findings confirm an overall modest but highly skewed data-center price premium. Across the full sample, the sales-weighted average residential price increase attributable to data centers amounts to around 0.6% over 2020–2024, translating into a direct headline CPI contribution of just 0.015pp. For the most exposed utilities, however, the same channel adds roughly 0.19pp to headline CPI, around thirteen times the full-sample figure.[26] Because these estimates capture only residential electricity and exclude commercial pass-through, they should be read as a lower bound on the true inflation impact, with the full effect on aggregate consumer prices likely materially larger once higher input costs for electricity-intensive sectors are reflected in final goods prices.

This premium is likely to grow as the investment wave continues. Data-center investment already grew by 32% in 2025 and is set to increase by a further 75% in 2026 alone.[27] As pending rate cases translate accumulated cost pressures into retail tariffs, our estimates point to an additional price effect of close to 14pps for the most exposed utilities over 2025–2026, equivalent to a further CPI contribution of around 0.34pp, nearly doubling the cumulative four-year effect in just two years. With infrastructure costs still largely socialized across all rate-payers rather than borne by the facilities driving demand, the question of who ultimately pays for the AI buildout is set to become increasingly central to the policy debate.

[26] Direct CPI contribution estimates assume an electricity share of 2.48% in the consumer price basket, based on BLS CPI relative importance weights (December 2024)

[27] Key Questions on Energy and AI - IEA



Preparing AI for energy and energy for AI: policy recommendations

The scale and pace of AI-driven power demand growth points to a set of policy priorities that are urgent, tractable and largely absent from current frameworks, which would help lay the foundation for AI ambitions that the grid can actually support.

1) Expand and accelerate

- **Reform interconnection queues.** Interconnection queues are the single most binding near-term constraint on both generation capacity and data-center deployment. Streamlining queue processes, introducing binding timelines for interconnection studies, strengthening financial penalties for speculative filings that inflate backlogs and delay genuine projects and prioritizing shovel-ready projects with firm power commitments would significantly accelerate the pace at which new capacity reaches the grid.

2) Govern & allocate

- **Improve transparency and demand tracking.** Mandatory disclosure of data-center energy consumption, water use and interconnection requests would materially improve capacity planning. Incorporating large-load interconnection requests into publicly accessible data systems would make the pace and geography of data-center expansion visible to regulators, utilities and the public in real time.

- **Address overclustering risks in saturated markets.** The concentration of approximately 42% of US data-center capacity in Texas and Virginia creates systemic risks that extend beyond local grid stress. Overclustered markets face compounding vulnerabilities across power supply, water availability, cooling infrastructure, and skilled labor, while their position as critical nodes in global AI value chains means that a regional grid failure or prolonged supply disruption could have disproportionate consequences for AI deployment globally. Policymakers and grid operators should assess concentration thresholds beyond which new approvals in saturated markets require enhanced scrutiny, consider incentive structures that actively redirect investment toward underutilized regions with available grid capacity, and prioritize locations with proximity to renewable energy resources to reduce long-run dependence on firm fossil fuel generation.

- **Require data centers to internalize their infrastructure costs.** The socialization of capacity costs onto existing ratepayers is both economically inefficient and politically unsustainable. Policymakers should establish frameworks requiring large new loads to bear a proportionate share of the transmission and capacity infrastructure their demand necessitates. Time-of-use and peak-pricing tariffs that reflect system stress would more accurately align data-center costs with the grid impacts they create, reducing the cross-subsidization of large loads by residential ratepayers. Where market-based pricing mechanisms are insufficient, direct compensation requirements, obliging data centers to offset the costs their demand imposes on host communities, provide an alternative route to addressing the equity dimension and reducing the political friction that is increasingly delaying project approvals.

3) Optimize & decarbonize

- Implement mandatory energy-efficiency standards. No binding energy performance standards currently apply to data centers in most jurisdictions despite their growing share of national electricity consumption. Minimum PUE thresholds, water-usage effectiveness requirements and mandatory energy intensity reporting would establish a baseline accountability framework, with stricter requirements for new hyperscale builds where best-in-class efficiency is already technically and economically achievable.[28]

- **Develop demand-flexibility frameworks.** Full curtailment of data center operations is economically impractical given facility capital intensity, but partial demand flexibility is more tractable and largely untapped. Batch workloads such as model training and non-time-sensitive inference can in principle be shifted to off-peak periods, reducing peak demand pressure and congestion costs without compromising operational continuity. Regulators and grid operators should develop demand-response frameworks that reward this flexibility through capacity market mechanisms or dynamic pricing signals.

- **Ensure AI powers the energy transition rather than slowing it.** The risk that AI-driven data-center growth locks in years of additional fossil-fuel generation is real and underappreciated. Behind-the-meter generation, power purchase agreements tied to additionality requirements, and prioritization of nuclear and long-duration storage for firm power needs would reduce this risk. AI infrastructure and decarbonization are not inherently in tension. The former should be designed to accelerate the latter.

[28] Data Centers and Their Energy Consumption - Congressional Research Service

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Allianz Research | 6 May 2026

US large banks: The peak of the cycle is not the time to be complacent

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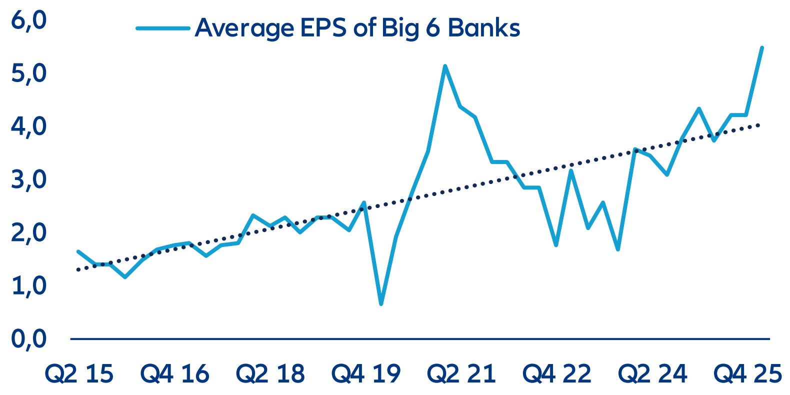
In Summary

- **US large-cap banks posted record Q1 2026 earnings, and both earnings growth and asset quality sits well above trend. Yet, investors seem wary about how long the good times can last, for at least four reasons.** Bank equities underperformed the S&P 500 and credit spreads widened versus non-financial peers. The market is looking beyond balance sheets and pricing a gap between reported solvency and true loss-absorption capacity.
- **First, buybacks offer short-term support but have limits.** The Big Six returned over USD110bn in 2025 and ~USD32bn in Q1 2026 alone via share repurchases, mechanically inflating EPS levels. But buybacks support EPS growth only as long as they continue: once excess capital is exhausted, the growth tailwind disappears with a cliff effect, leaving valuations resting entirely on business growth. Elevated trading income and the M&A cycle are both volatile and unlikely to persist at current levels.
- **Second, regulatory softening reduces buffers.** The March 2026 Basel III/GSIB/stress-test proposals will cut CET1 requirements for the largest banks by ~5–6% cumulatively. Banks have already positioned for this: CET1 ratios for the Big Six fell ~100bps y/y, and this direction will persist as big banks still have rich buffer of around 2pps against the minimum required CET1 ratio.
- **Third, SRTs artificially lift reported CET1.** Synthetic risk transfers (SRT) have grown five-fold since 2016 (~EUR800bn outstanding), allowing banks to shed risk-weighted assets and boost CET1 ratios without reducing actual exposure. "Organic" solvency, which is not at the mercy of high-risk-buyers, is lower than headline ratios suggest – an estimated gap of 43bps from the CET1 ratio, plus the procyclical and liquidity risk that have not been tested yet.
- **Last, long financial cycles analysis points to weaker fundamentals for the banking industry in the years ahead.** Based on BIS methodologies, potential financial strains can be detected by examining credit-to-GDP, real credit growth, and residential property prices over 8–30 year periods. This framework, which successfully identified the two last U.S. financial or banking crises (the S&L crisis and the GFC), helps spot out cycle reversals that typically precede stress, though timing remains uncertain. Current signals indicate the U.S. long financial cycle has passed its peak and entered a downturn phase between 2021 and 2023. Emerging strains may extend beyond banks to non-bank financial institutions, given their larger role in the economy. Although real corporate credit growth has been picking up recently, and credit standards have loosened, most other segments (such as mortgages) have been subdued for several years now, suggesting that the US economy is indeed entrenched in the downward phase of the long financial cycle**.**
- **The conclusion is asymmetric.** Equity prices may capture the upside created by continuous share buybacks in the short term. But the valuations, which already embed rate-driven net interest income (NII), buyback-fueled EPS growth and M&A super cycle tailwinds, are fragile and leave no room for disappointment. Credit spreads have widened modestly but do not compensate for structurally thinner capital buffers. The largest banks may be well-positioned to weather a downturn; the question is whether their investors are being paid enough for the risk that one arrives.

Is it the peak of the cycle for US banks?

America's large banks entered Q2 2026 in their strongest collective shape since before the financial crisis. Five out of the biggest six banks beat Q1 consensus estimates, supported by high trading income and the M&A and IPO cycle. Bank of America's equities trading income jumped 30%, representing its best trading quarter in 15 years. JPMorgan's markets revenue hit a record USD11.6bn, with fixed income up 21% and equities up 17%. Goldman Sachs' investment banking fees grew 48% year-over-year to USD2.8bn. However, Wells Fargo stood out, with the NII revenue miss sending its stock price down 5-6% on results day. Looking past the decade, the current cycle is clearly well above the long-term trend. Plotting the average earnings per share of the Big Six against its long-run trend shows that the group's average EPS now sits materially above trend at levels that have historically preceded compression rather than continuation.

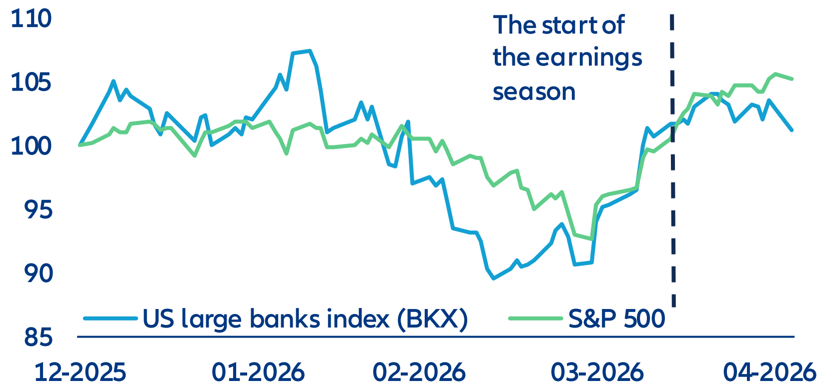
Figure 1: Earnings cycle goes above long-term trend



Sources: Bloomberg, Allianz Research

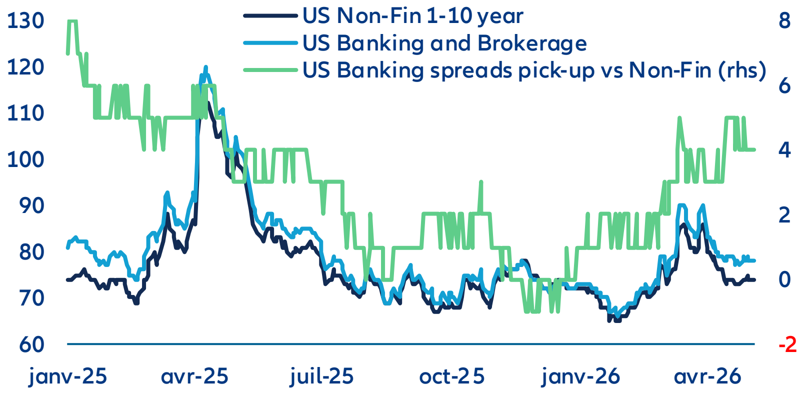
Nevertheless, markets are wary about how long the good times can last. The recent market reaction demonstrated more concerns than confidence. Since earnings season kicked off on 13 April, US large banks (KBW Bank Index) have dropped 0.6%, and underperformed the broader index (S&P 500) by more than 5pps. In credit, the divergence is subtler but more structurally revealing: US bank and broker spreads have widened relative to non-financial investment-grade issuers (keeping ratings and maturities equal). Investors have started to demand more compensation to hold bank credit than comparable corporates.

Figure 2: US bank stocks performance year to date (rebased to100)



Sources: Bloomberg, Allianz Research

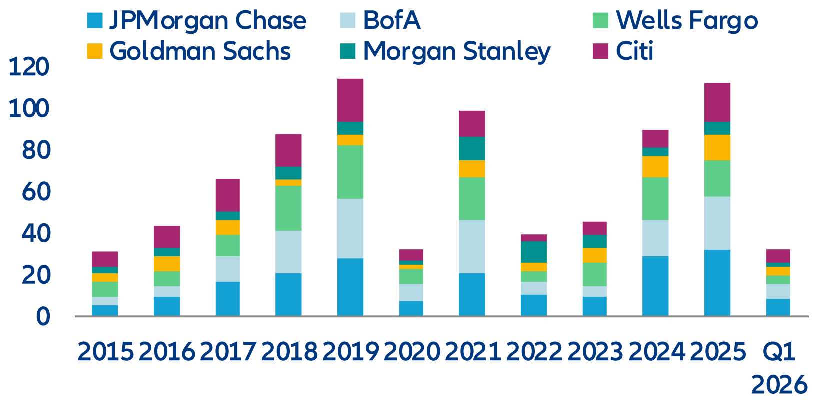
Figure 3: Credit performance - US banks credit premium is picking up



Sources: LSEG Datastream, Allianz Research

In the short term, the large scale of share buybacks is a cause for concern. America's six largest lenders returned more than USD110bn to shareholders in 2025, and Q1 2026 suggests the pace is accelerating with around USD32bn total repurchased. Share buybacks are, mechanically, among the most direct levers available to support earnings per share and the stock price: the fewer the shares outstanding, the higher earnings per share – an immediate signal of management confidence. But this is capital optimization, not capital creation. As buyback programs slow – whether from unstable trading revenue or simply from the pool of excess capital being exhausted – the mechanical EPS support they have provided will diminish. The equity story then rests entirely on the underlying earnings power of the business. At current multiples, that is a thin margin of safety because the elevated trading income and investment banking income is inherently volatile and not guaranteed to persist.

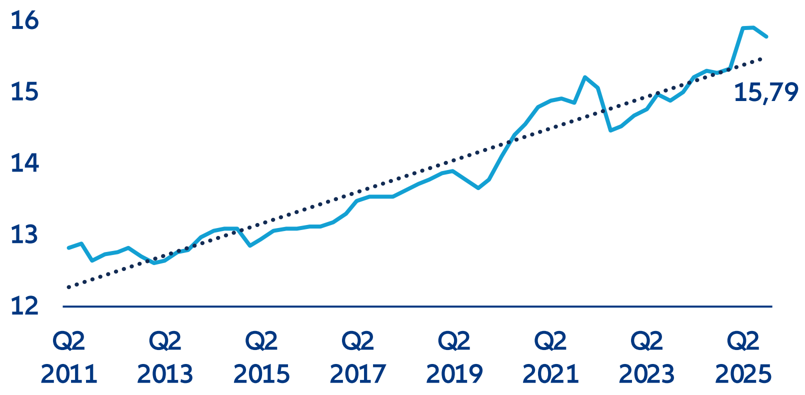
Figure 4: US Big Six banks distributed more than USD120bn capital in 2025 by share repurchases



Sources: Bloomberg, Allianz Research

For credit investors, it hurts more. The equity on a bank's balance sheet is the buffer that absorbs losses before bondholders are impaired. Each repurchase, executed at a premium to book, reduces that buffer in absolute terms. The Tier 1 capital ratio across the US banking system, which reflect banks' credit quality, is now turning for the first time after the interest rate shock in 2022. Tier 1 capital adequacy fell 13bps in the last quarter of 2025, reflecting the capital return program and signaling the peak of the credit cycle. And the new Basel III proposal announced last month was explicitly designed to push it further down.

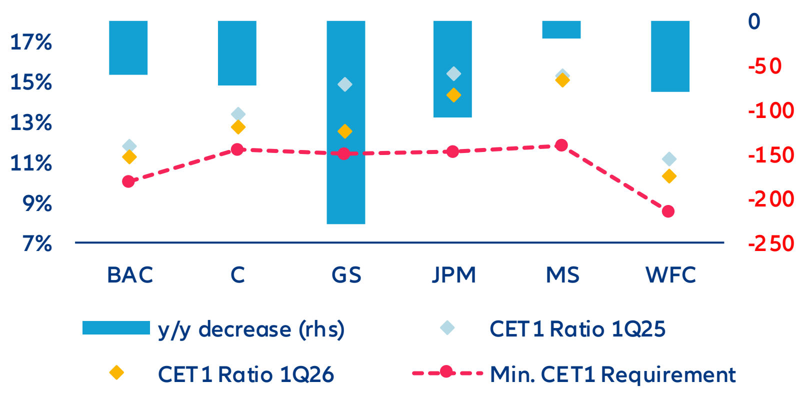
Figure 5: US banks regulatory Tier 1 capital to risk-weighted asset



Sources: LSEG Datastream, Allianz Research

In the medium term, deregulation is reducing buffers. The turning trend of bank Tier 1 capital adequacy brings us to the mid-term concern of deregulation. On 19 March, the Fed, OCC and FDIC jointly issued three proposals that cover Basel III Endgame, the GSIB surcharge and stress testing. The Basel III component revises capital requirements specifically for the largest banks (Category I and II banks) – setting higher requirements on trading activity while deliberately lowering risk weights on traditional lending. However, the GSIB surcharge revision reduces additional capital charges on the eight largest systemic institutions. Combined with proposed stress-testing changes, CET1 requirements for the largest banks will decline by 4.8% on a cumulative basis, while Tier 1 requirements will fall by approximately 6%. Banks had anticipated the direction and positioned accordingly. Ahead of formal confirmation, the Big Six had built an average CET1 buffer of 2pps above regulatory minimums to guard against the threat of the original, more punitive 2023 rules. With that threat now formally lifted, the incentive to maintain the excess is gone. CET1 ratios fell by an average of nearly 100bps y/y – Goldman Sachs and Wells Fargo led the compression – with excess capital flowing directly into the buyback programs detailed above.

Figure 6: Big Six CET1 Ratio Q1 2026 vs Q1 2025, y/y change in bps



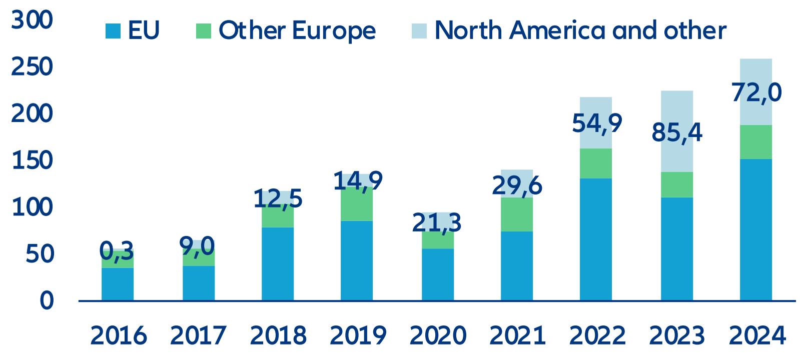
Sources: Bloomberg, Corporate reports, Allianz Research

The second engine is less visible but runs parallel. Even before deregulation formalized the relief, banks had been quietly optimizing regulatory capital through synthetic risk transfers (SRTs). Through these structures, banks transfer the credit risk of a tranche of their loan book to private investors – typically private credit funds, insurers or hedge funds – retaining the loans on their balance sheets while the CET1 ratio improves. Globally, SRT issuance has grown fivefold since 2016, with outstanding loan portfolios protected reaching almost EUR800bn as of end-2024[1]. European banks have traditionally dominated the market, but US banks expanded this market rapidly in 2023 and

[1] *International Association of Credit Portfolio Managers (IACPM) (2025): Global SRT Bank Survey 2016–2024.*

2024 and account for roughly 20% of global issuance volume by the end of 2024. [2] While SRTs volume is still small relative to bank balance sheet, procyclicality risk and liquidity risk that haven't been tested by a credit downturn are the core problems. In a broad economic downturn, banks may find credit protection expensive or unavailable when capital needs are greatest – forcing a simultaneous pullback in lending that amplifies the very downturn it coincides with. The liquidity dimension compounds this when open-ended funds investing in SRTs face redemption pressure in stress.

Figure 7: Underlying pool size by issuer region (EUR bn)

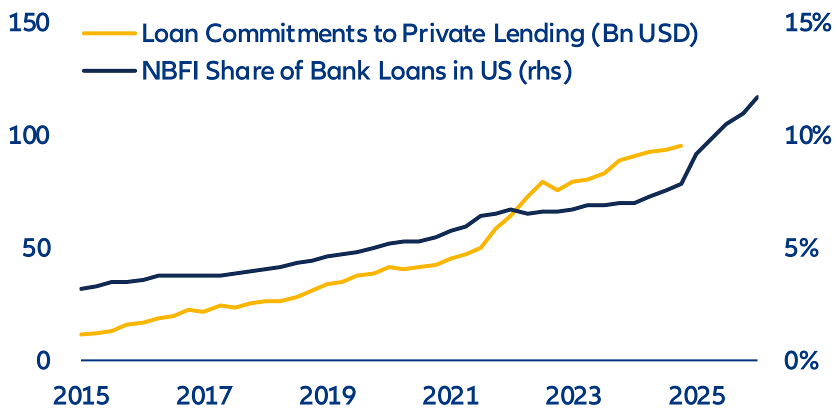


Source: Basel Committee on Banking Supervision (BCBS), Allianz Research

Regulatory relief lowered the floor, while SRTs removed the risk from headline ratios. Combining those two engines, the financial system is not as well-capitalized as its headline ratios suggest: everything that has been moved into the shadows should also be deducted. When stress surfaces, the gap that is invisible on paper will become visible in prices.

The opacity of fast-growing private lending has created another risk in the longer term. In response to growing market concerns on private credit, the Big Six have voluntarily disclosed more detail on their non-bank financial institution (NBFI) lending in recent quarters. Without a certain reporting standard, banks brought similar reassurance – NBFI exposure is growing slowly but the size is still relatively small.

Figure 8: Increasing amount of US bank loans to NBFIs



Source: FED, Allianz Research

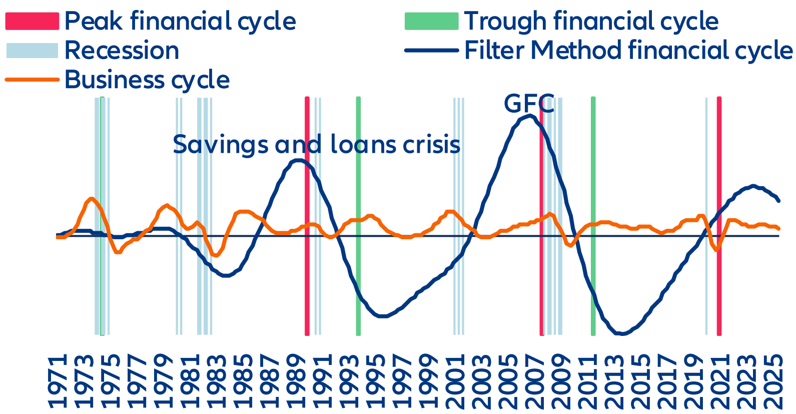
[2] *Basel Committee on Banking Supervision (BCBS) (2026): The rise and risks of synthetic risk transfers.*

Detecting downturn risks

Long financial cycle analysis offers valuable insights to detect potential banking strains before they occur. We build on the work from the BIS[3] which seeks to characterize empirically the booms and busts of the financial cycle since the 1970s. The analysis combines two analytic approaches: turning-point analysis and frequency-based filters. It looks at three macro-financial variables that are found to best capture the medium-term financial cycle, lasting between 8 and 30 years: the credit-to-GDP ratio, the growth of real credit and the growth of real residential property prices. Only credit to the private credit is included. When the statistical analysis identifies a consistent turning of the cycle across these macro-financial variables, including past the trough, financial strains are likely to occur in the next couple of quarters or years – the timing being uncertain. The long financial cycle has been a good indicator to anticipate the two banking crises in the US over the past 40 years: the L&S savings crisis starting in 1988-89 and the Global Financial Crisis (GFC) starting in 2007. Unfortunately, a big caveat is that there have been only two occurrences of banking or financial crises over the past couple of decades, so the empirical regularity of the signal has to be taken cautiously. Nevertheless, the long financial cycles have also been good predictors of banking and/or financial strains in other markets, which leads us to take these signals seriously.

The financial cycle now signals weakness for the finance industry ahead. Figure 9 depicts the financial cycle for the US since the 1970s. It shows that we are past the peak of the cycle recently, with both signals converging, and signaling the beginning of a downturn phase in the cycle, which had begun between 2021 (turning-point analysis) and 2023 (filter method). Figure 10 shows the evolution of the three underlying variables. Since 2021, the year-on-year growth slows down, or even becomes negative since 2024-25. The strains looming in the finance industry are not necessarily bank-driven. The analysis looks at aggregate measures of credit. Supply of credit by banks makes up only 30% of the total supply of credit. Hence, strains could also emerge in non-bank financial institutions (NBFIs) rather than banks, though the links between the two are closely tied.

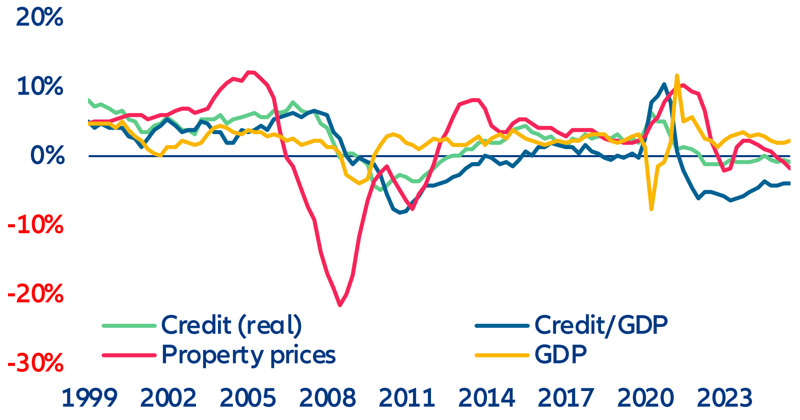
Figure 9: Long financial cycle in the US



Sources: BIS, Allianz Research

[3] *Characterizing the financial cycle: don't lose sight of the medium term!, BIS Working Papers No 380, June 2012.*

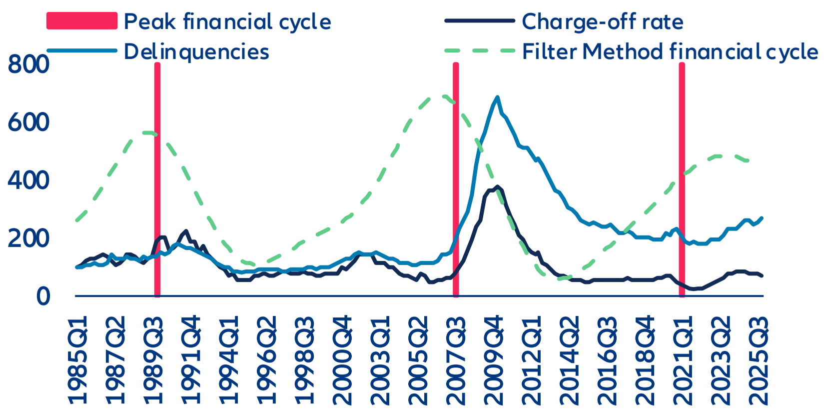
Figure 10: Growth y/y (%) of the components of the financial cycle and GDP



Sources: BIS, Allianz Research

The turning of the long financial cycle has just begun and signals further rises in delinquencies and charge-offs ahead. The turning of the financial cycle reflects the combination of potentially several factors but are mostly determined by rising Fed interest rates when the central bank strives to reduce inflationary pressures or financial exuberance (loose credit standards and increased supply of credit to risky products). Higher interest rates are gradually feeding into higher rates to the private sector and tightening credit standards as borrowers become weaker amid rising interest expenses. In turn, curtailed supply of credit and higher interest rates lead to decelerating real property prices, pressuring down the value of collateral for households and some corporates, which further exacerbate the credit downturn. This is related to the financial accelerator theory. Eventually, it means typically slower growth of banks' revenues, and increasing costs related to provisions as more borrowers struggled to replay their debt. Figure 11 shows that after the financial cycle turns down (with the turning point method sending a signal before the filter method in the last episode), delinquencies and charge off rise. In the current environment, private credit defaults would appear much higher if borrowers held to public market standards. Figure 12 shows that including distressed exchanges realized default rates on leveraged loans would indeed have been much higher, corresponding well to the current trend of the financial cycle.

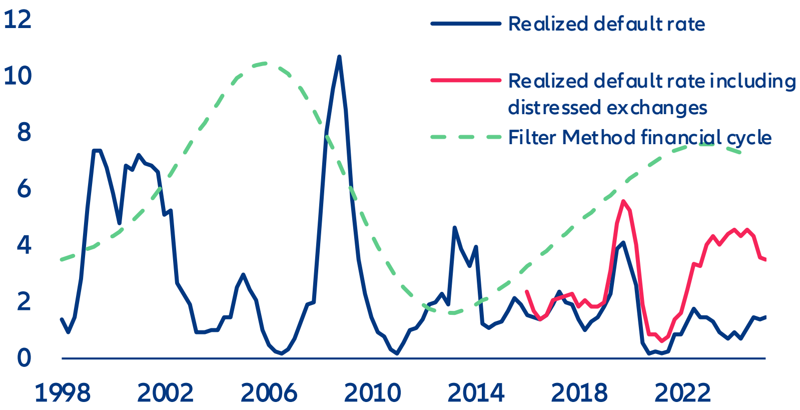
Figure 11: Financial cycle compared to the evolution (basis 100) of the charge-off rate (on all loans) and delinquencies (on all loans and leases, all commercial banks)



Sources: FED, BIS, Allianz Research

It is important note that this analysis captures long-term trends: within this long cycle, we are currently witnessing easier credit standards and a pickup of credit creation, suggesting that strains are not imminent. For instance, the Fed SLOOS survey indicates that banks have loosened credit standards for corporate loans since early 2023 and that real credit growth has accelerated from end-2025. But those are unlikely to be sustained if the downturn phase of the long financial cycle is entrenched. Our analysis suggests that this short term cycle will reverse soon.

Figure 12: Financial cycle compared to default rates on leveraged loans including distressed exchanges



Sources: FED, PitchBook, BIS, Allianz Research

Note: The default rate on leveraged loans including distressed exchanges is calculated as the number of issuers in default or distressed exchange over the past 12 months divided by the total number of issuers at the beginning of the 12-month period. These loans are issued to highly leveraged companies and are more sensitive to changes in the financial cycle.

What does this mean for investors?

US large-cap banks have entered mid-2026 in their strongest operational position since 2008 – earnings at cycle highs, balance sheets well-capitalized, capital returns at a record pace. However, as discussed above, the forces pushing the cycle turning are cumulating. Beyond the cycle, structural headwinds form from digitization, where low-cost disrupters are manifold. The American Bankers Association warned that a stablecoin market scaling toward USD2trn could drain up to 10% of core bank deposits, raising funding costs by roughly a quarter of a percentage point – a modest but directionally unwelcome addition to an NII outlook that is already plateauing[4]. At the same time, current equity valuations and credit spreads offer investors almost no cushion against disappointment.

Share buybacks and high ROEs will keep benefiting equity holders in short-term. Current valuations price in three tailwinds continuing simultaneously – rate-driven NII, a buyback-fueled EPS ramp-up and a M&A super-cycle. But none are guaranteed in the long-term. The market has already paid for the good news but has not priced in the medium-term and long-term cycle turning pressure.

With tight spreads, the upside is more limited for credit investors. Downside risk would be amplified if the large amount of capital distributed to shareholders and the persisting downtrend of solvency has been billed to them, as lower required capital means a thinner equity buffer sitting above bondholders. Though solvency is still healthy and far from triggering concerns even after the proposed capital release, credit investors are not being asked to absorb such structurally lower protection at historical thin compensation. Senior bank spreads have widened relative to non-financial investment-grade peers, but only modestly, and not in a way that reflects the structural shifts.

The banks themselves, in launching CDX Financials index, have built another instrument to make the shadow system they created legible and hedgeable — and declined to include themselves in its perimeter. The largest banks, at their peak, are still well-positioned to weather the downturn. **The question the market is beginning to ask in the gap between record earnings and cautious share prices is whether their investors are being adequately paid to stand in the rain with them.**

[4] *Huther, J. and Wang, Y. (2025). "How stablecoins could affect borrowing costs for the government, businesses and households." ABA Banking Journal, August 2025.*

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Automotive: Will the Middle East crisis supercharge EV momentum?

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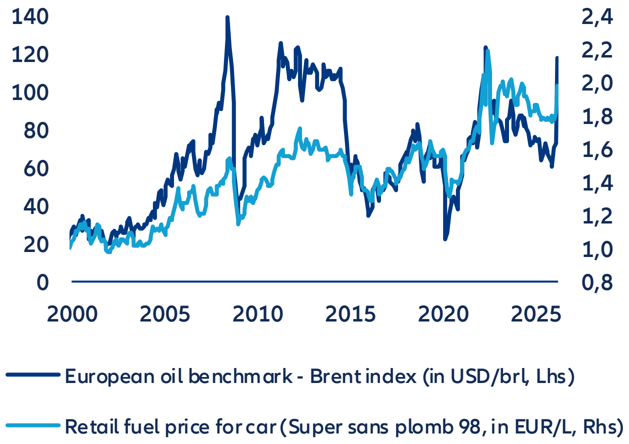
In Summary

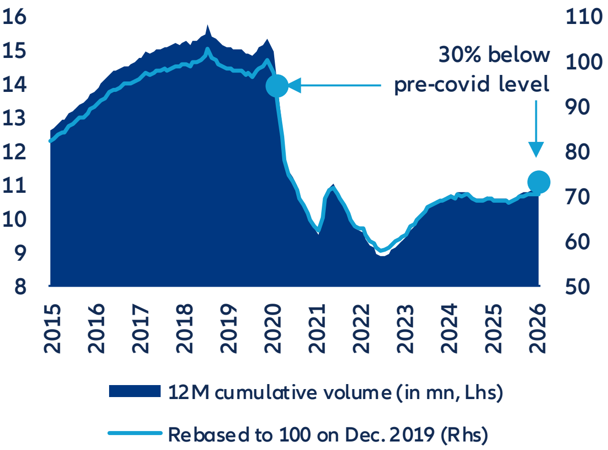
- **An electric tilt boosted by energy volatility.** After a bruising 2025, Q1 2026 data reveal a striking reversal. BEV market share hit 19% EU-wide (+4pps vs; Q1 2025) and surged to 28% in France and 23% in Germany. The 30% oil spike fueled by Middle East tensions has sent pump prices back above EUR 2.0/L, reviving the cost-of-ownership debate and accelerating a shift that was already structurally underway. This is notably thanks to price compression over other powertrain models on primary markets and rising supply on second-hand markets.
- **Consumers are more sensitive than ever before to energy costs.** Fuel costs are hitting European households already squeezed by a post-Covid cost spiral across all car-related services with repair, maintenance and parts costs up between 20-30% since 2021, 10% higher than the increase in average disposable income. Car-usage costs absorb 7–8% of disposable income on average in France and Germany but rise to 11% for lowest-income quintiles during periods of volatility (e.g. 2022), making fuel-consumption costs a critical issue for low-income households. Currently, switching to BEVs would offer material energy consumption savings – equiv. to a 4–5% purchasing-power gain per capita on average in Western Europe – with an energy-cost differential range estimated at 5-7x regardless of price volatility.
- **But turning current momentum into a durable energy shift will require firing on all four policy cylinders in parallel: increasing local battery production, building sufficient grid infrastructure implementing effective carbon pricing and consistent subsidies.** While there have been positive signs of structural progress (e.g. average battery range crossing 500km psychological barrier, faster charging times), Europe remains exposed to China's technological dominance on batteries and powertrains. Its infrastructure deficit gets structural, with only 1.1mn charging points in Q1 2026 (mostly concentrated in 4 countries) – far from the European Commission's 3.5mn target by 2030. Moreover, AI-driven data-center expansion will compete directly with EV electrification for constrained grid capacity: EU consumption is projected to grow from 70TWh in 2024 to 115TWh by 2030. To reach its electrification targets by 2030, Europe needs carbon pricing that makes the fossil-fuel cost signal permanent, purchase subsidies that bridge the affordability gap for mass-market buyers, technological acceleration that makes BEVs the cheaper option without government support and grid decarbonization that ensures electrification delivers on its promise.
- **We find that even the most generous subsidy scenario (at least EUR5,000) reaches only 70% BEV share by 2030.** Carbon pricing under NDC commitments lifts the EU BEV share from roughly 29% to 42% by 2030 – meaningful, but still not enough to be on the Net Zero target which estimates 79% of BEV share in the EU by 2030. On the technology side, battery costs have fallen -93% since 2010 and are heading toward USD 60–70/kWh by 2030, the level at which EVs become cheaper than combustion vehicles in most segments without any grant. But ultimately an EV is only as clean as the electricity it runs on: even full fleet electrification falls short of its environmental potential without a cleaner grid. Moving Europe's low-carbon electricity share from 70% to 80% by 2035 would cut per-vehicle EV emissions by over 40%.
- **What if energy volatility continues beyond Q2?** EV resiliency is likely to be tested, notably as the progressive phase-out of subsidies reduces public support buffers against energy shocks. However, relatively strong purchasing power among EV buyers, ongoing technology improvements and intensified competition from Chinese OEMs should continue to underpin demand. The main downside risk lies upstream: renewed semiconductor supply disruptions could disproportionately impact EV production and pricing, given their higher chip intensity.

An electric tilt boosted by energy-price volatility?

Another energy crisis or another reason to switch to EVs? The recent 30% spike in Brent crude triggered by the Middle East crisis has sent retail fuel prices in several major European economies like France, Germany or Netherlands surging back above EUR 2.0/L, approaching historical peaks last seen when Russia's invasion of Ukraine unleashed an unprecedented shock across oil, gas and power markets. For most European economies – structurally exposed as net energy importers – such volatility is far from abstract: A decade of data reveals a stubborn negative correlation between rising pump prices and new car registration volumes in Germany, confirming that fuel costs remain a decisive purchasing factor. Each price shock becomes a powerful advertisement for electrification. Indeed, in Q1 2026, Europe's BEV market share reached a record 19%, up 4pps year-on-year, confirming a structural shift already underway. While Nordic countries lead the EV penetration trend – Norway records almost 100% – some large markets like France and Germany have seen a strong impulse for EVs, with market share increasing by +10% and +6%, respectively, over the past 12 months to 28% and 23% in Q1 2026. Indeed, Germany hit the 1mn milestone in EV sales amid the energy crisis of 2022 and just crossed the 2mn mark in early 2026. Nevertheless, overall demand remains relatively moderate (in absolute terms) and fully electric vehicles account for only around 3% of the total European fleet. Moreover, the momentum remains highly uneven: Nearly half of EU countries still post BEV shares below 10%, while only a small core – around 15% of members – exceeds one-third penetration. The growing popularity of EVs also reflects a broader decline in ICE appeal, a movement amplified by energy price volatility but rooted in deeper market dynamics. Falling retail prices, driven by intensified competition – particularly from Chinese OEMs – and a growing secondary market fueled by off-lease vehicles are easing affordability constraints and stabilizing residual values, unlocking new demand segments. Still, headline gains should be interpreted cautiously: they occur in a stagnant market where total registrations remain roughly 30% below pre-Covid levels, meaning higher shares partly mask weak volumes. Even leading BEV markets such as Norway, Belgium or the Netherlands have recently seen declining sales in absolute terms.

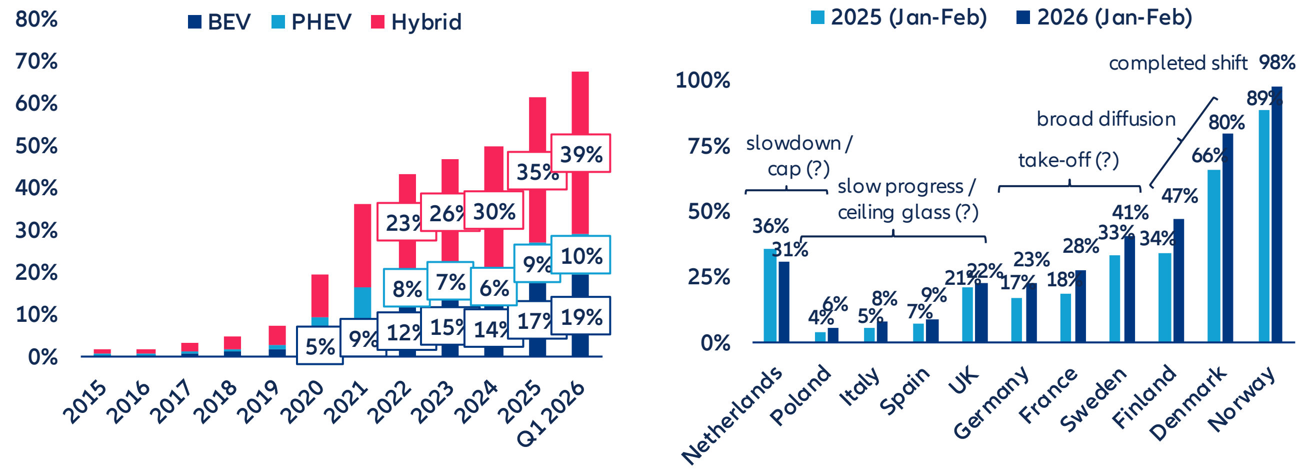
Figure 1 & 2: Oil price benchmark in Europe (Brent) vs retail car fuel price in France / 12-month trailing cumulative registration of new passenger cars in EU-27





Sources: INSEE, ACEA, LSEG Datastream, Allianz Research

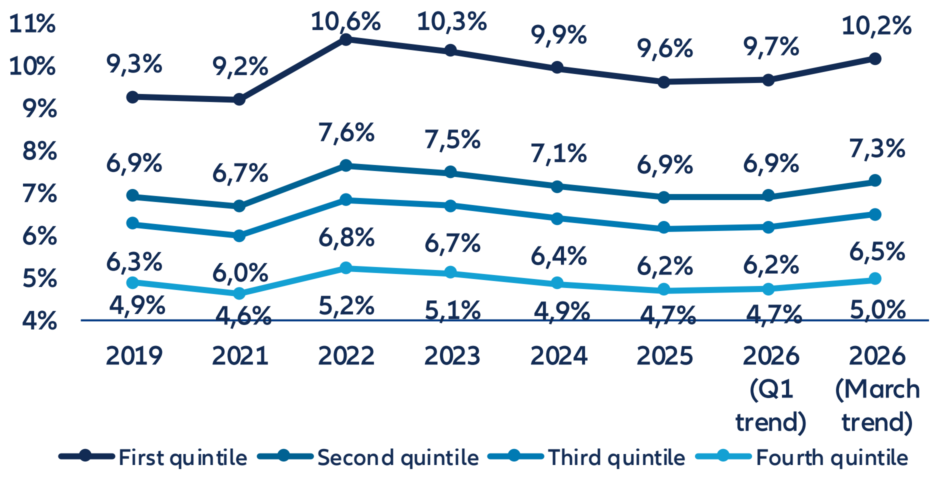
Figure 3 & 4: Annual powertrain car market share in EU-27 / BEV market share in top EU markets (Q126 vs. Q125)



Data as of Q1 2026. Sources: ACEA, Allianz Research

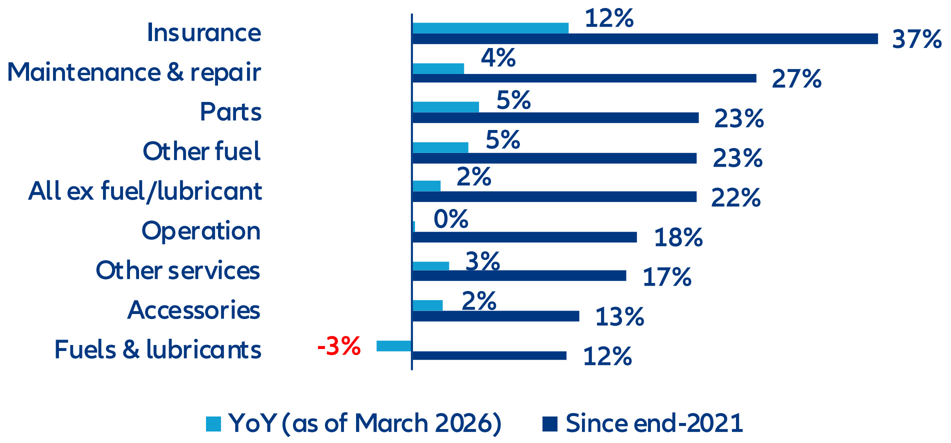
Car-running costs have become a genuine financial burden for European households – and the current oil-price spike is exposing once again how structurally vulnerable consumer budgets remain to energy volatility, which could accelerate the shift to electrification. In France and Germany, car-usage spending (fuel, insurance, maintenance – excluding the purchase price) accounted for 7-8% of annual disposable income on average, with the lowest earners hit hardest (11% of annual disposable income). This share visibly surged during the 2022 energy crisis and is trending sharply upward again in early 2026 (Figure 5). Car-usage spending is among the heaviest fixed-cost categories in the family budget and one of the least compressible, given the structural dependency on personal mobility in both urban peripheries and rural areas, notably for commuting to work. In the past, social movements such as the Yellow Vest protests in France erupted due to the rising burden of energy costs on personal transport. The pressure has intensified after the pandemic and the 2022 crisis as car-related services were entangled in a broader cost spiral: Across the EU-27, insurance prices have climbed +37% since end-2021, while maintenance, parts and repairs are up 20-27%, driven by the increasing electronic complexity of modern vehicles and the costlier battery components embedded in new powertrains (Figure 6). The 2021-2022 period has left a permanent scar in the mindset of European households, with inflation being by far the most deterrent downside risk in the main European consumer barometers since then.

Figure 5: Share of annual disposable income allocated to car usage spending* in France, per income distribution quintile



**This includes all costs except car purchase (fuel & lubricant, maintenance, repairing, insurance.). 2025 and 2026 figures are estimates calculated on the basis of HICP statistics for fuel, repairing & maintenance and insurance for personal transport. Sources: SDES (Chiffres clés des transports, April 2026), Eurostat, Allianz Research*

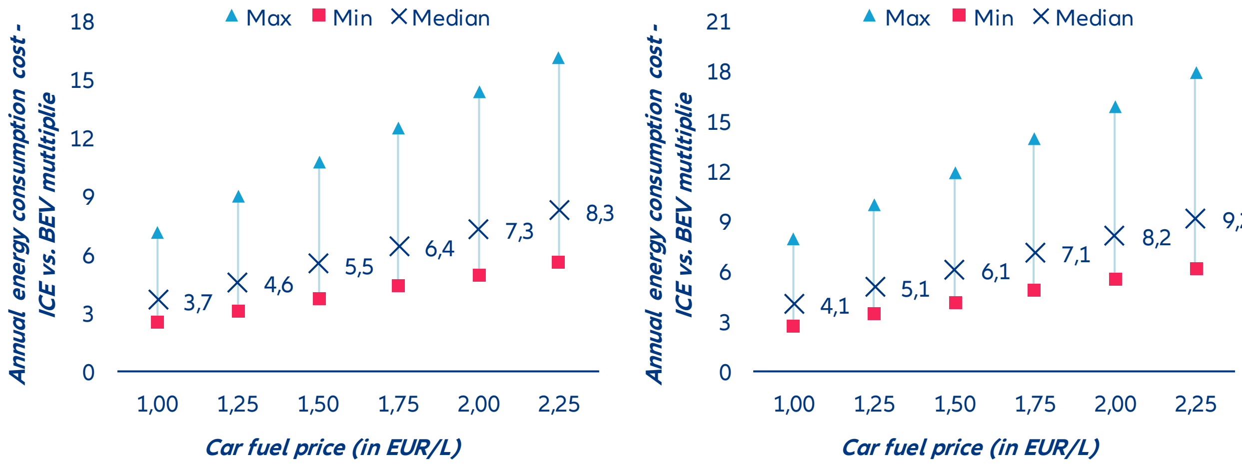
Figure 6: EU-27 HICP index for personal transport index (2015 = 100), trailing 12-month average growth



Sources: Eurostat, Allianz Research

Powertrain technology arbitrage offers a 4-5% purchasing power boost. At today's fuel price of around EUR 2.0/L across Western Europe, the annual energy bill for an ICE car runs roughly 7–8 times higher than for an equivalent BEV – 7x based on the most efficient models available, rising to 8x when taking into account the current European fleet average, of which EVs still account for less than 3%. That spread is wide by any historical standard but even before the Middle East crisis, when fuel hovered between EUR1.50 and EUR2.00, the multiplier was already sitting at 5–7x – enough to settle the budget debate in favor of EVs regardless of geopolitical tailwinds. Country dispersion adds another layer: Depending on local electricity tariffs, the cost advantage ranges from a modest 5x to a striking 14x, meaning the weakest case for EVs still represents a substantial saving. When translated into purchasing power, that energy differential amounts to a 4–5% income boost for the average Western European driver – not negligible in an inflationary environment. Add the narrowing total cost of ownership (BEVs now show roughly twice fewer recalls than ICE models per ADAC's latest figures, offsetting higher upfront battery costs) with public purchase subsidies closing much of the residual price gap and the picture is clear: the structural economics of the technology shift do not hinge on an oil-price spike. The total cost of ownership equation is quietly, but decisively, tilting toward electric.

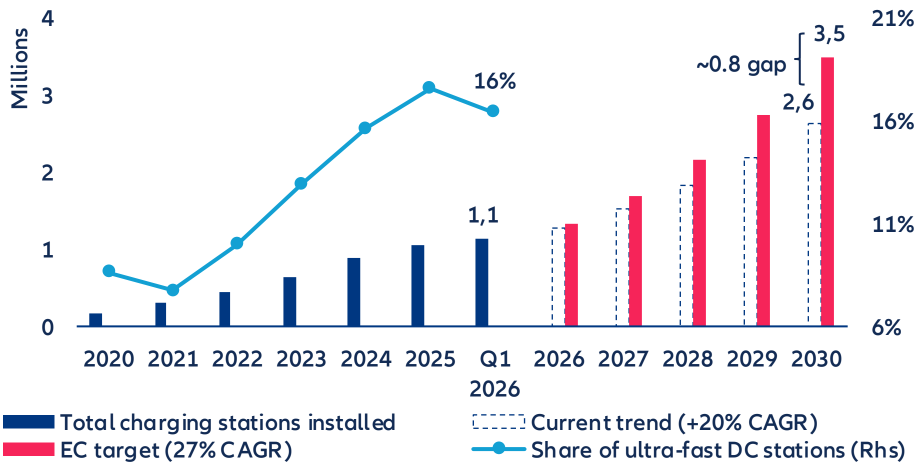
Figure 7 & 8: Annual car energy consumption cost comparison between ICE and BEV models in Europe* (based on most advanced energy-efficient model and average consumption of current fleet in 2025)



** based on a benchmark of 10 European countries (Belgium, Denmark, France, Germany, Hungary, Italy, Netherlands, Poland, Spain & Sweden). Calculations based on average power price in April in each country (as of 29 April 2026) and assumption of an average car distance of 12000 km per inhabitant (ACEA). Sources: ACEA, EPEX, Alternative fuels observatory (European Commission), Allianz Research*

Infrastructure deficit and grid pressure: The invisible brake on Europe's EV momentum. Beyond technology and price dynamics, the charging infrastructure gap remains one of the most tangible and underestimated structural constraints on EV mass adoption across Europe. Figure 12 illustrates the scale of the challenge starkly: despite a consistent ~20% CAGR in charging stations installed since 2020, reaching 1.1mn points in Q1 2026, the current trajectory falls materially short of the European Commission's 27% CAGR target – projecting a gap of approximately 0.8mn stations by 2030 relative to the EC's 3.5mn objective (Figure 9). Critically, only 16% of installed stations qualify as ultra-fast DC chargers, the very infrastructure that would most directly address consumer range anxiety and charging convenience – arguments that, as discussed, weigh heavier than price premium in many consumer resistance surveys. Dismissing the infrastructure deficit as a consequence of still-insufficient EV penetration misreads the causality: it is the perceived inconvenience of charging that actively deters consumers from making the technology shift in the first place. The distribution problem is equally structural: four countries – the Netherlands, France, Germany, and Belgium – account for roughly 65% of all EU charging stations, with France and Germany alone concentrating 40% of ultra-fast DC points. Rural areas across the broader EU remain critically underserved, amplifying adoption barriers precisely where private vehicle dependency is highest. Compounding these infrastructure pressures is an emerging and formidable competitor for Europe's electrical grid: the rapid expansion of data center capacity driven by AI investment. The IEA estimates EU data center electricity consumption at 70 TWh in 2024, with projections pointing toward 115 TWh by 2030 – a near 65% surge that will compete directly with transport electrification for grid capacity and investment priority. EU electricity demand growth remains comparatively modest, at around 1.1% in 2025 before a modest acceleration to 1.5% in 2026, leaving limited headroom to absorb simultaneously the electrification of mobility and the capex-heavy buildout of AI infrastructure. In this context, grid modernization and targeted infrastructure investment are not secondary enablers of Europe's EV transition – they are its most urgent prerequisite.

Figure 9: Total number of charging stations installed across the EU and projections based on current trend and European Commission target for 2030



Data as of Q1 2026. Sources: Alternative fuels observatory (European Commission), Allianz Research

From crisis to structural shift: Four pillars for a durable EV transition in Europe

There is robust evidence that fossil fuel prices influence the adoption of electric vehicles. In a study of California between 2014 and 2017, Bushnell et al. (2022)[1] show that when consumers decide whether to go electric, the price of gasoline matters four to six times more than the price of electricity, suggesting that what pushes people toward EVs is less the promise of cheap charging than the pain of expensive fuel. The pattern holds in China: Fei et al. (2025)[2] track monthly sales across 36 cities and find that a 1 CNY/L rise in gasoline prices lifts overall EV sales by 4.67%, with pure battery electric vehicles surging by 9.04%. Most recently, Zhang et al. (2026)[3] confirm the relationship in a European context, showing that a 1% increase in gasoline prices raises EV registrations by 0.85% across Denmark, Finland, Norway and Sweden, with budget-friendly models benefiting the most.

Yet the historical record shows that without structural reinforcement, oil-shock-driven shifts in vehicle preferences tend to reverse. During the 2007–08 oil spike, US demand for SUVs and pickup trucks fell sharply, before recovering in subsequent years as fuel prices declined and macroeconomic conditions improved[4]. The pattern repeated after 2014: When crude prices collapsed from USD115 per barrel to below USD30 by early 2016, lower fuel prices coincided with renewed consumer demand for SUVs and pickups, encouraging manufacturers to prioritize those segments[5]. Analysis by Resources for the Future confirms that gasoline prices accounted for a substantial share of the shift toward smaller vehicles between 2003 and 2007, yet that trend reversed once prices fell post-2014[6].

Crucially, where EV adoption remained resilient through past oil price downturns, strong policy frameworks appear to have played a decisive role. The International Council on Clean Transportation (ICCT) observed that during the 2014 crash, EV sales continued to grow in Norway, the UK and China, markets that combined robust purchase incentives, charging infrastructure deployment and regulatory support, while markets without comparable policy frameworks showed stagnation[7]. The IEA's Global EV Outlook 2025 reinforces this point looking forward: lower oil prices reduce the fuel-cost savings of EVs, and this effect is particularly pronounced in countries with low fuel taxation[8]. The distinction between cyclical and structural displacement is critical: the demand lost during the pandemic was temporary and rebounded once economies reopened, whereas displacement driven by EVs becomes durable only when underpinned by lasting changes in technology, policy and consumer behavior.

This is precisely where carbon pricing enters the picture as a mechanism to make the fossil-fuel cost signal permanent and predictable. While geopolitical shocks produce sharp but temporary price spikes, a well-designed carbon tax raises the floor price of fossil fuels structurally, removing the cyclicality that has historically allowed consumer preferences to revert. Under such a regime, every barrel of oil carries its environmental cost regardless of what OPEC decides or whether a Middle Eastern conflict escalates or de-escalates.

To explore how different carbon pricing trajectories might shape the pace of BEV adoption in Europe, we consider three oil-price scenarios over the 2025–2035 horizon, each reflecting a distinct policy ambition, following the NGFS framework. Under a Baseline scenario – current policies, no additional carbon pricing – oil prices stabilize around USD77–86/barrel through 2035, despite short-term possible price fluctuations related to exogenous factors such as geopolitical instabilities. EU pump prices remain flat at approximately EUR1.73–1.95/L, and BEV adoption grows primarily through the autonomous trend at roughly 1.75 pps per year, driven by technology improvement rather than price signals[9].

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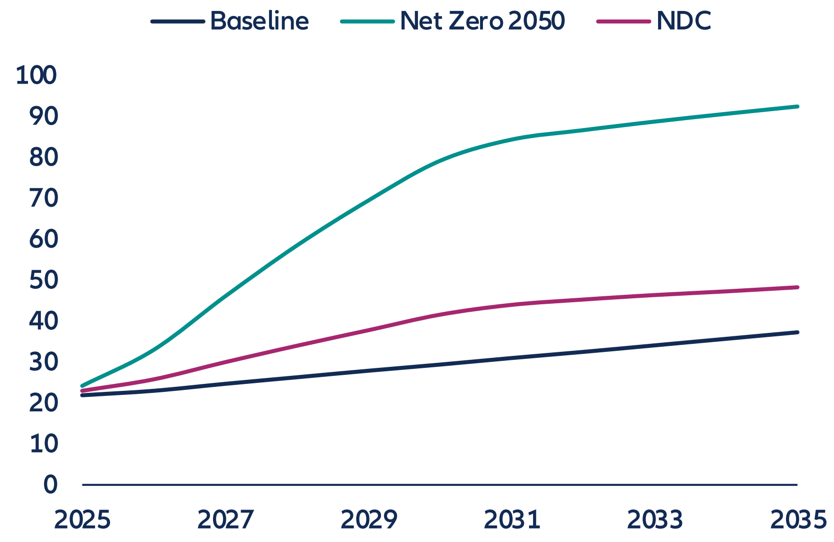
[1] Energy Prices and Electric Vehicle Adoption | NBER
[2] Does high gasoline price spur electric vehicle adoption? Evidence from Chinese cities - ScienceDirect
[3] Electric vehicle adoption and energy prices: Empirical evidence from four Nordic countries - ScienceDirect
[4] Causes and Consequences of the Oil Shock of 2007-08 | NBER
[5] Automobile manufacturers, electric vehicles and the price of oil - ScienceDirect
[6] How Do Gasoline Prices Affect New Vehicle Sales?
[7] 2014 fuel price turbulence didn't pull the plug on EVs - International Council on Clean Transportation
[8] Global EV Outlook 2025 – Analysis - IEA

Under a Nationally Determined Contributions (NDC) scenario, carbon border adjustments and emissions trading begin to bite, pushing oil toward USD100–114/barrel by the early 2030s and EU pump prices to approximately EUR2.10–2.55/L, enough to make fuel expenditure a meaningful push factor, with BEV share reaching approximately 50% in Germany by 2030. Under a Net Zero 2050 scenario, aggressive carbon pricing drives oil above USD190/barrel by 2028–2031 and pump prices to EUR3.50–5.60/L, at which point ICE ownership becomes economically untenable for most households and the mass switch to BEVs accelerates sharply. Figure 10 shows the different pathways of BEV share related to a permanent shift in oil prices.

Figure 10: BEV adoption in EU-26 under different carbon pricing pathways (%)



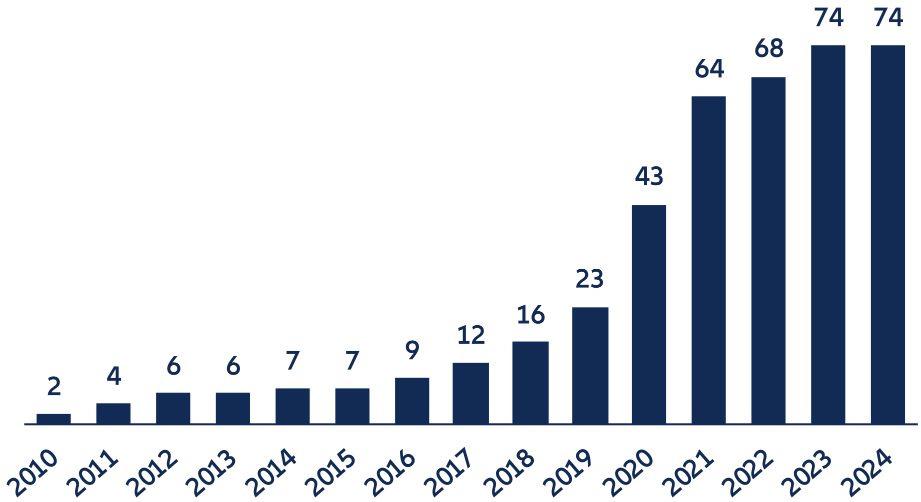
Sources: NGFS, Allianz Research

Carbon pricing raises the long-run cost of fossil-fuel driving. But making the old technology expensive is not the same as making the new one affordable, particularly for low- and middle-income households, who are the mass market. These are distinct economic problems, because carbon pricing internalizes the emissions externality, while purchase subsidies address the separate barriers – such as upfront cost, learning-by-doing, network effects around charging infrastructure – that hold back adoption of a technology most consumers still perceive as novel and risky[10]. The two instruments are often best deployed as complements rather than alternatives. European policymakers appear to have absorbed this logic, as the number of active national EV incentive policies across EU-27 countries has grown from just two in 2010 to 74 by 2024, with a sharp acceleration after 2019 as governments layered purchase grants, tax exemptions, charging infrastructure support and fleet incentives on top of one another (Figure 11).

[9] This estimate is derived from a linear probability model of BEV sales share across EU-26 countries over 2010–2024: ev_sales_share = $\alpha(u)$ + β_1·pump_price + β_2·ln_income + β_3·(ln_income × price_range) + β_4·year_trend. The autonomous trend coefficient ($\beta_4 = 0.0175$, $p < 0.001$) captures the annual increase in BEV market share attributable to technology diffusion and shifting consumer preferences, independent of price and income dynamics. The pump price coefficient ($\beta_1 = 0.161$, $p < 0.001$) implies that a EUR0.10/L increase in fuel prices lifts BEV share by approximately 1.6 percentage points, while a EUR0.50/L spike — comparable to the 2022 energy crisis — adds roughly 8.1 percentage points. Income enters negatively at the baseline ($\beta_2 = -0.292$, $p = 0.027$) but is moderated by an interaction with within-year price volatility ($\beta_3 = 0.0075$, $p < 0.001$), indicating that higher-income consumers respond more strongly to fuel price uncertainty when choosing EVs.

[10] Carbon pricing in climate policy: seven reasons, complementary instruments, and political economy considerations - Baranzini - 2017 - WIREs Climate Change - Wiley Online Library

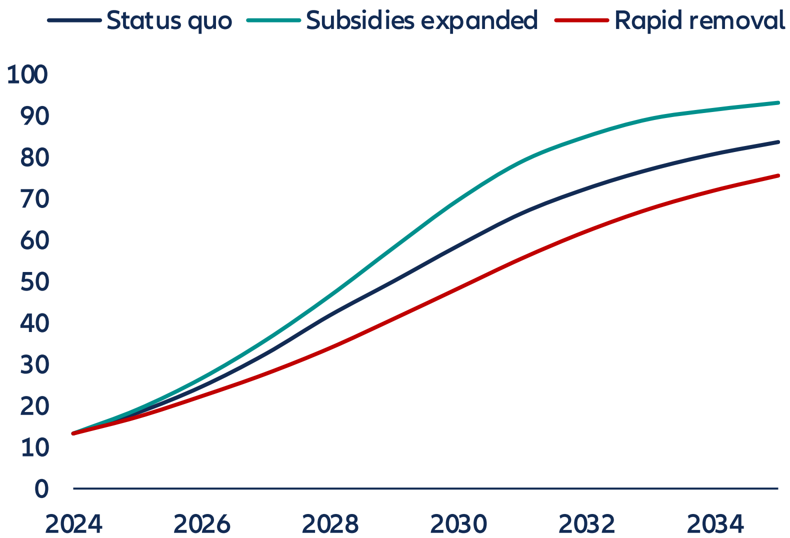
Figure 11: Number of EV subsidies schemes in the EU



Sources: IEA, Allianz Research

How far can subsidies alone take Europe? To answer this, we model BEV adoption across EU-27? countries under three policy scenarios, calibrating the subsidy effect from Martins et al. (2024)[11]. Their estimates show that a modest tax exemption worth around EUR1,000 adds 3.2% annual growth to BEV market share, a moderate grant below EUR5,000 adds 11.6% and a generous grant above EUR5,000 adds 26.4%. To analyze the BEV pathways as a response to subsidies, we built different scenarios (Figure 12). In the most optimistic scenario, every EU country raises grants to at least EUR5,000 through 2031, then gradually phasing them out by 2035 as BEV-ICE cost parity approaches, i.e., market share reaches 70% by 2030 and 93% by 2035. If instead governments simply maintain current programs through 2028 and then let them wind down, BEV share reaches only 59% by 2030. If political backlash or fiscal pressure leads to abrupt subsidy removal by 2029, adoption stalls at just 48% by 2030, slowly recovering to 76% by 2035 as the market falls back on cost economics and regulation alone. Therefore, even the most generous pathway falls 10pps short of the 80% BEV share by 2030 that would be consistent with meeting European decarbonization targets.

Figure 12: BEV adoption in EU-26 under different subsidies pathways (%)



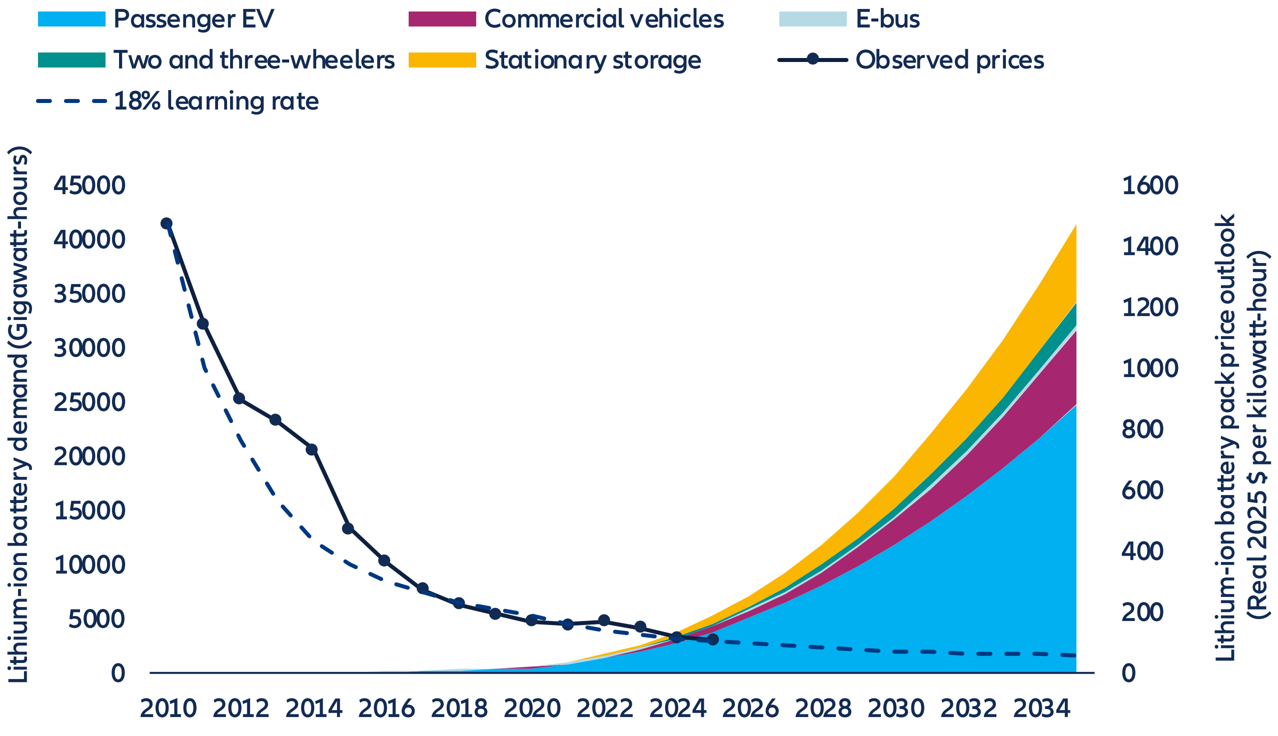
Sources: Allianz Research

[11] Assessing the effectiveness of financial incentives on electric vehicle adoption in Europe: Multi-period difference-in-difference approach - ScienceDirect

The evidence shows that subsidies accelerate adoption, but fiscal constraints might make it unrealistic for several economies to sustain large-scale purchase incentives through the full duration of the transition. Muehlegger and Rapson (2022)[12] show that low- and middle-income households are highly price-sensitive when it comes to EVs: a 10% price reduction can increase sales in this segment by roughly 21%. But no government can subsidize its way to 80% market share indefinitely. The fiscal cost escalates as adoption scales, and as Archsmith et al. (2022)[13] argue, what ultimately sustains mass adoption is not the subsidy itself but what they call "intrinsic demand", the point at which vehicle quality, model availability, charging convenience and total cost of ownership make the EV the obvious choice without a grant, which would dependent on speeding up the technological transition.

The technological case for EVs rests, above all, on one number: the price of a lithium-ion battery pack. Figure 13 shows that price has fallen from USD1,474 per kilowatt-hour in 2010 to USD108 in 2024, a 93% decline in 14 years. This trajectory follows a remarkably stable learning rate of approximately 18%: every doubling of cumulative manufactured capacity reduces costs by roughly a fifth. What makes this learning rate so consequential is that global battery demand is now entering an exponential phase. Passenger EVs already dominate, but demand from electric buses, commercial vehicles, two- and three-wheelers and stationary storage is compounding rapidly. It is projected that total demand will rise from roughly 1,000 GWh today to over 5,000 GWh by the early 2030s. Each of these segments feeds the same manufacturing base, driving cumulative production volumes higher and pulling costs down further regardless of what happens in any single market. If the 18% learning rate holds, pack prices are on track to reach approximately USD60–70/kWh by 2030 and could fall below USD55 by the mid-2030s. At those levels, the upfront purchase price of a BEV falls below that of an equivalent ICE vehicle in most segments without any subsidy at all. This is the point at which "intrinsic demand" takes over: the EV becomes the default choice not because governments pay people to buy it, but because it is simply the cheaper, better car.

Figure 13: Lithium-ion battery pack prices and demand



Sources: New Energy Outlook BNEF 2025, Allianz Research

[12] Subsidizing low- and middle-income adoption of electric vehicles: Quasi-experimental evidence from California - ScienceDirect

[13] Future Paths of Electric Vehicle Adoption in the United States: Predictable Determinants, Obstacles and Opportunities | NBER

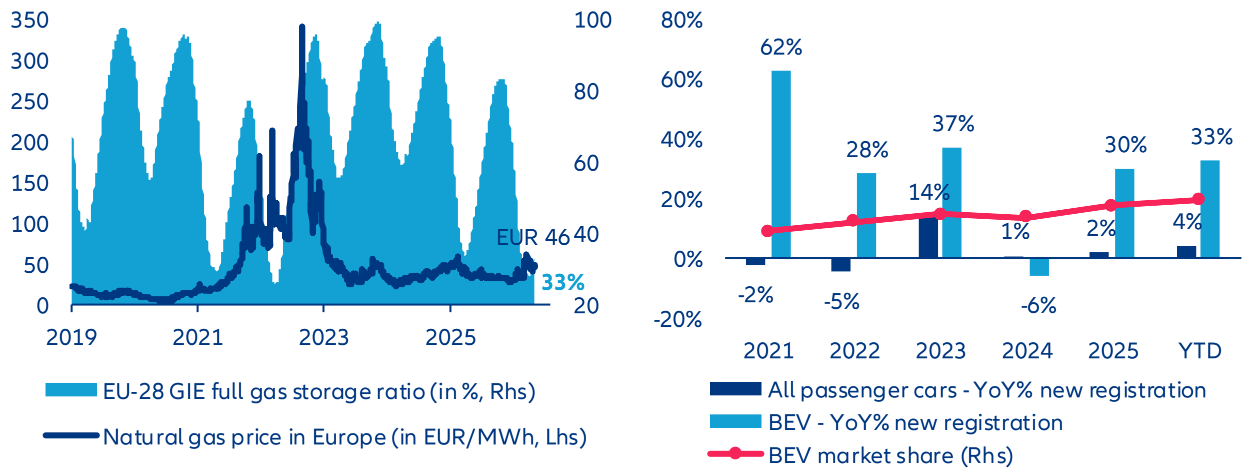
There is, however, a final condition that determines whether all of this – the carbon pricing, the subsidies, the falling battery costs – actually delivers on its environmental promise. An EV is only as clean as the electricity it runs on. Across the EU, grid carbon intensity has fallen from 0.45 tCO_2/MWh in 2005 to 0.22 in 2025, driven by a low carbon energy share that has grown from 47% to 70% of total generation (renewables grew from 16% to 50%). Under the EU NDC scenario, that trajectory continues: carbon intensity drops to 0.18 by 2030, 0.13 by 2035 and 0.07 by mid-century as low-carbon sources rise from 70% to 86% of generation. Each reduction compounds the emissions benefit of every EV already on the road. Moving from today's low-carbon electricity share of 70% to roughly 80% by 2035 would, on its own, reduce the per-vehicle carbon footprint of an EV by over 40%.

Germany illustrates the benefits of the grid transition. With a low-carbon share of 57% in 2025, an EV driven 14,000 km per year emits approximately 0.98 tCO_2, already 46.6% less than a new ICE car. But the gains from grid greening are steep: as Germany's low-carbon share rises to 64% by 2030 and 76% by 2035 under NDC commitments, the annual CO_2 saving per vehicle climbs from 0.85 tCO_2 today to 1.05 by 2030 and 1.39 by 2035, a 75.6% reduction relative to ICE. Therefore, grid decarbonization alone, without any change in driving behavior or vehicle efficiency, cuts the German EV's carbon footprint by 54% over a single decade. This creates a powerful feedback loop: the more EVs on the road, the stronger the case for grid investment; the greener the grid becomes, the larger the climate return on every vehicle already switched. Conversely, stalling grid decarbonization erodes the credibility of the entire transition. An 80% BEV fleet running on a 'brown' grid delivers far less than a 50% BEV fleet running on a clean one. Grid decarbonization is a critical complement of green transport in Europe.

Downside scenario: what if Middle East effects extend beyond Q2?

#1 - What if power price spikes? European EV demand proved unexpectedly resilient during the European energy crisis of 2022, with registrations rising ~28% while the total car market declined. This resilience reflected two temporary buffers: aggressive public intervention to cap electricity prices and still-generous purchase subsidies. Today, both buffers are weaker, notably subsidy schemes that broadly phased out over 2022-2023 in large markets like France and Germany. Fiscal space is more constrained and subsidy schemes are being scaled back or better targeted. Meanwhile, gas storage levels and geopolitical uncertainty—particularly around the Strait of Hormuz—reintroduce upside risks to power prices. In such a scenario, EV total cost of ownership (TCO) advantages could narrow, especially for households relying on spot-priced electricity. That said, structural mitigants are stronger than in 2022. Europe's power mix has shifted toward renewables, nuclear availability has improved (notably in France), and occasional negative wholesale prices in 2025 highlight underlying capacity surplus. In addition, diversified gas supply from Norway, the U.S. and LNG markets reduces extreme tail risks. Net-net, EV demand is no longer insulated but remains less exposed than often assumed—suggesting downside risk is real but likely contained rather than systemic.

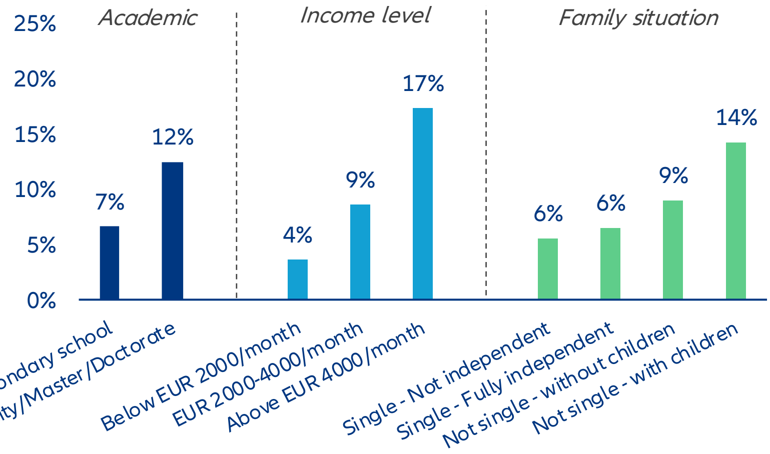
Figure 14 & 15: Natural gas price and storage rate in Europe / Annual passenger car registration trend & BEV market share in EU-27 since 2021



Market data as of 1rst May 2026. Sources: LSEG Datastream, ACEA, Allianz Research

#2 - What if consumer sentiment deteriorates? EV demand has historically shown relative resilience during macro slowdowns, largely because early adopters skew toward higher-income households with lower marginal sensitivity to energy and financing costs. This partially explains why EV penetration continued to rise even as broader European car demand remained roughly 30% below pre-COVID levels in recent years. However, this resilience has limits. EVs remain a discretionary, high-ticket purchase and global uncertainty—geopolitical tensions, energy volatility or tighter monetary policy—typically triggers postponement behavior. Rising interest rates directly affect monthly leasing costs, a key driver of EV affordability in Europe, while weaker confidence can delay fleet renewals by corporates. Policy can smooth the cycle but is becoming less supportive. With subsidy schemes being reduced or retargeted, the burden increasingly shifts to private demand elasticity. Even if new instruments (e.g. social leasing in France) provide targeted support, they are unlikely to fully offset a broad-based confidence shock. The key risk is not a collapse in penetration, but a slower-than-expected ramp-up, as consumers defer adoption despite improving technology fundamentals. In this context, EV demand should remain relatively more resilient than ICE demand in a downturn—but not immune, so a potential extension of the conflict in Iran could downsize EV sales growth to 10-15% this year (vs. almost -35% in Q1 2026) while total registration would contract again.

Figure 16: Profile ID of EV purchasers in Europe (2025)



Sources: EU drivers' view on electric cars : European Alternative Fuels Observatory Consumer Monitor 2025, ACEA, Allianz

#3 - What if automakers are lacking chips? EVs are structurally more exposed to semiconductor supply disruptions than ICE vehicles. Semiconductor content is roughly 2x higher in EVs—and up to 3x for advanced models integrating L2+/L3 functions—driven by power electronics, battery management systems and centralized computing architectures. This amplifies sensitivity to any disruption across the semiconductor value chain. A key vulnerability lies in upstream inputs such as helium, essential for advanced lithography. A supply shock linked to Middle East tensions could disrupt production at major foundries like TSMC or Samsung Electronics, with immediate spillovers to automotive chip supply. This risk compounds an already tight environment, as AI-driven demand continues to absorb leading-edge capacity. The consequences would be twofold: higher input costs and renewed production bottlenecks. Unlike 2021–2022, OEMs have improved inventory management and sourcing diversification, but buffers remain limited for advanced nodes and power semiconductors. In a severe scenario—particularly if disruptions coincide with tensions around the Strait of Hormuz—EV supply could tighten more than ICE, widening price differentials at the retail level. Based on a price sensibility matrix, we estimate that on average BEV retail prices would be twice as high as ICE models if chip price double or triple whatever state incentives allocate to boost sales (assuming a maximal subsidy rate at EUR 10,000). Bottom line: semiconductor risk is asymmetric. Any supply shock would disproportionately weigh on EV output and margins, reinforcing short-term downside risks to adoption.

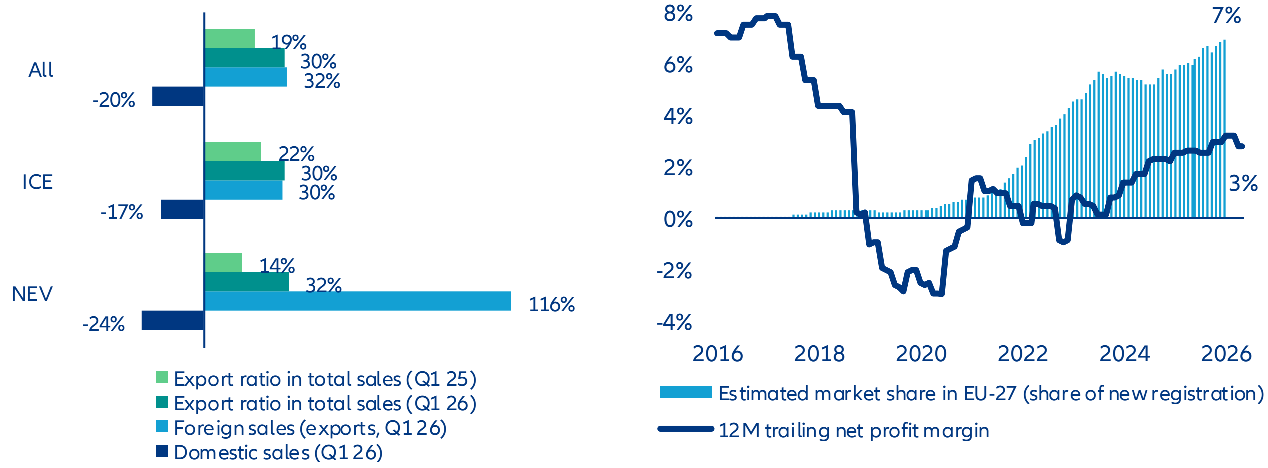
Table 1: BEV vs. ICE retail price spread matrix based on chip price volatility and state EV subsidy policy

BEV vs ICE - average retail price spread (End-2025 = ~30%)		Chip price - Multiplier ratio				
		2.0	2.5	3.0	4.0	5.0
EV subsidies on new vehicle purchase (in EUR)	1000	114%	118%	122%	129%	136%
	2000	109%	114%	118%	125%	132%
	3000	105%	109%	113%	121%	128%
	4000	100%	105%	109%	117%	124%
	5000	96%	101%	105%	113%	121%
	6000	92%	96%	101%	109%	117%
	7000	87%	92%	96%	105%	113%
	8000	83%	87%	92%	101%	109%
	9000	78%	83%	88%	97%	105%
	10000	74%	79%	83%	92%	101%

Sources: Allianz Research

#4 - What if China slows down? An extended Middle East conflict could push energy costs higher in Asia, weighing on China's already fragile domestic demand. Combined with intense price competition and margin compression among OEMs, this would likely accelerate the international expansion of Chinese manufacturers. Europe stands out as the primary target market, especially as access to the U.S. remains restricted. Chinese OEMs have already increased their European market share to ~7% in 2025, leveraging strong cost competitiveness and increasingly high product quality. In a context of domestic overcapacity, exports become a key adjustment variable. Trade policy remains the swing factor. While the European Commission has initiated anti-subsidy investigations, the current regulatory framework still allows significant market access. Notably, local content requirements remain partial, with battery sourcing rules tightening only progressively toward the end of the decade. If no additional tariff barriers emerge, a China-led supply push could compress EV price premiums versus ICE, accelerating mass-market adoption in Europe. Paradoxically, a negative external shock (China slowdown) could therefore support EV penetration in Europe through intensified competition and lower prices—at the cost of increased pressure on domestic OEM margins.

Figure 17 & 18 : Chinese automakers' passenger car sales breakdown (domestic & foreign market) / Net profit margin rate & estimated market share in EU-27 (all powertrains)



Sources : CAAM, Eurostat, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
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Allianz Research | 30 April 2026

Happy Labor Day? How geopolitics, immigration and AI will reshape work

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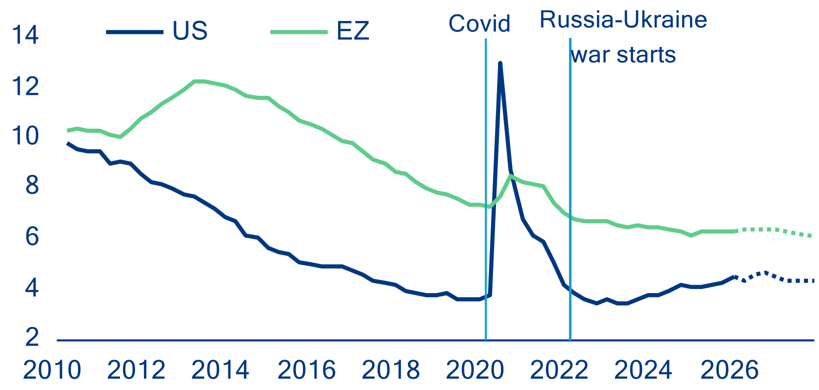
In Summary

- **US and European labor markets appear to be in good shape, with headline unemployment rates near historic lows. But three strong undercurrents (immigration policy, energy-price shock and AI) are churning beneath the calm surface.** Demand for workers had already begun to soften before the shock of the Middle East crisis: vacancy rates were easing and hiring rates cooling from uncertainty and the trade war. The shift in immigration policy in the US, the UK and Germany is adding to the mix: Fading immigration inflows have quickly translated into lower employment numbers. In the US, immigration has gone from contributing over half of job creation in 2024 to near-absent in 2025 – weighing on potential growth and risking sectoral labor shortages, while offering a short-term offset against rising unemployment.
- **The Iran shock will land unevenly on labor markets depending on countries' energy exposure and the duration of the conflict. Still, we see only a limited increase in unemployment rates in the range of 0.1-0.3pp.** If the crisis is resolved by end-May, unemployment rates would rise only in the most exposed European economies, and by +0.1pp at most, with 102,000 jobs lost in the Eurozone and half that in the US. Unlike in 2022, when labor hoarding led to falling unemployment, leaner buffers today could see any shock frontloaded into unemployment. In case of a prolonged closure of the Strait of Hormuz, the energy shock would hit Europe harder than the US, though higher labor protection standards provide some cushion: an estimated 225,000 jobs would be lost (0.13% of employment) in the Eurozone versus 126,000 (0.08%) in the US, with higher losses in Germany, Poland, Italy, France and Spain due to high energy exposure.
- **AI's labor-market effects are emerging, pointing to a K-shaped pattern, with youth and mid-level white-collar workers most at risk.** Early evidence shows pressure on younger and less experienced white-collar workers in routine cognitive tasks, while gains accrue to higher-skilled, AI-complementary roles. Since late 2022, higher AI adoption has been associated with larger increases in youth unemployment, with AI exposure explaining about 40% of cross-country variation (excluding high-unemployment economies). AI may therefore appear first not as job loss but as fewer entry points, weaker wage growth, and sharper polarization, with reallocation driven mainly by shifts in job composition rather than wage pressures in labor-intensive activities predicted by Baumol.
- **In the medium term, the AI labor-market impact will be substantial, uneven across countries and unprecedented in the scale of workforce reorganization required**. Over the next 1-3 years, AI is expected to affect 23.3% of jobs across major economies, with reorganization (10.4% of jobs) dominating augmentation (5.3%) and outright displacement (7.6%). The share of jobs affected ranges from 9.2% in Italy to 28.7% in the US, with the UK (17.7%), Germany (16.2%), France (14.7%) and Spain (12.4%) in between. That is equivalent to 52.5mn jobs in the US and 21.8mn across the major European economies. Our analysis does not account for potential AI-related job growth, which we expect to at least partially offset adverse employment effects. However, job displacement is likely to outpace job creation in the medium term, as firms adjust faster than workers, creating a temporary gap. Ultimately, whether – and how quickly – AI leads to job losses, reorganization, or new job creation will depend less on technology than on policy choices. AI-proofing policy frameworks will be critical: labor-market policies, including re- and upskilling, active labor-market programs, and social protection, will shape worker transitions. Taxation (including the relative treatment of labor and AI capital), firm incentives, and competition policy will determine whether AI is deployed to augment or replace labor and how broadly productivity gains are shared.

The buffer has thinned: labor markets and the new energy shock

Labor markets on both sides of the Atlantic have so far remained resilient despite successive shocks. In the Eurozone, the unemployment rate remained close to historical lows at 6.2% in March 2026, reflecting the job-rich post-Covid recovery and the strong employment response relative to modest growth. In the US, labor-market conditions have been weaker, with net job creation grinding to close to zero since early 2025, after dynamic outturns in 2024. However, stalling job creation mostly reflects sharply weakening labor supply amid very tight immigration policy, though plunging labor demand for federal employees has contributed, too. As a result of lower immigration inflows, the US unemployment rate has not crept up by much: from 4.1% in January 2025 to only 4.3% in March 2026.

Figure 1: Unemployment rate (%) including Allianz Research baseline forecasts.

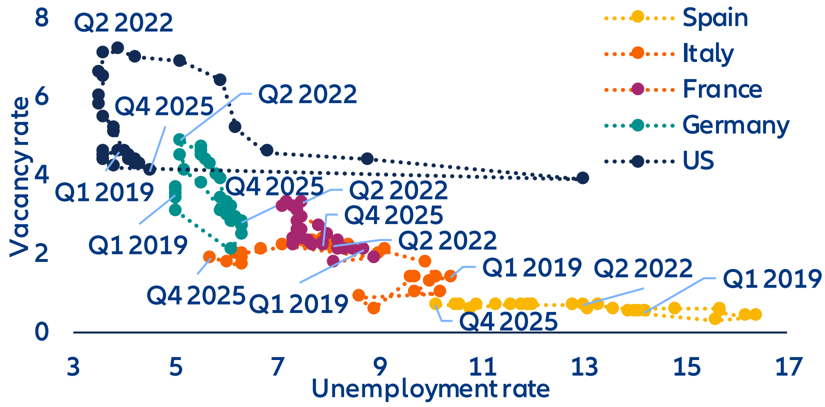


Sources: LSEG Datastream, Allianz Research

However, beneath the headline resilience, Eurozone labor market buffers have been progressively eroding. Even before the latest energy shock, labor demand had already cooled visibly (Figure 2): vacancy rates declined from 3.4% in Q4 2022 to 2.3% in Q4 2025; reported labor shortages eased across sectors, with firms seeing labor as a less limiting factor to production, and adjustment increasingly took place through hours worked rather than new hiring. Compared to the previous energy shock in 2022, firms no longer face acute rehiring constraints, reducing incentives to hoard labor. As a result, the Eurozone enters the current shock with a labor market that is less tight and with less buffers, despite still-low unemployment. Against this backdrop, the cooling of labor-market conditions should reduce the risk of second-round effects from higher commodity prices, dampening underlying wage and price pressures.

The US labor market also looks fragile, though the US economy is more insulated from surging energy prices. As a net energy exporter, the US has been far less exposed to energy-import-driven cost shocks since 2022, limiting the need for margin or labor adjustment and allowing firms to preserve buffers to a much greater extent. US corporates retain large cash buffers, structurally higher profit margins and a historically low interest-coverage ratio. Even so, labor demand looks increasingly soft. The job-hiring rate recently dipped to its lowest level since the pandemic (at 3.1%), signaling increasing hiring caution in an environment of persistent uncertainty, while the rollout of AI may have also deterred new hiring. Besides, the bulk of job creation has been concentrated in non-cyclical sectors, i.e. education & healthcare. Private jobs excluding education & healthcare have dipped by a cumulative -420k since January 2025.

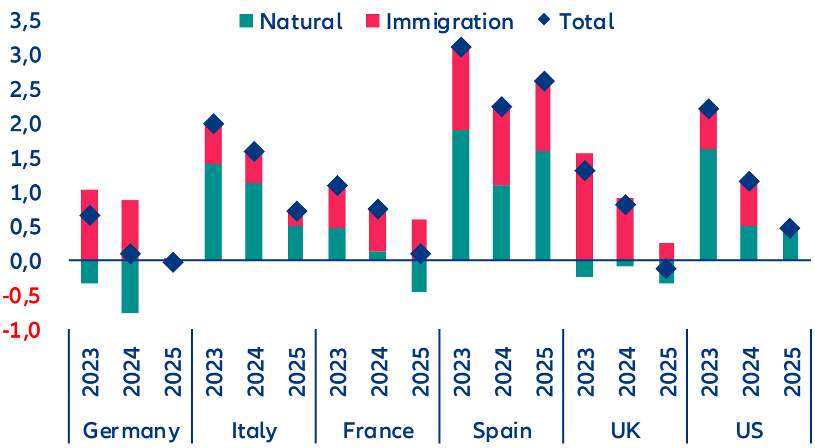
Figure 2: Beveridge curve (%)



Sources: LSEG Datastream, Allianz Research

Job growth was heavily supported by immigration over the past few years, but this dynamic is fading quickly, particularly in the US. Lower labor supply offers a short-term reprieve against rising unemployment rates but will weigh on potential growth. Figure 3 depicts the percentage annual change in employment (net job creation) since 2023 in the largest European countries and the US, with a breakdown of the contribution from immigration vs the resident. In Germany and the UK, employment of residents has decreased and net job creation has been concentrated among the new immigrants. Spain has seen strong employment growth of residents but new immigrants have contributed almost as much. In total, elevated immigration inflows have contributed to positive job creation in many European countries, alleviating labor shortages. However, job creation has stalled significantly in 2025, particularly in the UK, France and Italy. Firms have prioritized efforts to regain productivity rather than hiring to produce output. Consistent, harmonized working immigration inflows data are not available in European countries, but national data suggest a fading contribution in Germany, France and the UK. The US has seen a similar pattern of rapidly weakening job-creation momentum since 2025, driven almost entirely by reduced immigration inflows. Softer immigration inflows reduce labor supply and will thus weigh on potential growth. However, in the short term, scarcer labor is probably acting as a near-term brake on unemployment, even as it drags on growth potential.

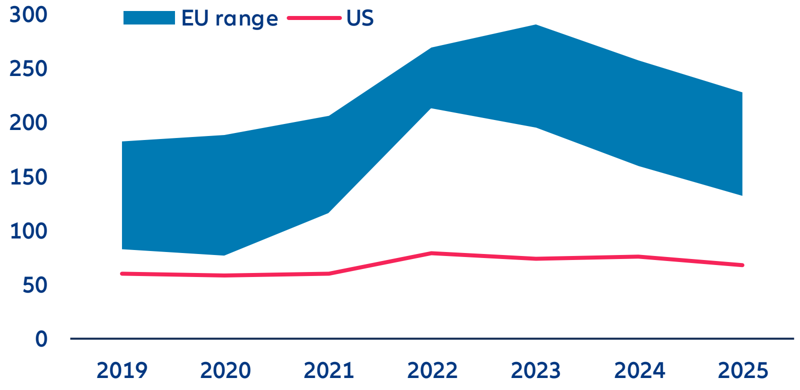
Figure 3: Employment change (% employed)



Sources: OECD, Census Bureau, Brookings, Allianz Research. Note: Permanent migration is defined by legal status, not duration of stay by the OECD. It includes migrants with a pathway to remain indefinitely such as refugees, family migrants, and some labor migrants, while those without such rights (e.g. students or seasonal workers) are not considered

Initial conditions are important for understanding how the renewed rise in energy prices propagates through labor markets. Although energy prices have fallen since their 2022 highs, industrial electricity prices in Europe remain significantly higher than pre-crisis levels and those in the US (Figure 4), even after adjusting for recent wholesale price stabilization. In contrast, US industrial electricity prices have remained stable over time due to abundant gas supply. Consequently, energy costs pose a persistent structural challenge for European firms rather than a temporary cyclical shock.

Figure 4: Industrial electricity prices, EUR/MWh including all taxes



Sources: EIA, Eurostat, Allianz Research. Notes: EU price range for consumption bands above 2,000 MWh.

There is far less margin to absorb renewed cost pressures than in 2022. Total energy costs are high in manufacturing, utilities and transport due to the scale of these sectors. The most energy-intensive industries, such as chemicals, steel and metals, paper, glass and ceramics, and cement, are concentrated in large industrial economies, particularly Germany, followed by France, Italy and Poland. Since margins have not fully recovered from the last energy crisis, firms have limited capacity to absorb new shocks through profitability alone.

As a result, the labor-market adjustment mechanism is shifting. While the 2022 energy shock was largely absorbed through margins, working-time reductions and labor hoarding in an overheated labor market, firms today face stronger incentives to adjust earlier if high energy prices persist. This time it might be different due to labor markets in a looser state where less labor hoarding will be seen. The immediate risk is not a sharp unemployment spike but a gradual erosion of labor-market resilience, in particular in countries that are suffering from structural economic weakness such as Germany.

Consistent with this gradual erosion, baseline projections already point to non-negligible job losses driven by broader macroeconomic headwinds. For the Iran war, we consider two scenarios (Figure 5). In our baseline scenario, traffic through the Strait of Hormuz gradually normalises by the end of May. This implies employment losses of around 102,000 in the Eurozone (with German, France and Spain already accounting for 68% of the total), and 51,000 in the US by 2026 – so in both cases the impact on the unemployment rates stemming solely from the Iran war under the baseline scenario remains very small (<0.1pp). These baseline dynamics provide the reference against which additional energy-related labor-market effects should be assessed.

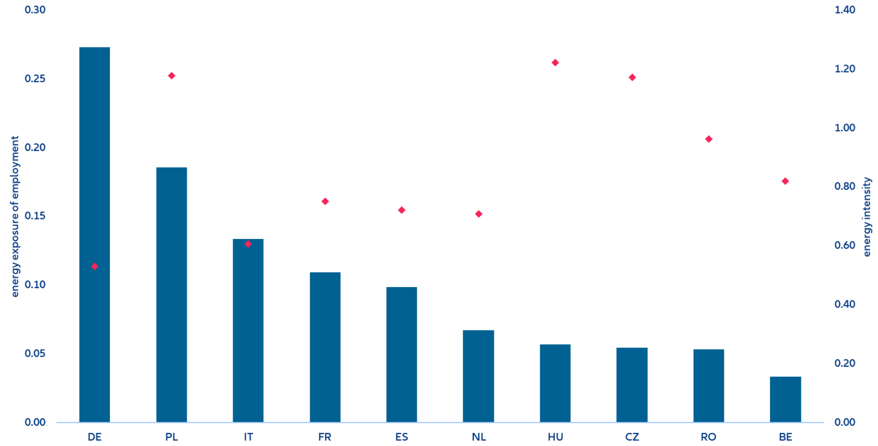
Figure 5: Middle East escalation scenario

		Feb 28th	Baseline (45% probability): Prolonged conflict with less than three months of global oil and gas disruption ***	Downside (35% probability): Long sustained escalation with prolonged disruption of global oil and gas supply ***
			2026	2026
Description			A deal allowing for a progressive normalization of oil and gas trade in May. In between partial re-routing, additional production (US, Russia) and strategic reserves being tapped covers >50% of Hormuz gap in oil and gas; no material destruction of oil/gas industry.	Hormuz closed for more than three months. Oil, gas and other commodity supply shortages (fertilizers, helium) transmits to confidence & equity shock exacerbating downturn.
Oil Brent (USD/bbl), Q2 average / eoy 2026		73$	90$ / 78$ ($140)*	130$ / 85$ ($180)*
Gas prices, annual avg.		32€	49€ (150€)*	65€ (200€)*
GDP	EZ	1.3%**	0.8%	0.2%
	US	2.6%**	2.1%	1.5%
Inflation	EZ	1.9%**	3.0% (3.4%)*	3.9% (4.9%)*
	US	2.5%**	3.2% (3.7%)*	4.0% (4.6%)*
EURUSD		1.18	1.15 (1.11)*	1.12 (1.05)*
Monetary Policy	ECB	2.0	2.25	2.75
	Fed	3.75	3.75	4.25
10y	DE	2.6	2.8 (3.3)*	3.1 (3.7)*
	US	4.0	4.5 (4.8)*	5.0 (5.7)*
Equity market	EZ	6	5 (-10)*	-25 (-30)*
	US	1	6 (-10)*	-20 (-25)*

Sources: Allianz Research. (*) max daily draw-down in brackets, (**) macro forecast for 2026 before the start of the Iran war, (***) probabilities add up to 80% with remaining upside and downside tail-risks (≤5%) omitted

Country-level exposure further shapes how energy shocks translate into labor-market outcomes. Energy intensity – measured as energy use relative to gross value added (GVA) – varies widely across Europe and is not, on its own, a reliable indicator of employment risk. Some highly energy-intensive economies, such as Hungary, the Czech Republic and Romania, account for a relatively small share of total Eurozone employment, despite requiring more energy per unit of output. By contrast, larger economies such as Germany, France, Italy and Spain combine more moderate energy intensity with much larger employment shares, making them more exposed to labor-market adjustment driven by energy prices (Figure 6). Poland stands out in this respect, combining a sizable employment share with high energy costs relative to GVA, which increases its vulnerability. This composition explains why employment risks are likely to materialize unevenly across countries, rather than through a synchronized rise in Eurozone unemployment.

Figure 6: Energy exposure of employment (index) and energy intensity (MWh per current 1,000 EUR), by country

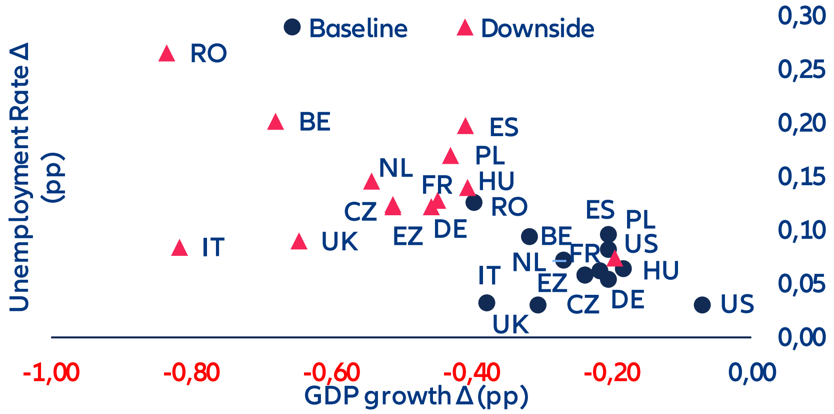


Sources: Eurostat, SBS, Allianz Research. Notes: Energy exposure are energy costs relative to gross value added (GVA) relative to EU employment shares by country. Energy intensity is calculated as energy use divided by gross value added; it gives an indication how many MWh are needed to generate 1,000 EUR of GVA.

In a downside scenario involving a prolonged closure of the Strait of Hormuz (Figure 6), employment effects would intensify but remain gradual. Rising fossil-fuel prices would weigh on value added through energy dependencies and supply bottlenecks in intermediate goods. Eurozone unemployment rates would rise by around +0.14pp on average, with the largest increases in Spain and Belgium (around +0.20pp). Eastern European economies – Czechia, Poland and Romania – would experience the steepest output losses, with unemployment rising by +0.12pp, +0.17pp and +0.27pp respectively. Germany and Italy, despite accounting for a large share of Eurozone employment, would see more contained increases (+0.12pp and +0.09pp), reflecting their comparatively higher energy efficiency. In absolute terms, the Eurozone would lose an estimated 225,000 jobs (0.13% of employment) versus 126,000 (0.08%) in the US. This is the subset of job losses attributed to the current energy shock in a downside scenario, compared to job losses due further economic factors, such as uncertainty, structural country-specific challenges and the ongoing trade war, to mention only a few, in our baseline scenario quantified in our latest insolvency report[1]. For comparison, for the US, about 11.9% of the total expected job losses of 428,000 in 2026 may be attributable to the energy-price shock in the baseline scenario, while for Germany it would be 12.4% of the 209,000 job losses and for France 7.1% of the 283,000.

The relatively narrow dispersion in unemployment outcomes, however, masks a wider divergence in how the shock actually lands on workers. More energy-intensive economies – particularly in Eastern Europe – will face sharper inflation and steeper margin compression, but much of their labor-market adjustment will occur through real wage erosion rather than outright job losses. Unemployment figures alone therefore understate the true dispersion of worker hardship: where prices rise fastest and margins are thinnest, it is purchasing power, not headcount, that takes the first hit.

Figure 7: Change in GDP and unemployment rate in a downside Iran war scenario compared to the baseline, in pp



Sources: Oxford Economics, Allianz Research

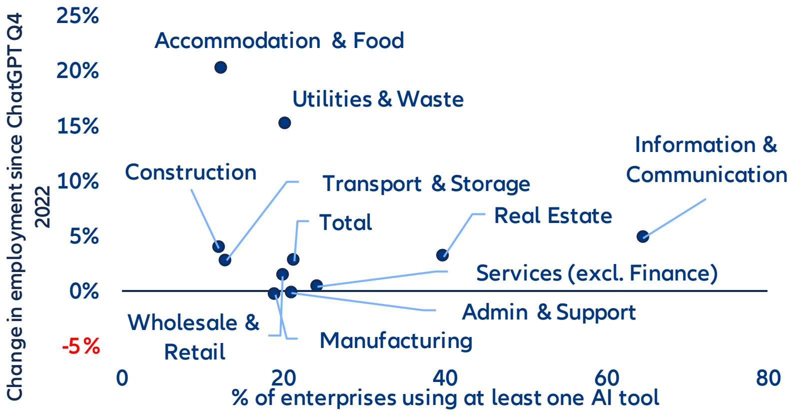
AI is reshaping adjustment pathways, not driving the shock

AI is altering how firms adjust to the energy shock, not replacing energy costs as its source. With AI already embedded in production processes, firms have gained greater scope for internal substitution, particularly in routine cognitive tasks and entry-level and mid-level white-collar roles . In a context of elevated energy costs and softer demand, this lowers the threshold at which firms move from margin and working-time adjustment to labor reallocation, primarily through hiring restraint and weaker entry-level absorption rather than outright job destruction. Energy prices determine the need to adjust; AI shapes the speed, intensity and distribution of that adjustment.

Accordingly, more AI-intensive sectors have displayed muted or zero employment growth in the Eurozone since late 2022, consistent with adjustment taking place through subdued hiring and task reorganization and slower entry-level inflow rather than outright job destruction (Figure 8). This reinforces the view that AI dampens measured employment dynamics even as underlying restructuring progresses.

[1] See also: Allianz Trade Insolvency report 2026

Figure 8: Eurozone employment growth by sector and AI usage

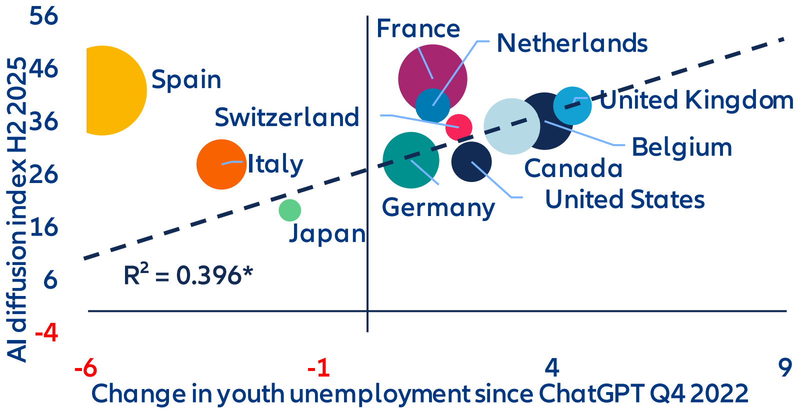


Sources: Eurostat, Allianz Research.

From a Baumol perspective, uneven productivity growth tends to generate upward pressure on relative wages in less automatable, labor-intensive activities. In the current environment, however, this channel is muted: with AI dampening aggregate labor-demand growth and firms adjusting primarily through hiring restraint and task reorganization, rebalancing is taking place mainly through employment composition rather than prices. Wage adjustment is therefore asymmetric, with compression in AI-exposed white-collar and entry-level roles, while less-AI-intensive sectors absorb labor without broad-based wage acceleration.

AI has not yet raised aggregate unemployment, but it is already reshaping labor-market adjustment in a k-shaped manner. Since late 2022, countries with higher AI diffusion have experienced larger increases in youth unemployment, with AI exposure explaining around 40% of the cross-country variation (excluding economies with structurally high youth unemployment), even as overall unemployment rates remain contained. At the same time, countries with comparable levels of AI uptake display markedly different labor-market outcomes, underscoring the decisive role of institutions, labor-market structures and adjustment channels. This divergence suggests that AI diffusion does not mechanically translate into net job losses, but neither is it benign: it shapes who adjusts, how quickly and at which margin, depending on how easily firms can substitute tasks, how flexibly workers can be reallocated and how effective retraining and activation policies are. Where adjustment mechanisms are rigid and transition support is weak, the risk is not an immediate rise in aggregate unemployment but persistent scarring at the margin, concentrated among younger workers and in routine-intensive entry-level occupations. The policy challenge therefore lies less in protecting aggregate employment than in shortening transitions and reducing polarization, through portable skills, flexible wage-setting and active labor-market policies that allow economies to capture the productivity upside of AI without incurring lasting distributional costs.

Figure 9: Global AI adoption and youth unemployment rate



*Sources: Microsoft Global AI Adoption Index, OECD, Allianz Research. *The regression line and R^2 exclude Spain and Italy. Both economies have structurally high youth-unemployment levels since the GFC and the sovereign-debt crisis. Bubble size corresponds to total unemployment rate.*

AI will have a more substantial impact on the labor market over the medium term

As AI moves from initial deployment to broader integration in business processes, the focus shifts to how it shapes labor markets in the near- to medium-term. While it remains difficult to predict how the technology will evolve and diffuse, the next one to three years are likely to be critical in determining the extent to which current capabilities translate into tangible changes in jobs and tasks. Against this backdrop, we assess the potential impact of AI on the current workforce across the Eurozone's four largest economies – Germany, France, Italy and Spain – as well as the UK and the US.

AI job exposure varies significantly across sectors. As a starting point for the analysis, we identify AI exposure across sectors, measuring the share of tasks that AI can perform significantly faster or more efficiently. Based on the recent analysis of Falck and Röhe (2026)[2], it ranges from around 14% in agriculture and 22–25% in construction and manufacturing to over 40% in finance and real estate and nearly 48% in information and communication. These differences in AI exposure are driven by task composition and are broadly stable across countries. Sectors dominated by cognitive work are substantially more exposed, while manual and physical tasks remain less affected.

Table 1: Heatmap of AI exposure across sectors (% jobs exposed) and countries' sectoral employment composition (% total employment)

	AI Exposure by Sector	Employment Shares by Sector					
		USA	UK	France	Germany	Italy	Spain
Wholesale and Retail Trade	34,5%	11,8%	13,7%	12,6%	12,5%	13,9%	14,5%
Manufacturing	24,7%	6,9%	7,2%	10,8%	18,2%	18,2%	12,0%
Administrative and technical activities	34,0%	12,3%	17,5%	11,2%	10,1%	10,9%	10,9%
Public Administration and Defence	38,3%	25,5%	4,8%	8,3%	7,2%	5,0%	6,6%
Education, Health and Social Work	30,5%	15,2%	8,6%	7,7%	6,9%	7,0%	7,1%
Accomodation and Recreation	29,4%	9,3%	10,2%	6,4%	5,0%	7,9%	10,6%
Construction	22,3%	4,6%	5,1%	6,6%	6,3%	6,7%	6,8%
Transportation and Storage	28,7%	3,6%	5,2%	5,0%	4,8%	4,9%	5,6%
Financial Activities	41,3%	5,0%	5,1%	4,7%	4,0%	3,2%	2,8%
Information and Communication	47,8%	1,5%	4,3%	3,6%	4,1%	3,2%	3,8%
Other Services	27,0%	3,3%	1,9%	2,5%	3,1%	3,1%	2,4%
Agriculture	13,5%	0,4%	0,6%	2,3%	1,1%	3,4%	3,5%
Electricity, Gas, Water supply	32,0%	0,3%	1,1%	1,7%	1,6%	1,7%	1,2%
Mining	25,5%	0,3%	0,1%	0,1%	0,2%	0,1%	0,2%
Total Economy AI Exposure		33,1%	32,1%	30,5%	31,0%	29,5%	30,0%

Sources: Falck and Röhe (2026), BLS, ONS, Eurostat, Allianz Research.

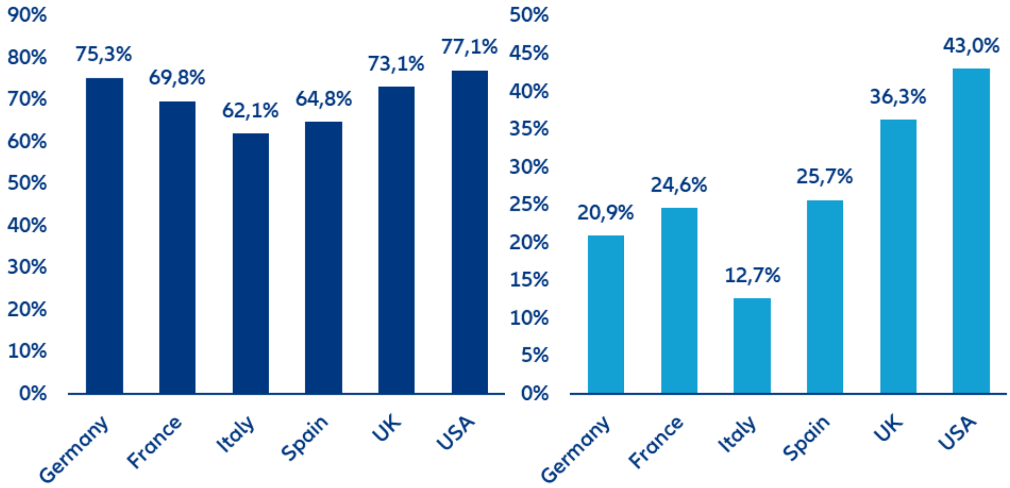
Cross-country labor market exposure to AI is driven by economies' sectoral employment composition (Table 1). While AI affects sectors similarly across countries, economies differ in how their workforce is distributed across those sectors. Translating sectoral exposure into country-level outcomes therefore depends on national employment structures, making economic composition the key driver of aggregate exposure. For example, high-exposure sectors such as ICT (48%), finance and real estate (41%), public administration and defense (26%), and health and education (15%) account for a larger share of employment in the US and partly the UK than in continental Europe. By contrast, Italy and Spain are more concentrated in lower-exposure sectors such as manufacturing (18% and 12%) and agriculture (3%), reducing aggregate exposure. Germany occupies a middle position among all countries, combining a sizable manufacturing base (18%) with more moderate service exposure. As a result, even with identical sectoral exposure, total AI impact differs substantially across countries, with service-oriented economies more exposed than industrial or agriculture-based ones.

Among AI-exposed jobs, we apply a job-transition framework that distinguishes between three adjustment channels: augmentation, reorganization and automation. These categories capture how firms are likely to

[2] For the estimates of the AI exposure across sectors, see: Falck, E., and O. Röhe (2026), "From potential to practice: AI exposure and adoption in European industries", SUERF Policy Brief 1358.

restructure tasks within the existing workforce rather than simply whether jobs disappear. Based on the analysis by Martin Richmond (2026), 22% of AI-exposed jobs are expected to be augmented, 44% reorganized and 33% automated.[3] Augmented jobs expand as productivity gains stimulate labor demand, reorganized jobs retain their core function but undergo task-level changes while for automated jobs humans are fully replaced by AI.

Figure 10: Workforce readiness for AI (left) and AI adoption among workers (right), in %

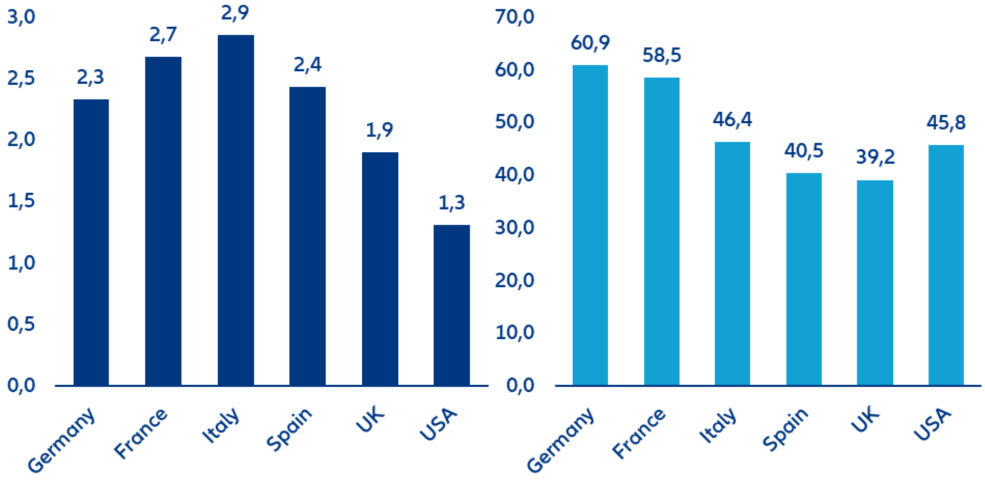


Sources: IMF, Eurostat, Bick et al. (2026), Allianz Research.

How these job-level effects translate into labor-market outcomes depends on country-specific conditions, which determine how quickly and to what extent AI exposure materializes over the next 1–3 years. We capture this through four scaling factors that amplify or cushion the impact. In the short term, AI adoption and workforce readiness for AI determine how quickly theoretical exposure causes actual impact (Figure 10). Countries such as the US and the UK lead both in current AI adoption and workforce readiness for AI, while Italy ranks the lowest among the countries considered, with Germany, France and Spain placed in the middle. In addition, we consider more structural factors such as unit labor costs and employment protection legislation that will shape firm incentives in adopting AI as well as influencing whether AI is used to complement or replace workers (Figure 11). High labor costs clearly increase the incentive to automate, while stronger employment protection legislation tends to slow displacement, at least in the near- to medium-term. For instance, the US has the weakest labor market regulations among the countries considered, leading to a higher expected pass-through of the theoretical AI exposure into actual labor market impact. By contrast, Italy has the strongest employment protection legislation, which shields the economy from the AI-related labor market disruption at least temporarily but also limits the economy's ability to benefit from AI-enabled productivity gains.

[3] These figures refer to AI-exposed jobs in the US economy. While we assume that the split between augmentation, reorganization and automation is broadly similar across countries, we adjust for country-specific structural factors, which results in lower estimated impacts in our analysis. Source: Martin Richmond, A. (2026), "The AI jobs transition framework: Mapping AI's near-term impact on jobs", OpenAI Research Paper.

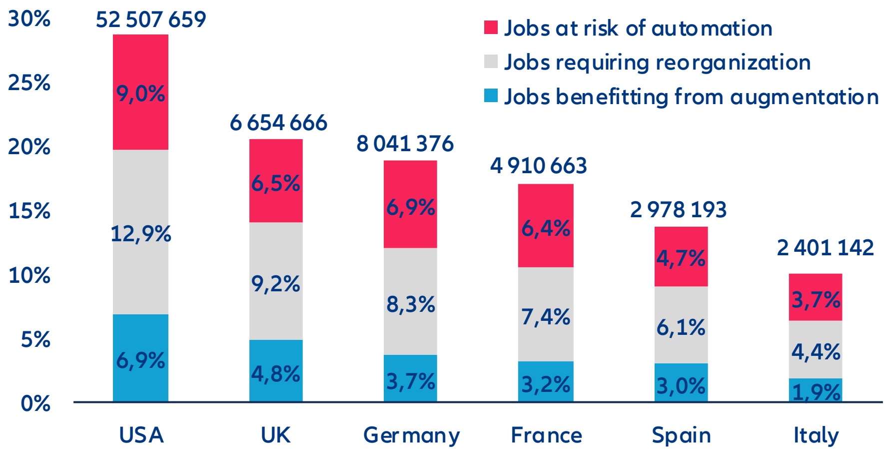
Figure 11: Employment protection legislation (indicator range from 0 (weak) to 6 (strong), left), and average annual hourly labor costs per employee (USD PPP, right)



Sources: OECD, ILO, Allianz Research.

The AI impact on the current workforce across major European economies and the US is substantial but far from uniform (Figure 12). To gauge the overall AI impact on the current workforce, we combine multiplicatively at the country level the sectoral AI exposure estimates, sectoral employment structures as well as the four scaling factors. The results are country-specific shares of jobs affected by AI, reflecting both technological potential and real-world constraints. Over the next 1–3 years, AI is expected to affect 23.3% of jobs across major economies, with reorganization (10.4% of jobs) dominating augmentation (5.3%) and outright displacement (7.6%). The share of jobs affected ranges from 9.2% in Italy to 28.7% in the US, with Spain (12.4%), France (14.7%), Germany (16.2%) and the UK (17.7%) in between. These differences closely mirror the distribution of employment across sectors. Economies with larger shares of finance, ICT and professional services exhibit substantially higher exposure than those with more manufacturing, construction or agriculture.

Figure 12: Medium-term AI impact on the current workforce across countries, in % and number of jobs



Source: Allianz Research.

Across all countries, the most important adjustment channel is job reorganization, while automation remains more material than augmentation. This reflects the fact that in most AI-exposed jobs, a human in the loop remains necessary, leading firms to reorganize tasks rather than replace workers, while only a smaller share of jobs meets the conditions for full automation and even fewer benefit from demand-driven expansion. In the US, 12.9% of jobs fall into the reorganization category, compared to 7–8% in Germany and the UK and 4–6% in Southern Europe. This finding is consistent with AI adoption currently targeting auxiliary or "side" tasks rather than core functions, reinforcing a complementary rather than substitutive dynamic. The share of jobs at risk ranges from 3.4% in Italy to

9.0% in the US, with most countries clustering around 5–6%. This gap between exposure and actual automation reflects the role of labor-market regulation, human-interaction requirements and demand effects, which limit the immediate substitutability of labor even where tasks are technically automatable. With respect to the expected real world labor impact of AI over the next 1-3 years, between 2% and 7% of jobs benefit from AI-driven expansion, thanks to augmentation, with the strongest effects in the US and UK. These gains arise where productivity improvements outweigh labor costs and hence increase employment.

The transatlantic divide in AI labor-market exposure is substantial and systematic: the US can expect more short-term AI-related labor market pain coupled with more long-term gain. In the US, 28.7% of jobs – equivalent to around 52.5mn positions – are affected by AI, compared to 14.5% (21.8mn jobs,) across the major European economies (Germany, France, Spain, Italy, UK). This pattern largely reflects structural differences in economic composition. The US economy is more heavily weighted toward high-exposure sectors such as information, finance and professional services, where AI can be deployed at scale, whereas Europe – particularly Southern Europe – retains a larger share of lower-exposure sectors such as manufacturing, construction and hospitality. The comparison between the US and Italy illustrates the extremes of this distribution: while the US combines high exposure (28.7%) with strong augmentation (6.9%), reorganization (12.9%) and automation (9.0%), Italy shows low exposure (9.2%) and correspondingly limited adjustment. As a result, the US faces greater transition pressures but also stronger potential productivity gains, whereas Europe's lower exposure provides short-term stability but risks slower productivity growth and weaker AI-driven job creation over time.

A key limitation of these results is that they focus on the current workforce and do not account for AI-induced job creation. Our estimates should therefore not be interpreted as direct changes in headline employment. Historically, technological change has followed a consistent pattern: job destruction by automating tasks is offset over time by job creation through new tasks and demand effects.[4] However, this adjustment is neither immediate nor evenly distributed. Job creation typically lags behind displacement due to reallocation frictions and skill mismatches. The time mismatch may be more pronounced for AI, as diffusion and adoption rates are unprecedented. Taken together, AI-driven job losses are likely to be partially offset over time, but with meaningful transition costs.

The scale of required job creation is material. Our results indicate that around 25mn jobs across the US and major European economies are subject to automation. Fully offsetting displacement would therefore require the creation of a comparable number of new jobs over the medium term. Early evidence shows strong growth in AI-related roles with some 1.6m new AI-related jobs like AI engineers, forward-deployed engineers and data annotators created over the past two years plus 600,000 jobs driven by the data center boom with the US the key beneficiary[5] However, diffusion across sectors and countries remains uncertain. Net employment outcomes will depend on the speed of job creation relative to displacement and on the efficiency of worker reallocation. In ageing economies, declining labor supply may additionally cushion the employment impact as labor shortages persist in many sectors.

Preparing policy frameworks for the AI age

AI does not determine labor-market outcomes on its own. The technology sets the direction, but policy and institutions determine the speed, scale and distribution of its effects including the balance between job augmentation, reorganization and automation.[6] What distinguishes the current wave of AI is not only its breadth across sectors, but the pace and depth of task-level transformation, amounting to one of the most significant labor-market transitions in recent decades. Existing policy frameworks, designed for slower, occupation-based change, are therefore ill-suited to the AI age, where jobs are continuously reconfigured. To remain effective, education systems, active labor-market policies, social protection systems, tax structures and firm-level incentives must be updated and rethought to become more forward-looking. The main challenge is therefore not preventing

[4] See: Acemoglu, Daron, and Pascual Restrepo (2019), "Automation and New Tasks: How Technology Displaces and Reinstates Labor", *Journal of Economic Perspectives,* 33 (2), 3-30; Hötte, K. Somers, M., and A. Theodorakopoulos (2023), "Technology and jobs: A systematic literature review", *Technological Forecasting and Social Change*, 194, https://doi.org/10.1016/j.techfore.2023.122750.

5 See: Linkedin (2026), linkedIn-labor-market-report-building-a-future-of-work-that-works-jan-2026.pdf

[6] For an excellent overview of the importance of policies and on how to make AI "pro-worker", see: Acemoglu, D., Autor, D., and S. Johnson (2026), "Building Pro-Worker Artificial Intelligence", NBER Working Paper 34854, https://www.nber.org/papers/w34854.

adjustment, but to shape its direction and distribution, ensuring that productivity gains translate into broad-based improvements in employment, wages and job quality rather than displacement or increased inequality.

First, labor-market policy will determine how smoothly workers move between tasks, roles and sectors. Traditional active labor-market policies will need to evolve from reactive systems focused on unemployment toward anticipatory, data-driven and task-based approaches. This includes early-warning systems to identify AI exposure, individual learning accounts, and rapid retraining mechanisms that can be deployed before displacement occurs. Public employment services should be strengthened with real-time labor-market intelligence and AI-enabled matching tools that connect workers to adjacent roles based on transferable skills. Wage insurance, portable benefits and unemployment support linked to retraining will be essential to reduce income risk and support mobility. Since most AI-exposed jobs are likely to be redesigned rather than eliminated, the priority is not protecting existing tasks, but enabling workers to transition quickly into new ones. Since most AI-exposed jobs are likely to be redesigned rather than eliminated, the priority is not protecting existing tasks, but enabling workers to transition quickly into new ones.[7]

Second, education and skills will become more binding constraints. As AI increasingly complements cognitive work, the ability to work alongside AI systems will determine whether workers benefit from productivity gains or face substitution.[8] This requires embedding AI literacy as a core competence across education systems, alongside stronger emphasis on critical thinking, problem-solving, adaptability and communication. Education systems will need to become more modular and continuous, with expanded mid-career training, short-cycle credentials and closer alignment with labor-market needs. Governments should also support sector-specific training ecosystems and local partnerships between firms, unions and education providers to accelerate adjustment in regions and occupations most exposed to disruption.

Third, firm incentives, including taxation and regulations will shape whether AI is used to replace labor or to augment it.[9] A key issue is that labor is typically taxed more heavily than capital, which creates a structural bias toward automation even when augmentation would be more socially efficient. Tax systems therefore need to be adjusted to incentivize augmentation rather than the substitution of labor. One proposal is a "robot tax," aimed at reducing this bias by taxing automation more in line with labor—for example, by limiting preferential tax treatment of capital that substitutes for workers or linking levies to displacement. Implementation challenges would need to be addressed, including defining what constitutes a "robot," avoiding disincentives to innovation, and limiting adverse competitiveness effects if policies are not coordinated internationally. Whereas broad-based taxation of automation may be difficult to implement in practice, more targeted approaches could be more effective. These include tax incentives to support augmentation, such as credits for training and reskilling, incentives for job redesign, and support for firms investing in human capital alongside technology. Frameworks can also encourage sharing of productivity gains through wages, profit-sharing, or reduced working time. Policies to support AI adoption among SMEs, strengthen competition, and prevent excessive concentration of productivity gains will also be critical.[10] Even small differences in these incentives can lead to large differences in business and labor-market outcomes.

Finally, the broader societal response will matter: Public concerns around economic disruption, trust and safety, and the distribution of gains may lead to calls for stricter AI regulation and shape how AI diffuses across economies.[11] Maintaining social legitimacy will therefore be critical. This requires stronger transparency in the use of AI in the workplace, appropriate safeguards around its deployment, and mechanisms for worker voice and participation in how AI is introduced and used. Ensuring that the gains from AI are broadly shared will also be central to sustaining support for adoption. This includes profit-sharing mechanisms and competition frameworks that prevent excessive concentration of AI-driven rents. A complementary priority is to ensure that AI adoption is broad-based rather than concentrated. This calls for policies that lower barriers to adoption, including support for SMEs, promote competition, and facilitate the diffusion of AI capabilities across firms and sectors. Such measures will be critical to ensure that productivity gains translate into wider economic benefits and employment growth.

[7] See: Martin Richmond, A. (2026), "The AI jobs transition framework: Mapping AI's near-term impact on jobs", OpenAI Research Paper.

[8] See: Jaumotte, F. et al. (2026), "Bridging Skill Gaps for the Future: New Jobs Creation in the AI Age", IMF SDN/2026/001.

[9] See: Acemoglu, D., Autor, D., and S. Johnson (2026), "Building Pro-Worker Artificial Intelligence", NBER Working Paper 34854, https://www.nber.org/papers/w34854.

[10] See: Ferrando, A. et al. (2026), "Adopting and Investing in AI: Evidence from Euro Area Firms", ECB Economic Bulletin, Issue 2/2026.

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